STUDIO

FirstPass STUDIO

USER GUIDE

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

VERSION 1 · OCTOBER 2026 · BY PIVT

FirstPass STUDIO

● LIVE FROM PLAYHQ

USER GUIDE

Graphics that follow the game.

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

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ABOUT THIS GUIDE

Graphics that follow the game.

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type during the game.

This guide starts with the way most nights run, one operator and Studio doing the rest, then covers running it by hand, setting up a competition, sponsors, athletes and the Stream Deck.

5 s

The live score, quarter and clock read from PlayHQ every 5 seconds while the game is scored.

31

Broadcast graphics, from the scoreboard and ladder to the sponsor wall, plus a test pattern.

17

Competition logo animations, and six intro and outro styles.

1

Operator on a followed night. Sequences run the run-up, the breaks and full time.

IN

PlayHQ

The fixture, ladder, line-ups and the live score from the scorer's tablet. No account, no key.

→

ON YOUR MAC

FirstPass Studio

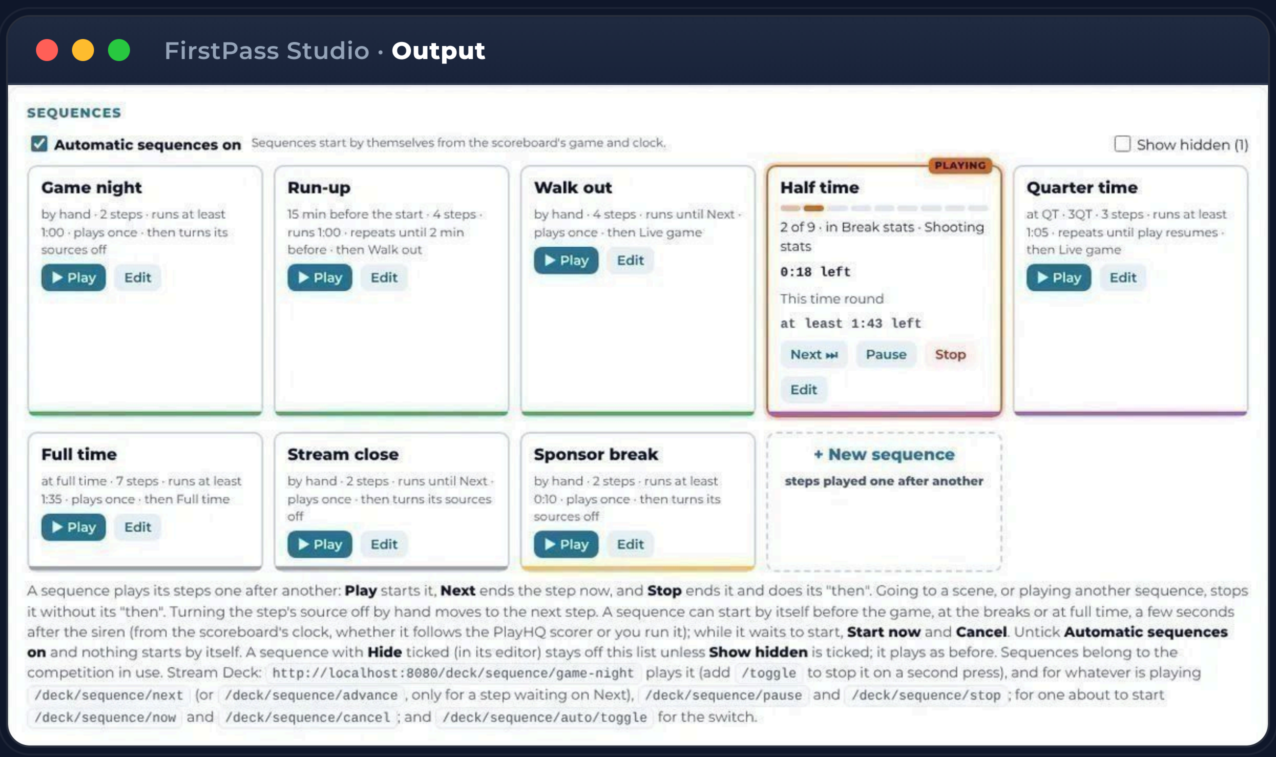
Builds one 1920 × 1080 scene from layers: the base scene, automatic layers, scenes and sequences.

→

OUT

Your switcher

A display, a Blackmagic device or NDI, with fill and key or a chroma colour over the cameras.



SEQUENCES

☑ **Automatic sequences on** Sequences start by themselves from the scoreboard's game and clock.

☐ Show hidden (1)

Game night
by hand · 2 steps · runs at least 1:00 · plays once · then turns its sources off

Run-up
15 min before the start · 4 steps · runs 1:00 · repeats until 2 min before · then Walk out

Walk out
by hand · 4 steps · runs until Next · plays once · then Live game

Half time
2 of 9 · in Break stats · Shooting stats
`0:18 left`
This time round
`at least 1:43 left`

Quarter time
at QT · 3QT · 3 steps · runs at least 1:05 · repeats until play resumes · then Live game

Full time
at full time · 7 steps · runs at least 1:35 · plays once · then Full time

Stream close
by hand · 2 steps · runs until Next · plays once · then turns its sources off

Sponsor break
by hand · 2 steps · runs at least 0:10 · plays once · then turns its sources off

+ New sequence
steps played one after another

A sequence plays its steps one after another: **Play** starts it, **Next** ends the step now, and **Stop** ends it and does its "then". Going to a scene, or playing another sequence, stops it without its "then". Turning the step's source off by hand moves to the next step. A sequence can start by itself before the game, at the breaks or at full time, a few seconds after the siren (from the scoreboard's clock, whether it follows the PlayHQ scorer or you run it); while it waits to start, **Start now** and **Cancel**. Untick **Automatic sequences on** and nothing starts by itself. A sequence with **Hide** ticked (in its editor) stays off this list unless **Show hidden** is ticked; it plays as before. Sequences belong to the competition in use. Stream Deck: `http://localhost:8080/deck/sequence/game-night` plays it (add `/toggle` to stop it on a second press), and for whatever is playing `/deck/sequence/next` (or `/deck/sequence/advance`, only for a step waiting on Next), `/deck/sequence/pause` and `/deck/sequence/stop`; for one about to start `/deck/sequence/now` and `/deck/sequence/cancel`; and `/deck/sequence/auto/toggle` for the switch.

The Output page runs the show; the super shot plays by itself.

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Every screenshot comes from Studio running the fictional Elite Netball League: eight clubs named after animals, three divisions, a live game and fictional sponsors.

CHAPTER 1 OF 11

What Studio is

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

1 The words used in this guide
2 The pages you work in
3 The sample competition in this guide

Studio does one job on stream night: graphics. Your cameras, switcher and streaming encoder stay as they are. Studio hands the switcher a picture with transparency (fill and key, or a chroma colour) so the graphics sit over the vision.

1.1 The words used in this guide

Studio	The whole thing: the Mac app, the pages you work in, and the graphics it puts on screen.
On air, off air	Studio is on air while a Studio window is open. On air it talks to PlayHQ, serves the graphics and answers the Stream Deck. Closing the last window takes it off air.
A graphic, or a page	One thing that can go on screen: the scoreboard, a ladder, a lower third, an animation. Each is a web page that Studio draws.
A source	A graphic placed in the scene, with its own position, options and fade. The scoreboard is one source; a ladder is another.
The scene	Studio's own 1920 × 1080 frame, built from sources switched on and off. It is what the switcher receives.
A scene (on the Scenes list)	A saved look: which sources are on and which are off. "Live game" and "Clear" are scenes.
A sequence	A list of steps played in order from one press: a run-up, a quarter-time break, a full-time wrap.
The competition	Everything that belongs to one league: its PlayHQ link, names, rules, colours, logos, sponsors and athletes. Studio keeps a setup per competition and uses one at a time.
The data folder	Where Studio saves everything, outside the app, so updates never touch it.

1.2 The pages you work in

Every Studio window has the same tab bar: **Output**, **Production Planner**, **Scoreboard control**, **Recordings**, **Socials**, **Competitions setup**, **Sponsors**, **Athletes**, **Stream Deck** and **Docs** (⌘1 to ⌘9, then ⌘0, in the app). Output is where the show is built and sent out; the Production Planner plans the court and the commentators for each night; Scoreboard control runs the board; Recordings keeps each PlayHQ game on air; Socials makes the night's pictures for social media; the rest are set up before the night.

> **Studio does one job on stream night: graphics.**
>
> CHAPTER 1 · WHAT STUDIO IS

1.3 The sample competition in this guide

The screenshots use a fictional league, the **Elite Netball League**, with eight clubs named after animals (Falcons, Panthers, Stingrays, Cobras, Grizzlies, Hawks, Rhinos and Wolves) across Premier, Men's and Under 21 divisions, playing at the Metro Arena. Its sponsors are fictional too. Everything you see is what Studio draws for any competition once its setup is in.

CHAPTER 2 OF 11

Getting started

Install the app, import or make your competition, and learn the one rule that matters on stream night: keep a Studio window open.

1 Install · 2 On air and off air · 3 The 1P menu · 4 Before you go on air

2.1 Install

01
Download `FirstPass Studio.dmg`, drag the app to Applications and open it. It runs on Apple silicon Macs (M1 and later).

02
The **1P** icon appears in the menu bar, off air. Choose **Open Studio**. A short splash shows, Studio goes on air and the Output page opens in the app's own window.

03
Go to **Competitions setup** and either **Import a template file** (your league's `.json`, with its logos and sponsors inside) or **New competition** and fill it in (Chapter 6).

04
Back on **Output**, add the graphics you need with **Add page**, and turn on the video output that feeds your switcher (Chapter 3).

2.2 On air and off air

Studio only runs while a Studio window is open. Opening one (Open Studio, any page in the 1P menu, or ⌘N) puts it on air; closing the last one takes it off air, and the graphics and Stream Deck get no answer until a window opens again. Closing the last window while on air asks first, and so does ⌘Q, so a stray keypress can't stop a stream. Leave the Output page open all night.

The status line under the Output page's title shows the game on the board, whether scores and clock follow PlayHQ, how long since PlayHQ last answered (amber when overdue, red after a minute of errors) and whether the video output is on. If anything is wrong, this line says so on the page you keep open.

2.3 The 1P menu

Open Studio, Scoreboard control, Competitions setup, Copy the Studio link, Devices…, Install the Stream Deck plugin, Back up… and Restore…, Settings folder (iCloud Drive), Open the settings folder, Open the debug log, Open the app log, Restart Studio, Open at login, Only this Mac can change things, What's new…, Check for updates, Stop Studio (while on air) and Quit. Updates download in the background and install when the app quits, never mid-stream; **Restart to update** installs one straight away.

LINKS Studio's links start with the Mac's Bonjour name, such as `http://matts-macbook-pro.local:8080/`, so the same link works from a phone, another computer or a Stream Deck on the same network. `http://localhost:8080` works on the Mac itself.

> **The one rule that matters on stream night: keep a Studio window open.**
>
> CHAPTER 2 · GETTING STARTED

2.4 Before you go on air

01
Open the app. The Output page opens and Studio goes on air. Keep a window open all night.

02
Right competition. Competitions setup shows the one in use at the top. Check its PlayHQ link, the court to follow and the scoring window.

03
Right game. On Scoreboard control, the picked game is tonight's first on your court and Follow PlayHQ is on.

04
Scores arrive. Once the scorer has started on PlayHQ, the control page says the game is being live scored and the board moves. Nothing is read before the scoring window opens, so a quiet board before then is normal.

05
Scene and output. On Output, turn on the display, NDI or Blackmagic output your switcher takes, check fill and key or chroma, and show one graphic to see it land.

06
Sound. Play a few seconds of music and an animation with sound, and check the sound only comes out where you want it.

07
Stream Deck. Press a Source key and watch the Sources list follow it. A score key switches scoring to Manual, so put it back to Follow PlayHQ after testing.

08
Sponsors. Tonight's spots and ads are on; the sponsor report starts counting when the output turns on.

CHAPTER 3 OF 11

Game night, the automated way

This is how most nights run: one operator, PlayHQ scoring the game, a base scene that stays on, automatic layers that fire by themselves, and sequences that handle the run-up, the breaks and full time.

1 How it fits together
2 Scoreboard control
3 The Output page
4 The base scene
5 Automatic layers
6 Scenes
7 Sequences
8 Music, playlists and mics
9 Video output
10 A night, start to finish
11 The Production Planner
12 Socials
13 Recordings

3.1 How it fits together

Four things do the work, and each one is set up once:

- **Follow PlayHQ.** The scoreboard reads the live score, quarter and clock from the same feed the PlayHQ game page uses, every 5 seconds by default (as often as every second), while someone live-scores the game on PlayHQ. Automatic game selection moves the board to each next game on your court as its electronic scoring starts.
- **The base scene.** A saved scene with the sources that stay on through the game: the scoreboard, the title banner, the branding logos and the score ticker.
- **Automatic layers.** Sources that stay on but only show when something happens: the super shot celebration, the hot streaks and game moments, the super shot tab on the scoreboard, the naming-rights sponsor's tab. They watch the board and play by themselves.
- **Sequences.** Steps played in order for the parts of the night that aren't live play: the run-up, the walk out, quarter and half time, full time and the close. A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.

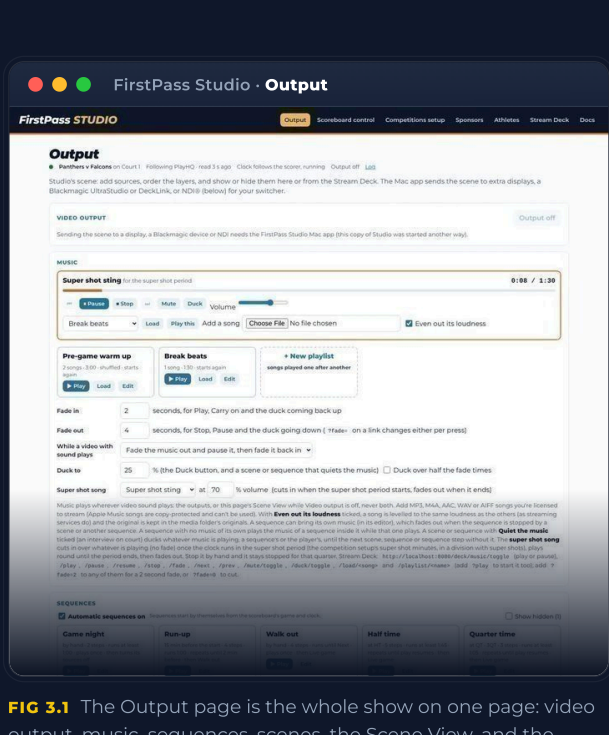


FIG 3.1 The Output page is the whole show on one page: video output, music, sequences, scenes, the Scene View, and the sources with their settings.

> **A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.**
>
> CHAPTER 3 · GAME NIGHT, THE AUTOMATED WAY

3.2 Scoreboard control

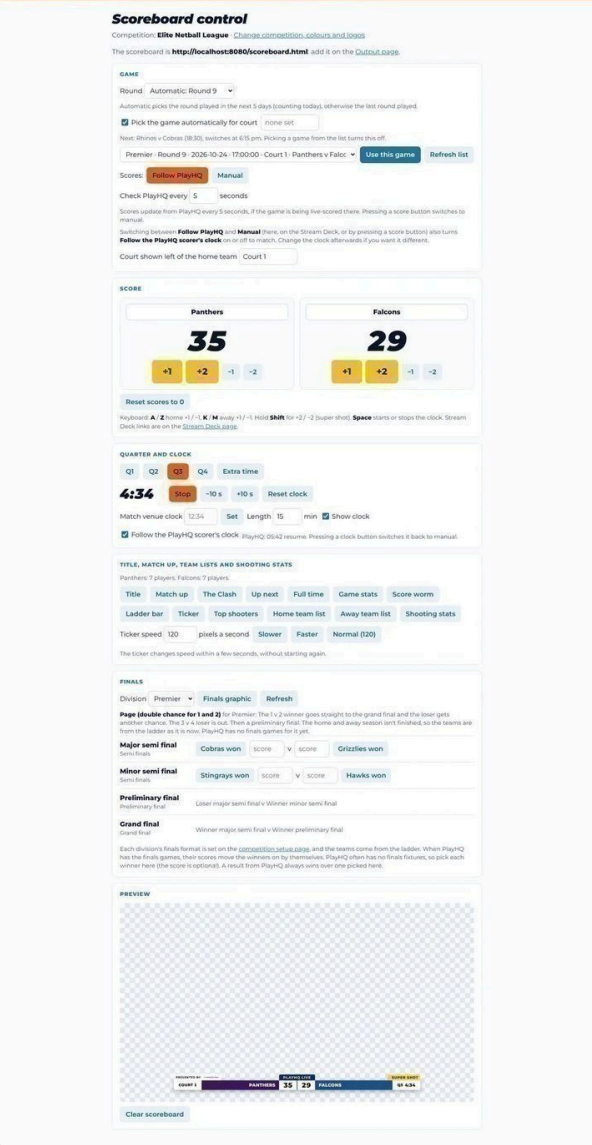


FIG 3.2 Scoreboard control: the game, the scoring mode, the score, the quarter and clock, quick links to the game graphics, finals winners, and a preview of the board.

ALWAYS A GAME ON

When Studio starts, it puts a game on the board with Follow PlayHQ on: the game on now on your court, otherwise the first game of tonight on that court. Games are picked by game night, not by round: the list holds every division's games that date, whatever round each is up to, so one division on Round 10 and another on Finals Round 1 sit side by side, each game showing its own round. Tonight is found by date (the first game night in the next 5 days, counting today), so a Tuesday night is found from the Friday before. You can pick another night from the **Game night** list, which names each night's rounds; its games on your court are then followed by their time (the one due now, otherwise the next to come). Set it back to Automatic afterwards.

AUTOMATIC GAME SELECTION

With **Pick the game automatically for court** ticked, the board shows the first game of the night on that court, then switches to each next game when its electronic scoring starts (**Electronic scoring starts N minutes before the game** on the setup page, 10 by default, 15 in the sample league), with the quarter back to Q1 and a full clock. In finals it keeps the final on until PlayHQ marks it finished, in case of extra time. Picking a game from the list by hand turns automatic selection off; tick it again to go back. When the Production Planner has a court for tonight, that court is set here when the night arrives, with **from the Production Planner** beside it. Changing it by hand still works, and the Planner flags it.

FOLLOW PLAYHQ

With **Follow PlayHQ** on, the score and quarter update every 5 seconds from PlayHQ's live scoring, and **Follow the PlayHQ scorer's clock** runs the clock from the scorer's clock. The clock matches the scores: switching to Manual, or pressing any score button, turns clock-following off as well, and the control page says so. Press a clock button to run the clock by hand while scores still follow PlayHQ. To line the clock up with the venue, type the venue's time in **Match venue clock**.

NOTHING BEFORE THE WINDOW Studio doesn't ask PlayHQ's live feed for a game until its electronic scoring window opens, whether the game was picked automatically or by hand. The control page says when it will start checking.

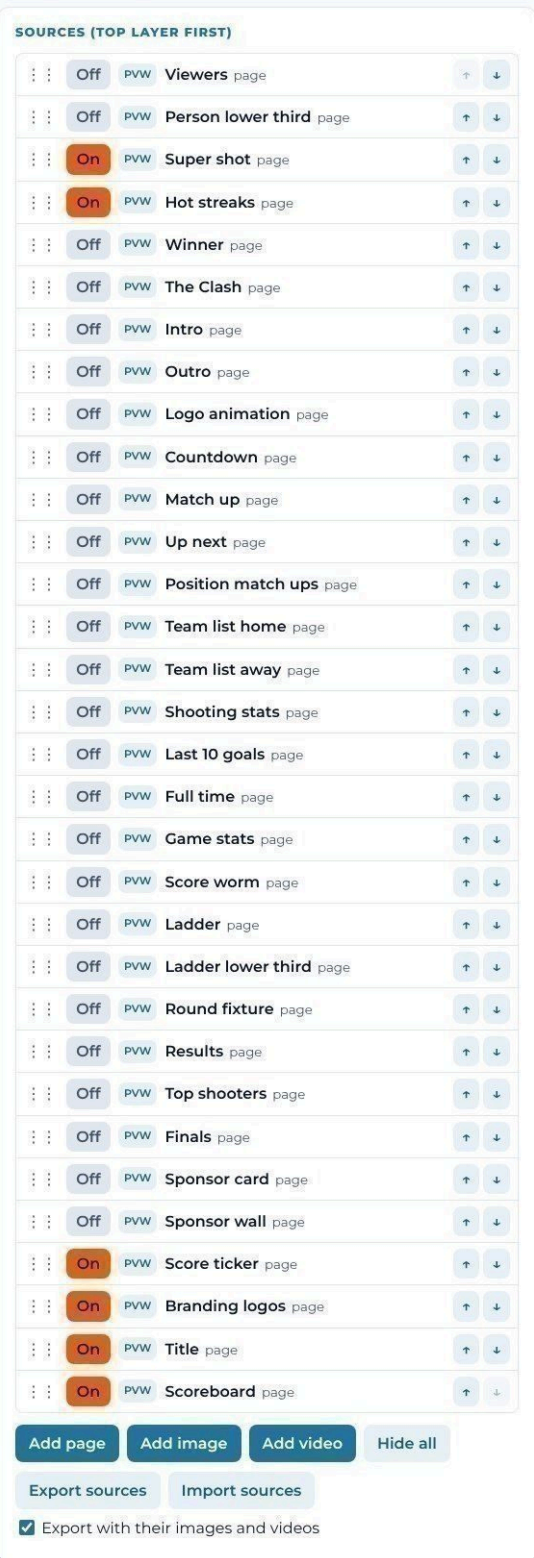
3.3 The Output page

The Output page builds Studio's own scene: sources drawn as layers on one 1920 × 1080 frame, top layer first, and sends it to a display, a Blackmagic device or NDI for your switcher.

Every section folds away with the arrow beside its title (or a click on the title), so the ones you don't use, say Music or Switcher, take one line. Studio remembers which are folded on that Mac, even after it restarts, and buttons in a folded section's title row, like **Output on**, still work. The longer explanations on the page wait behind **Expand for info**.

SOURCES

Add page adds any Studio graphic (or any web address), **Add image** a PNG, JPG, GIF, WebP or SVG, and **Add video** an MP4, WebM or MOV. A new source goes in hidden and selected, so its settings are ready to change. Drag a source up or down the list to move it between layers. **Add all unused pages** puts every Studio graphic in the scene at once, hidden.



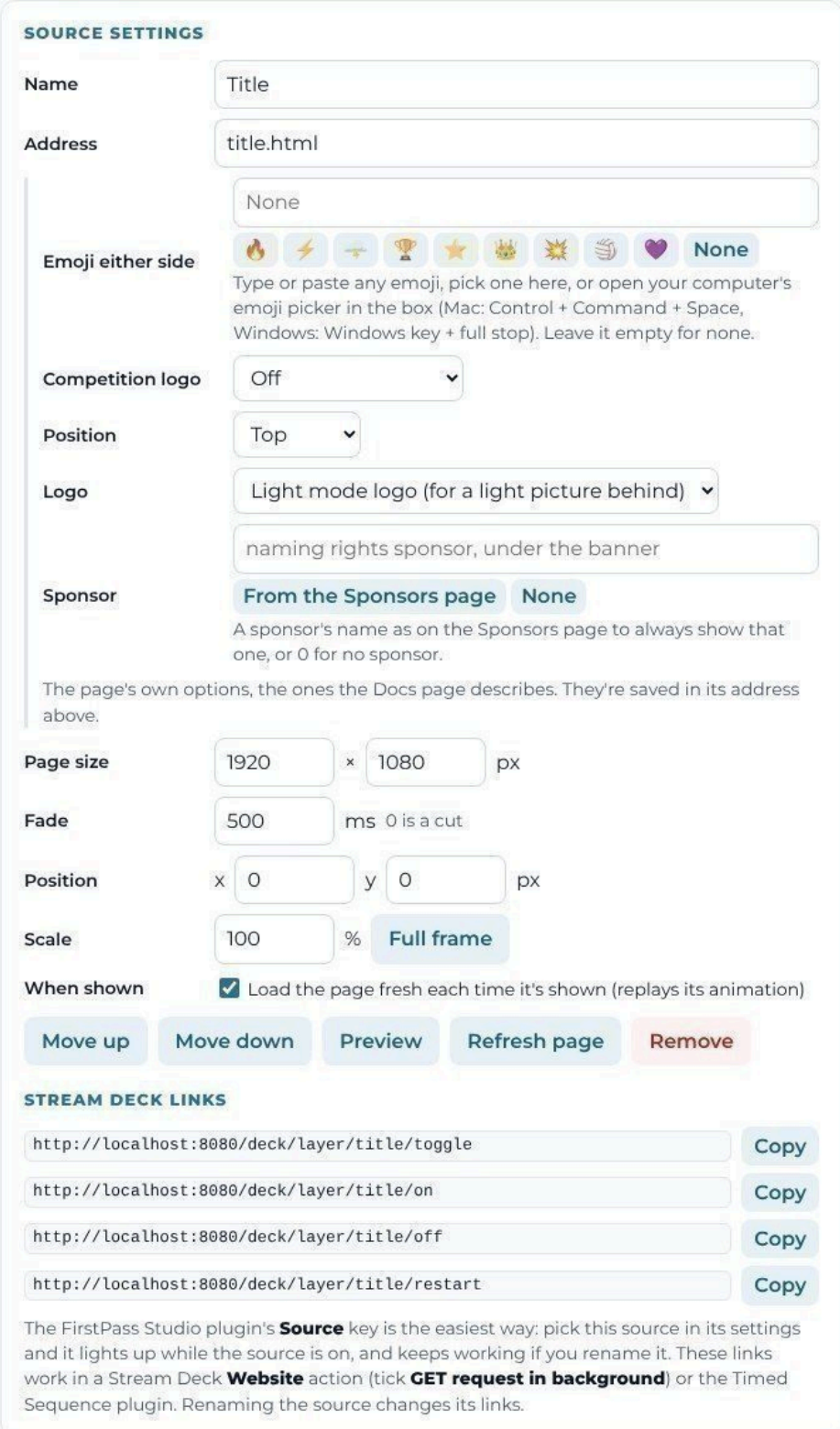
FirstPass Studio · Output

SOURCES (TOP LAYER FIRST)

Off PVW Viewers page
Off PVW Person lower third page
On PVW Super shot page
On PVW Hot streaks page
Off PVW Winner page
Off PVW The Clash page
Off PVW Intro page
Off PVW Outro page
Off PVW Logo animation page
Off PVW Countdown page
Off PVW Match up page
Off PVW Up next page
Off PVW Position match ups page
Off PVW Team list home page
Off PVW Team list away page
Off PVW Shooting stats page
Off PVW Last 10 goals page
Off PVW Full time page
Off PVW Game stats page
Off PVW Score worm page
Off PVW Ladder page
Off PVW Ladder lower third page
Off PVW Round fixture page
Off PVW Results page
Off PVW Top shooters page
Off PVW Finals page
Off PVW Sponsor card page
Off PVW Sponsor wall page
On PVW Score ticker page
On PVW Branding logos page
On PVW Title page
On PVW Scoreboard page

Add page · Add image · Add video · Hide all
Export sources · Import sources
☑ Export with their images and videos

FIG 3.3 The Sources list, top layer first. The orange On buttons are the base scene and the automatic layers; everything else waits for a scene, a sequence or a key.



FirstPass Studio · Output

SOURCE SETTINGS

Name: Title
Address: title.html
Emoji either side: None
🔥 ⚡ 🏆 ⭐ 👑 💥 💜 None
Type or paste any emoji, pick one here, or open your computer's emoji picker in the box (Mac: Control + Command + Space, Windows: Windows key + full stop). Leave it empty for none.
Competition logo: Off
Position: Top
Logo: Light mode logo (for a light picture behind)
Sponsor: naming rights sponsor, under the banner
From the Sponsors page · None
A sponsor's name as on the Sponsors page to always show that one, or 0 for no sponsor.
The page's own options, the ones the Docs page describes. They're saved in its address above.
Page size: 1920 × 1080 px
Fade: 500 ms 0 is a cut
Position: x 0 y 0 px
Scale: 100 % Full frame
When shown: ☑ Load the page fresh each time it's shown (replays its animation)
Move up · Move down · Preview · Refresh page · Remove

STREAM DECK LINKS

http://localhost:8080/deck/layer/title/toggle Copy
http://localhost:8080/deck/layer/title/on Copy
http://localhost:8080/deck/layer/title/off Copy
http://localhost:8080/deck/layer/title/restart Copy

The FirstPass Studio plugin's **Source** key is the easiest way: pick this source in its settings and it lights up while the source is on, and keeps working if you rename it. These links work in a Stream Deck **Website** action (tick **GET request in background**) or the Timed Sequence plugin. Renaming the source changes its links.

FIG 3.4 Source settings for the title banner: its options are fields, and the Stream Deck links for this source are copied from here.

SOURCE SETTINGS

Select a source to change it. A graphic's link options (position, heading, division, round, animate away and so on) show as fields under its address, so there is nothing to look up. **On / Off** fades it in or out over its own **Fade** time (0 is a cut). **Position and scale** place it on the frame. **Load the page fresh each time it's shown** (on by default) reloads a page as it fades in so its animation replays; untick it to keep a page running so it shows instantly. **Preview** (and the small PVW button) opens the source on its own, half size, on a checkerboard.

Videos play from the beginning each time they're shown, can loop, and fade out when they finish unless looping; a video that fades out switches its source off, so its key goes dark and the next On plays it again. A ProRes 4444 `.mov` is converted to a transparent WebM when it is added, keeping its alpha. **Export sources** saves the whole scene with its media as one zip to keep or move to another Mac; **Import sources** puts one back.

3.4 The base scene

The base scene is what viewers see for most of the game. In the sample league it is the **Live game** scene: the scoreboard at the bottom, the title banner at the top with the PIVT and FirstPass Studio logos on the left and the competition logo on the right, and the score ticker along the bottom edge with the other courts' scores.

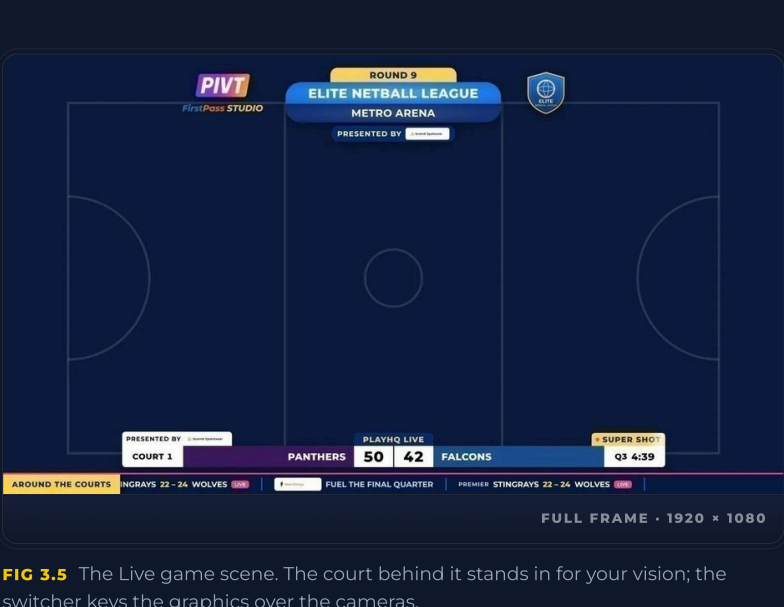


FIG 3.5 The Live game scene. The court behind it stands in for your vision; the switcher keys the graphics over the cameras.

- **Scoreboard.** Court in a pink box on the left, both teams in their club colours, the score at screen centre, quarter and clock on the right. "PlayHQ live" shows while scores follow PlayHQ; a "Presented by" tab carries the naming-rights sponsor.
- **Title.** Round, competition and venue, with the naming-rights sponsor's pill under it.
- **Branding logos.** PIVT and FirstPass Studio left of the title, the competition logo right of it, so other graphics can be turned on and off around them.
- **Score ticker.** Scrolling scores from tonight's other games, led by a Partner-and-up sponsor's tagline. Its speed is set on the control page or from the Stream Deck.

Build it once: turn those sources on, then press **Save what's on now** under Scenes and name it. From then on, one click (or one Stream Deck key) brings the show back to this look.

3.5 Automatic layers

These sources are left on in the base scene and play by themselves from the scoreboard. They add no PlayHQ requests of their own; they ask Studio, which already has the score.

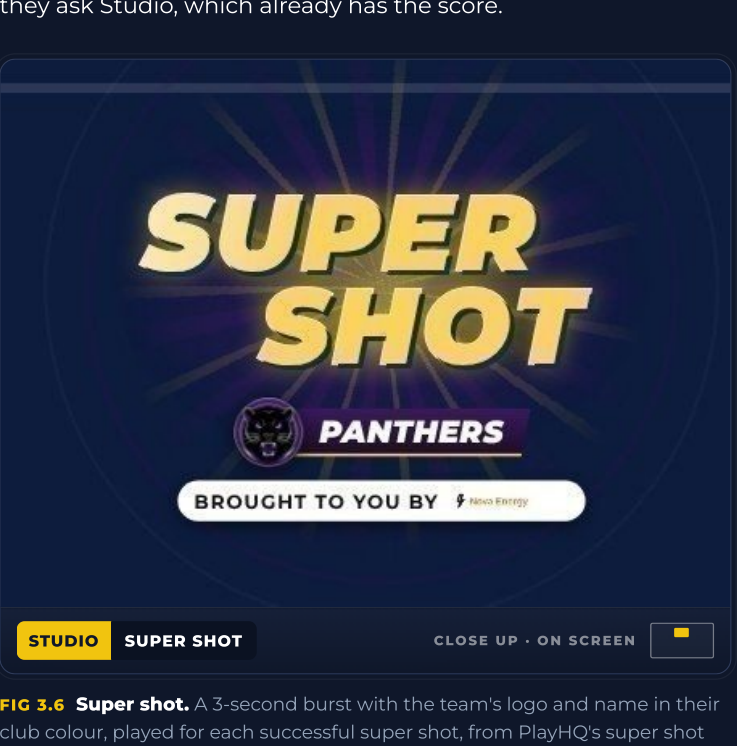


FIG 3.6 **Super shot.** A 3-second burst with the team's logo and name in their club colour, played for each successful super shot, from PlayHQ's super shot count or a +2 press. It plays small, centred just under the title.

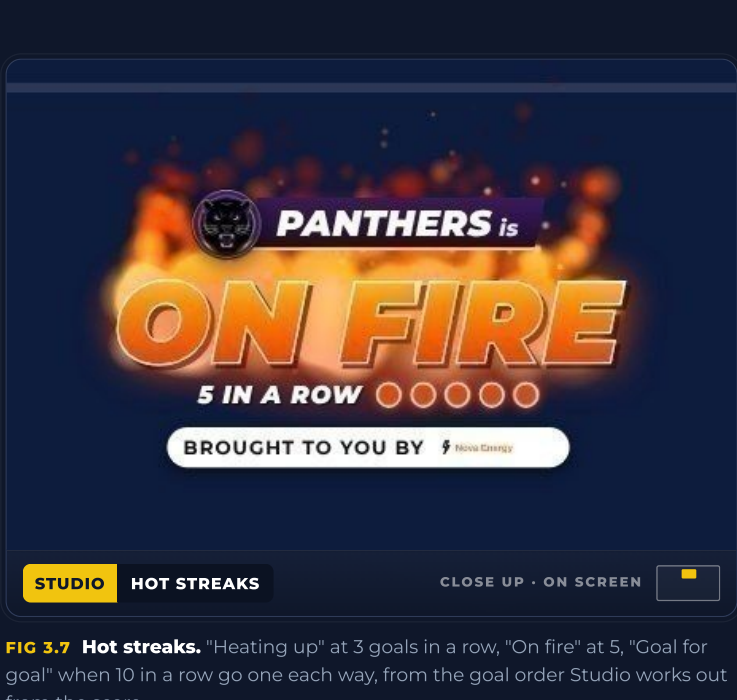


FIG 3.7 **Hot streaks.** "Heating up" at 3 goals in a row, "On fire" at 5, "Goal for goal" when 10 in a row go one each way, from the goal order Studio works out from the score.

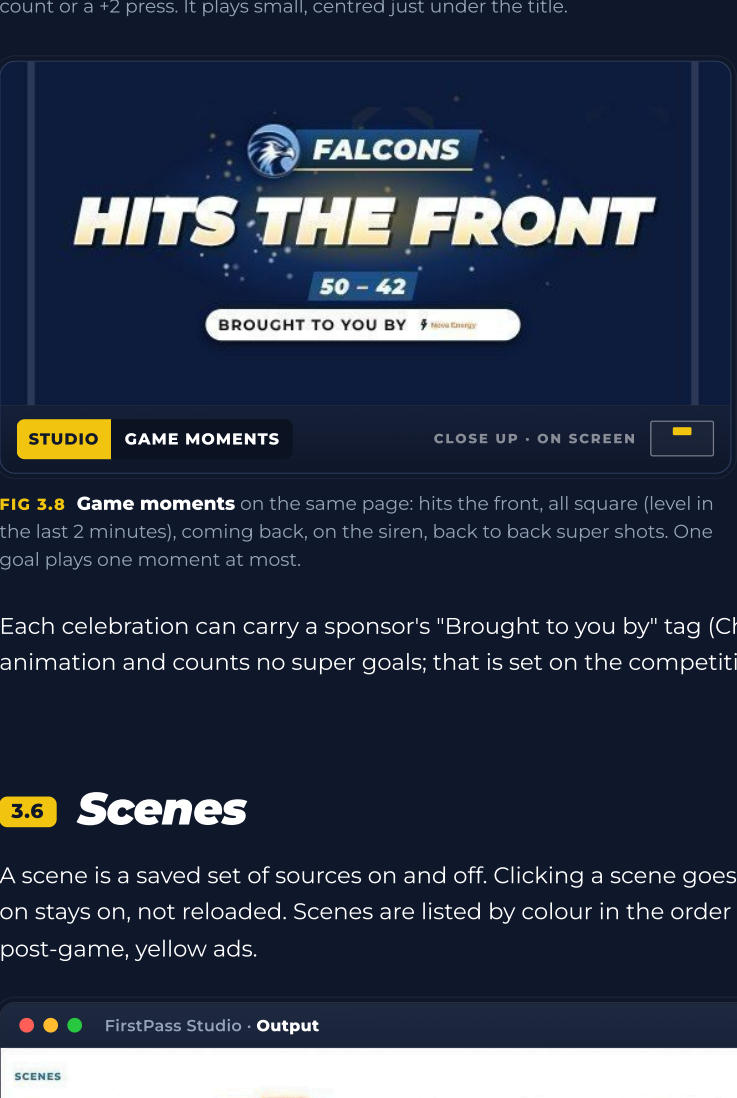


FIG 3.8 **Game moments** on the same page: hits the front, all square (level in the last 2 minutes), coming back, on the siren, back to back super shots. One goal plays one moment at most.

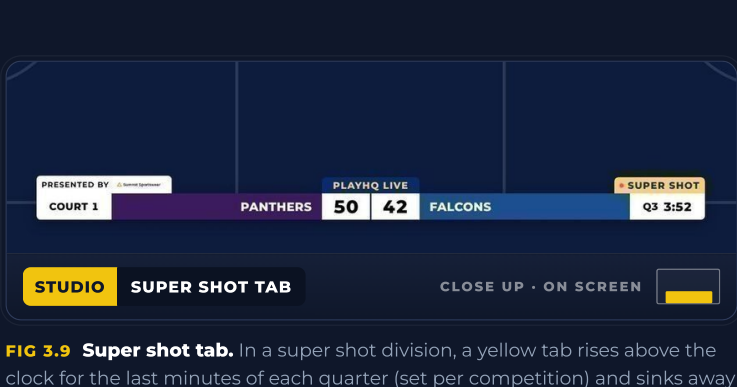


FIG 3.9 **Super shot tab.** In a super shot division, a yellow tab rises above the clock for the last minutes of each quarter (set per competition) and sinks away at the siren.

Each celebration can carry a sponsor's "Brought to you by" tag (Chapter 7). A division without super shots plays no super shot animation and counts no super goals; that is set on the competition setup page, per division if needed.

3.6 Scenes

A scene is a saved set of sources on and off. Clicking a scene goes to it, turning sources on and off to match while anything already on stays on, not reloaded. Scenes are listed by colour in the order of a stream: green pre-game, blue live, purple quarter time, grey post-game, yellow ads.

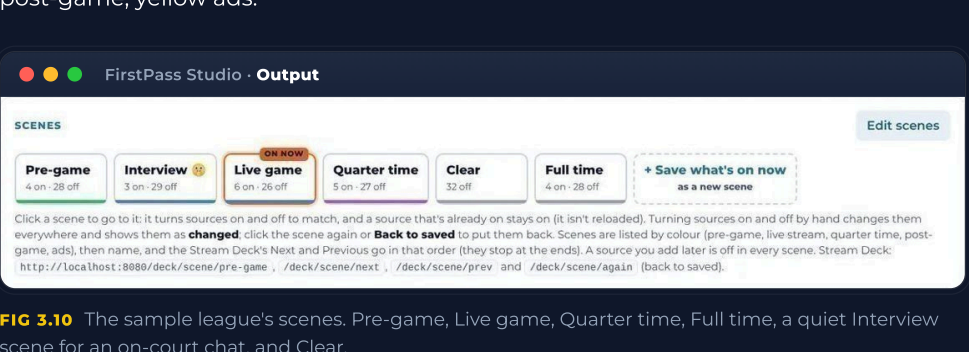


FIG 3.10 The sample league's scenes. Pre-game, Live game, Quarter time, Full time, a quiet Interview scene for an on-court chat, and Clear.

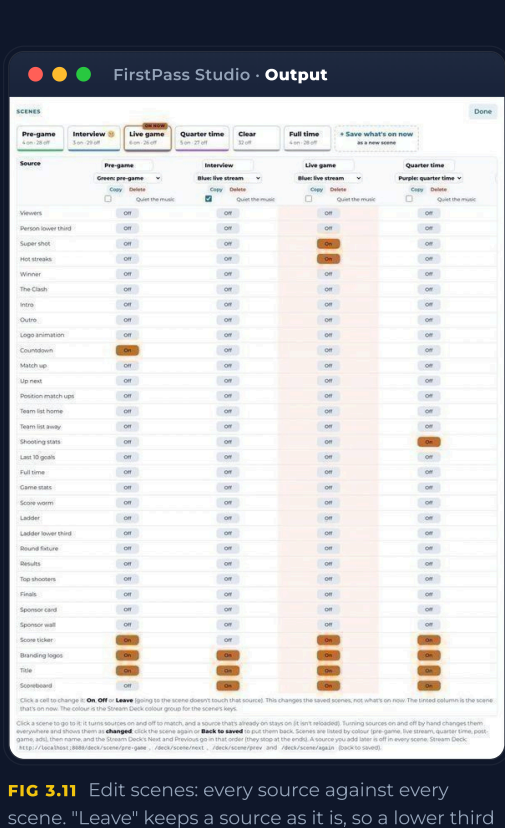


FIG 3.11 Edit scenes: every source against every scene. "Leave" keeps a source as it is, so a lower third can stay up through a scene change.

Turn a source on or off by hand after going to a scene and the scene is marked **changed**; the bar under the scenes offers **Back to saved**, **Save over** it, or **Save as new scene**. **Edit scenes** shows a grid, sources down and scenes across, to set each one On, Off or Leave, rename, colour, copy or delete scenes without changing what's on. A source added later is off in every scene.

QUIET SCENES Tick **Quiet the music when this starts** on a scene for an interview on court: whatever music is playing ducks and keeps playing until the next scene or sequence without the tick.

3.7 Sequences

A sequence plays the scene's sources one after another. Each step turns on the sources you pick and stays for a set time, until it finishes (its videos end or its animation animates away), until the game moves on (the clock starts, or the next break or full time), or until you press Next. As a step ends, sources the next step doesn't use go off with their own fades.

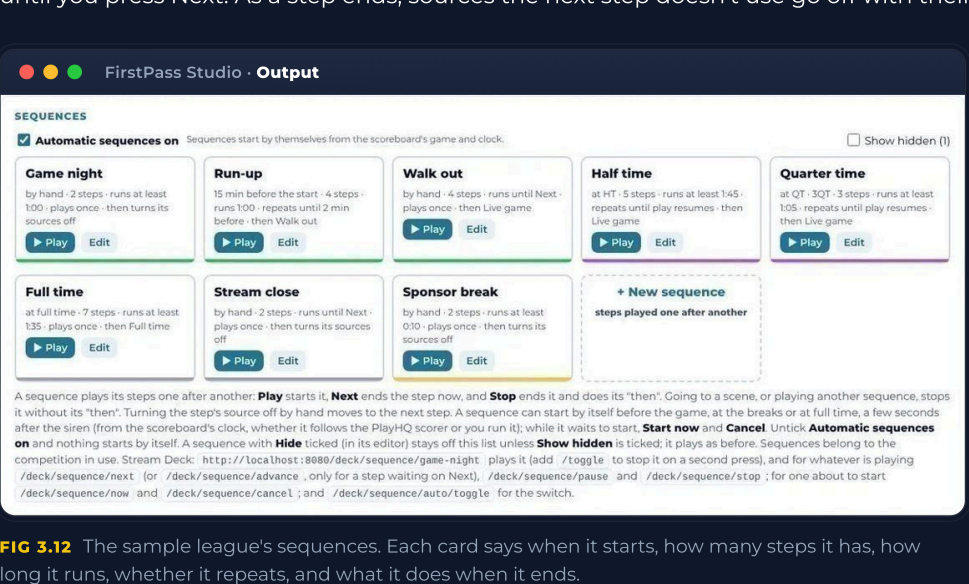


FIG 3.12 The sample league's sequences. Each card says when it starts, how many steps it has, how long it runs, whether it repeats, and what it does when it ends.

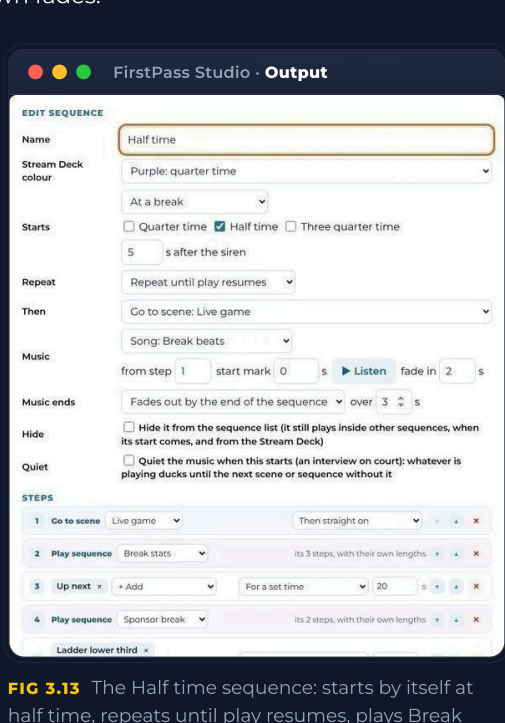


FIG 3.13 The Half time sequence: starts by itself at half time, repeats until play resumes, plays Break beats, and goes back to the Live game scene when it ends. Its steps are a scene step, two mini sequences and two source steps.

THREE KINDS OF STEP

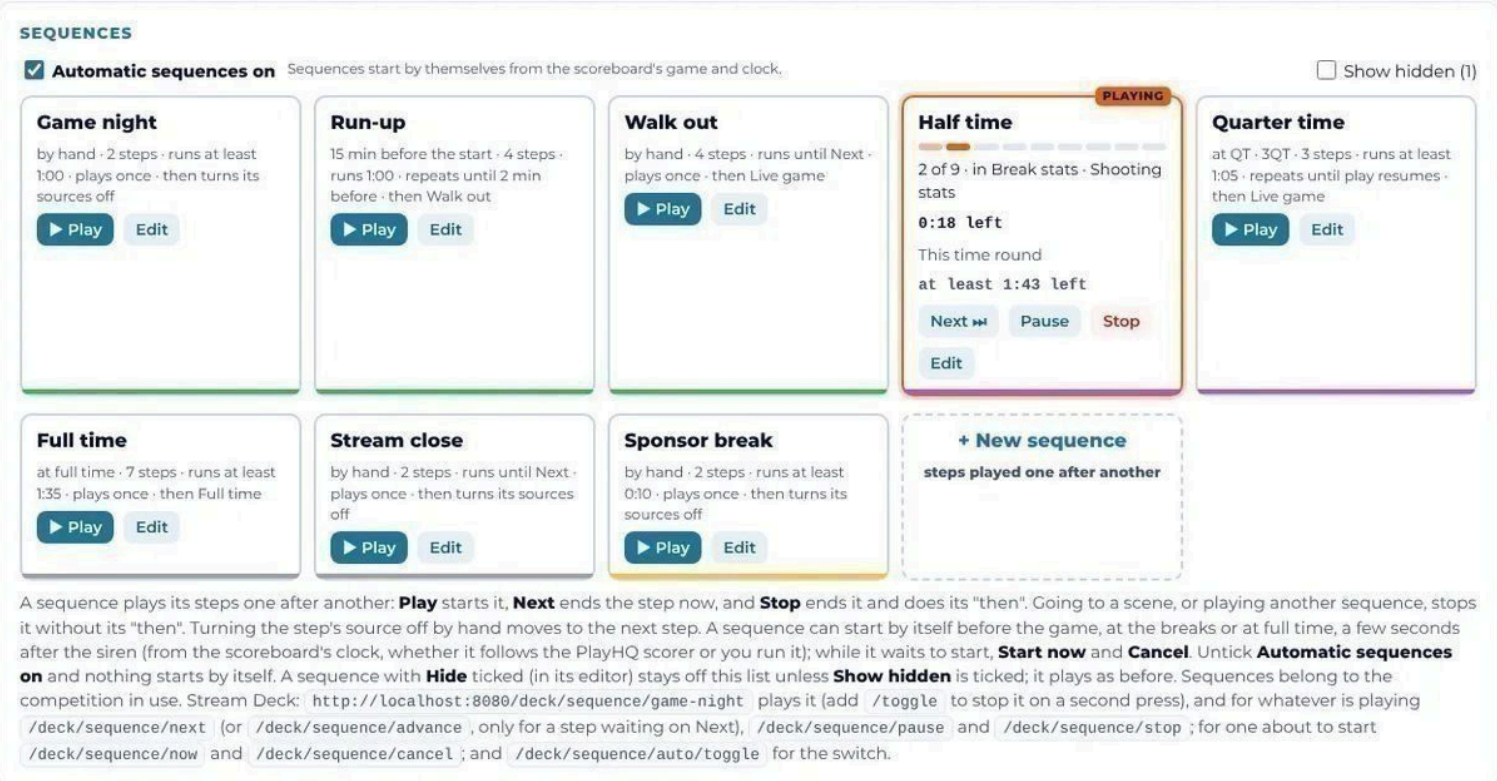
- **Sources** (**+ Add sources**): one or more sources on together, for a length.
- **A scene** (**+ Add scene**): a starting point or a reset point. What the scene turns on stays on under the next steps and after the sequence. Start a break sequence with the Live game scene and the scoreboard stays up under the stats.
- **Another sequence** (**+ Add sequence**): its steps play in place. Build a mini sequence once (the sample league's **Break stats** and **Sponsor break**) and use it inside Quarter time, Half time and Full time. A sequence made only of sequences is marked **Master**. Tick **Hide** on a mini sequence to keep it off the list; it still plays inside the others and from the Stream Deck.

+ Add sponsor ads adds a step that plays the next one to five sponsor ads in rotation (Chapter 7).

WHEN A SEQUENCE STARTS, REPEATS AND ENDS

SETTING	CHOICES	USED FOR
Starts	By hand · N minutes before the picked game's start (only if Studio was running then; opened later, play it by hand) · at the breaks you tick (QT, HT, 3QT) · at full time, a few seconds after the siren	Run-up 15 minutes out; the breaks and full time from the scoreboard's clock (the PlayHQ scorer's or yours).
Repeat	Play once · until stopped · until play resumes · until N minutes before the start	A break sequence loops until the clock starts again, finishing the step on screen.
Then	Go to a scene · turn off what it turned on · leave everything · play another sequence (one for each output it plays on)	Breaks go back to Live game; Run-up hands over to Walk out.
Music	None · a song or playlist from a step, from a start mark, with a fade in; ends by fading, cutting or carrying on into the next sequence	Warm-up music through the run-up that carries into the walk out.

A sequence that starts by itself waits a few seconds as **Starting**, with **Start now** and **Cancel**: press **Cancel** to hold the break back for a shot after the siren and start it by hand when the ball is dead, or **Stop** (or a scene) to come back from a break early. **Automatic sequences on** above the cards is the one switch for all of that, and it is remembered. Only one sequence plays at a time on each output; going to a scene, playing another sequence or hiding everything stops it without its Then. A sequence only ever turns off sources it turned on itself.



FirstPass Studio · Output

SEQUENCES

☑ Automatic sequences on — Sequences start by themselves from the scoreboard's game and clock. ☐ Show hidden (1)

Game night by hand · 2 steps · runs at least 1:00 · plays once · then turns its sources off. ▶ Play · Edit

Run-up 15 min before the start · 4 steps · runs 1:00 · repeats until 2 min before · then Walk out. ▶ Play · Edit

Walk out by hand · 4 steps · runs until Next · plays once · then Live game. ▶ Play · Edit

Half time PLAYING · 2 of 9 · in Break stats · Shooting stats · 0:18 left · This time round at least 1:43 left · Next · Pause · Stop · Edit

Quarter time at QT · 3QT · 3 steps · runs at least 1:05 · repeats until play resumes · then Live game. ▶ Play · Edit

Full time at full time · 7 steps · runs at least 1:35 · plays once · then Full time. ▶ Play · Edit

Stream close by hand · 2 steps · runs until Next · plays once · then turns its sources off. ▶ Play · Edit

Sponsor break by hand · 2 steps · runs at least 0:10 · plays once · then turns its sources off. ▶ Play · Edit

+ New sequence steps played one after another

A sequence plays its steps one after another: **Play** starts it, **Next** ends the step now, and **Stop** ends it and does its "then". Going to a scene, or playing another sequence, stops it without its "then". Turning the step's source off by hand moves to the next step. A sequence can start by itself before the game, at the breaks or at full time, a few seconds after the siren (from the scoreboard's clock, whether it follows the PlayHQ scorer or you run it); while it waits to start, **Start now** and **Cancel**. Untick **Automatic sequences on** and nothing starts by itself. A sequence with **Hide** ticked (in its editor) stays off this list unless **Show hidden** is ticked; it plays as before. Sequences belong to the competition in use. Stream Deck: `http://localhost:8080/deck/sequence/game-night` plays it (add `/toggle` to stop it on a second press), and for whatever is playing `/deck/sequence/next` (or `/deck/sequence/advance`, only for a step waiting on Next), `/deck/sequence/pause` and `/deck/sequence/stop`; for one about to start `/deck/sequence/now` and `/deck/sequence/cancel`; and `/deck/sequence/auto/toggle` for the switch.

FIG 3.14 While a sequence plays its card shows a dot per step, the time left on the step and for the whole sequence, with Next, Pause and Stop.

Sequences belong to the competition in use and change with **Use this one**; the sources and scenes are shared. The break lengths on the setup page (quarter time, half time, three quarter time) let the editor say how many times a break sequence fits in each break.

3.8 Music, playlists and mics

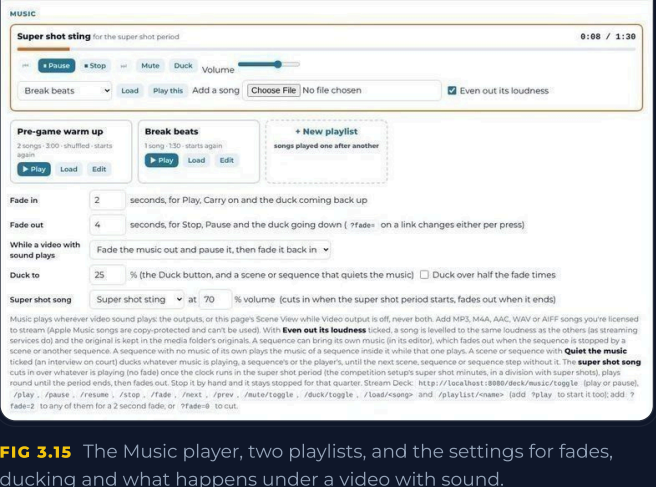
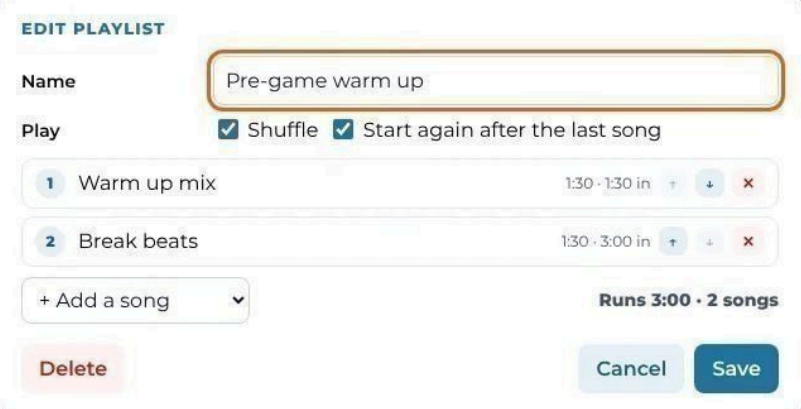


FIG 3.15 The Music player, two playlists, and the settings for fades, ducking and what happens under a video with sound.



FirstPass Studio · Output

EDIT PLAYLIST

Name: Pre-game warm up
Play: ☑ Shuffle ☑ Start again after the last song
1 Warm up mix 1:30 · 1:30 in
2 Break beats 1:30 · 3:00 in
+ Add a song
Runs 3:00 · 2 songs
Delete · Cancel · Save

FIG 3.16 A playlist: songs in order, shuffle, and whether it starts again after the last.

The Music player plays songs from the media folder (MP3, M4A, AAC, WAV or AIFF; use music you're licensed to stream), one song or a playlist, shuffled or not. Add a song with **Even out its loudness** ticked and it is levelled to match the others. Music plays wherever the video sound plays: through the output, or in the Scene View while Video output is off, never twice.

- **Play** fades in; **Pause** and **Stop** fade out (slower by default). **Duck** dips it low and keeps it running; **Mute** keeps it running silently.
- While a video with its sound on plays, the music fades out and pauses, then fades back in from where it was (or lowers, or carries on: a setting). A themed intro or outro does the same.
- A sequence can bring its own music, and a sequence inside it plays its own while it runs. Going to a scene fades a sequence's music out.
- The **super shot song** cuts in when the super shot period starts in a super shot division, loops, and fades out when the period ends.

Mics. Wireless effects mics on court and crowd mics go out with the music: plug their receivers into the Mac (directly or through an audio interface) and add each one under **Mics**, with its input and channel (two receivers on one stereo input are two mics, left and right). Each mic's **Goes to** picks the outputs it's mixed into (Output 1 to start), heard while they're on, never in the Scene View, so a speaker near a mic can't feed back. The first time, the Mac asks whether FirstPass Studio may use the microphone.

- A mic can be plugged into the switcher instead (an ATEM Mini's Mic 1 or Mic 2): the switcher mixes it and Studio moves its fader for the same level, mutes and dips. Move the fader on the switcher and that becomes the mic's level; Studio never moves it back.
- Each mic has a level, a mute, a **Low cut** for wind and rumble and a meter; **Level** and **Mute all** cover every mic, and a limiter stops a shout clipping the stream.
- The mics **dip** while music plays, while a video with its sound plays and during a scene that quiets the music (each a tick box), and by hand with **Dip**. A mic marked **Has an external mixer** never dips: the mixer does the fading.
- Stream Deck: `/deck/mics/mute/toggle`, `/deck/mics/dip/toggle`, `/deck/mics/up`, `/deck/mics/down`, and one mic with `/deck/mic/court-mic/mute/toggle`.

3.9 Video output

The **Video output** section at the top of the Output page sends the scene to your switcher. Pick one of three routes, then press **Output on**. While the output is on its settings are locked; turn it off to change them. `/deck/output/on`, `off` and `toggle` do the same from a Stream Deck, and the outputs go dark when Studio goes off air.

ROUTE	WHAT IT NEEDS	WHAT THE SWITCHER GETS
Display	One or two extra displays on the Mac (HDMI into the switcher), set to exactly 1920 × 1080, not HiDPI or scaled	**Chroma**: the scene over a solid colour on one display (green by default, blue or any colour). **Fill and key**: fill on one display, key (white where the scene shows) on the other. A plain M1 drives one extra display; Pro and Max chips drive more.
Blackmagic	An UltraStudio or DeckLink with Blackmagic Desktop Video installed, and the device free	**Fill and key** through the device's external keyer (fill on one SDI out, key on the other), or fill only, at the switcher's 1080p format (1080p50 is picked first). The fill is premultiplied: turn the switcher's Pre Multiplied Key on for clean edges, and leave Straight off (it leaves a light fringe round logos and text). **Sound** goes in the SDI with the fill (48 kHz stereo, in time with the picture): turn that switcher input's sound on. It can go on the Mac's sound output instead, or both.
NDI®	NDI's free runtime for Mac, and a wired network	Fill and key as two sources, chroma as one, or one source with transparency for receivers that key on NDI's alpha. They appear as this Mac's name with the output's name in brackets (or a name of your own).

HD OR 4K

The **HD** and **4K** buttons at the top of Video output set the picture for every output at once. In 4K each NDI and Blackmagic output sends 2160p at the frame rate you picked for it (2160p50 for 1080p50). Nothing else changes: scenes, sources, Stream Deck keys, links, NDI names and the switcher key stay as they are, and pressing HD puts everything back exactly as it was.

- **Sharp where Studio draws.** The graphics keep their 1920 × 1080 layout and are drawn at twice the density, so text, shapes, logos drawn by Studio and the animations' sparks and confetti are sharp. Pictures and videos are only as sharp as their files: a 1080p video or a small logo looks softer.
- **Only for 4K gear.** An HD switcher can't take 2160p. A Blackmagic device without the 2160p format (an HD UltraStudio) stays at 1080p and its card says so; a display is left alone, since a 4K screen set to look like 1920 × 1080 already draws at 4K.
- **Pick a 4K format directly.** In 4K a Blackmagic output's video format list also has the device's own 4K formats, such as 2160p50. In HD they're hidden, and the output sends the 1080p format at the same rate.
- **Any graphic's link works at 4K too.** Opened in a browser source or window set to 3840 × 2160, a graphic draws itself at twice the size, with nothing added to its link.
- **Check the Mac keeps up.** A 4K output is about four times the work of an HD one; the performance line under the output says whether it's keeping up.

MORE THAN ONE OUTPUT

One output is all most nights need. To give a camera its own graphics, press **Add an output** beside Output on (up to four). Each output is its own picture with its own route and its own on/off, so a second output can feed an upstream keyer tied to a camera and its graphics cut with that camera, while Output 1 stays the downstream key over everything. Name an output in its card ("Wide camera", say) and that name is used everywhere, including its Stream Deck links.

- **Sources are shared, on/off is per output.** Each source in the list has a coloured pill per output (orange, pink, blue, green for Outputs 1 to 4). Press a pill to show or hide that source on that output only. A new output starts with everything off. The **Show** filter above the list narrows it to one output.
- **Every output plays its own sound.** Each plays the sound of the videos and animations on it. The music plays on the outputs ticked in the music player's **Music plays on** (Output 1 to start), and a sequence's music on every output playing the sequence. Each mic's **Goes to** picks its outputs, so a commentators' feed can have the effects mics without the commentators' own voices.
- **Scenes and sequences are shared; each output plays its own.** Pick an output's tab over the Scene View and clicking a scene or sequence plays it on that output (the sections say which). **Hide all** clears that output only (or the one the Sources list's **Show** is set to), and says which, so clearing a monitor never blanks the stream. Each output has its own current scene and plays one sequence at a time, so the same scene or sequence can be on two outputs at once; coloured dots on scene cards, and on sequence cards each output's tick with how the sequence is there (Playing, Paused, Starting or Stopped (it ended there, played out or cut short); a hollow dot and "off" while that output isn't sending), show where each one is on. A sequence's **Plays on** ticks (on its card and in its editor) pick the outputs it plays on together: started by itself, or played on any of them, it starts on all of them at once, and each output has its own **Then** for when it ends there, so a quarter-time sequence can go back to the live scene on Output 1 while the commentary monitor goes on to its stats loop. Played on an output it isn't ticked for, it plays there alone (`/deck/output/2/sequence/<name>/only` does that on a ticked one). Stop on a card stops it on that output only. Remove every output it plays on and the sequence stops starting by itself until you tick another, so it never lands on air by surprise. A Sponsor ads step off Output 1 plays its ads in that output's own ad sources. `/deck/scene/<name>` and `/deck/sequence/<name>` play on Output 1; `/deck/output/2/scene/<name>`, `/deck/output/2/sequence/<name>`, `/deck/output/2/scene/next` and `/deck/output/2/sequence/stop` (and the rest) play on Output 2.
- **Sponsors count on Output 1, and on outputs you tick.** An extra output's card has **Counts for sponsors**, off to start. Tick it only when everything on that output is seen on the stream, like a second downstream key. Leave it off for a camera's upstream key or the commentary monitor: Studio can't tell when they're on air, and the report shouldn't claim time nobody saw. Ads there show whose turn it is without using it up, so the order on air stays fair.
- **The Scene View has a tab per output**, coloured to match, so you can check what each one is sending.
- **Routes can't be shared.** Two outputs can't use the same display, the Blackmagic device or the same NDI name; the earlier output keeps it and the later one says so on its card. Two HDMI displays give either two chroma outputs or one fill and key output. Several NDI outputs can run at once, each with its own name, which suits feeding a switcher's inputs over the network.
- **Stream Deck.** `/deck/output/2/toggle` (or the output's name) turns Output 2 on or off, `/deck/output/all/off` turns them all off, and `/deck/output/2/layer/<source>/toggle` shows or hides a source on Output 2. Source settings lists these links for each output.

EACH OUTPUT COSTS Every extra output runs its pages once more on the Mac. Try the number of outputs you plan to use on your own Mac before a stream night: while an output is on, its card shows its CPU, memory and frame rate (amber when it can't keep up), with Studio's total under the status, and the log notes them once a minute.

A COMMENTARY MONITOR

Commentators can have graphics of their own: a monitor at the commentary desk with the live stats they talk to, never seen on the stream. The **Commentator stats** page is made for it: one full screen with both team lists on the left (each team's time since its last goal, and each athlete's games played this season), the game stats big in the middle (each team's logo with its ladder position, record and form under it, head to head under the names, then shooting %, quarters won, best quarter, super shots, runs, biggest lead and lead changes), down the right the game clock with the super shot window, who is on the mic (from the Production Planner) and up next, then a list of the latest moments (lead changes, comebacks, super shots, hot streaks), newest at the top, each with its quarter, game time and how long ago it happened, and across the bottom the last 10 goals, the current run in big numbers and the score worm. Its stats refresh themselves every 15 seconds and when the score changes (untick **Auto-refresh** in its options, or pick another frequency), and the clock, moments and goals follow the game as it happens. Other graphics can go on the commentary feed too: the last 10 goals and the scoreboard follow the game live, and the rest show what they read each time they come on.

- **On the same Mac.** Add an output for the commentary monitor (a display, or NDI to a screen on the network) and turn sources on for it with its pills or its Stream Deck keys, such as `/deck/output/commentary/layer/ladder/toggle`. Play scenes and sequences on it from its Scene View tab or its own Stream Deck links while the stream carries on as normal.
- **On a second Mac.** Run Studio on another Mac with the same competition and game, and Follow PlayHQ on. It reads PlayHQ's live scoring for itself, so it needs nothing from the stream Mac. Its Output 1 goes to the commentary monitor, and everything on it is for the commentators.

On the commentary output (or the second Mac), make a **commentaryLoop** sequence that steps through what commentators find useful (game stats, shooting stats, the last 10 goals, the ladder) with **Repeat** set to until stopped, and give the commentators a Stream Deck of their own:

- A key per graphic they want on demand, each going to a scene with that graphic (on the same Mac, `/deck/output/commentary/scene/ladder`). Going to a scene stops the loop on that output only, so their pick stays up and the stream is untouched.
- A **Loop** key to go back to looping when they're done: `/deck/output/commentary/sequence/commentaryloop` on the same Mac, or `/deck/sequence/commentaryloop` on a second Mac. With the FirstPass Studio plugin, use Scene and Sequence keys with **Output** set to the commentary output; they light pink while it's on.

Each output's sound plays through the Mac's audio output, except on a Blackmagic output, where it goes in the SDI with the fill (or on the Mac too, as its **Sound** says). The **Scene View** on the Output page shows the scene as it is right now, on a checkerboard so see-through parts show; **Pop out** opens it in its own window to move to another screen, with a floating list of On/Off buttons. The Scene View plays the sound of the output it shows only while that output is off, so it never plays twice.

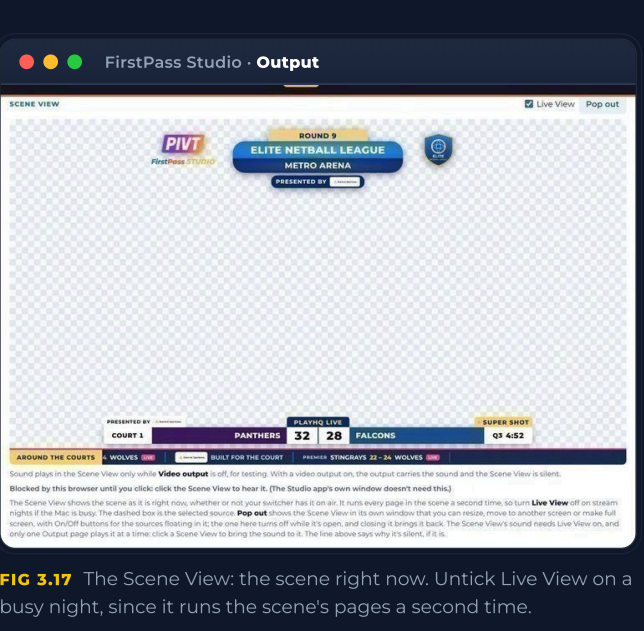


FIG 3.17 The Scene View: the scene right now. Untick Live View on a busy night, since it runs the scene's pages a second time.

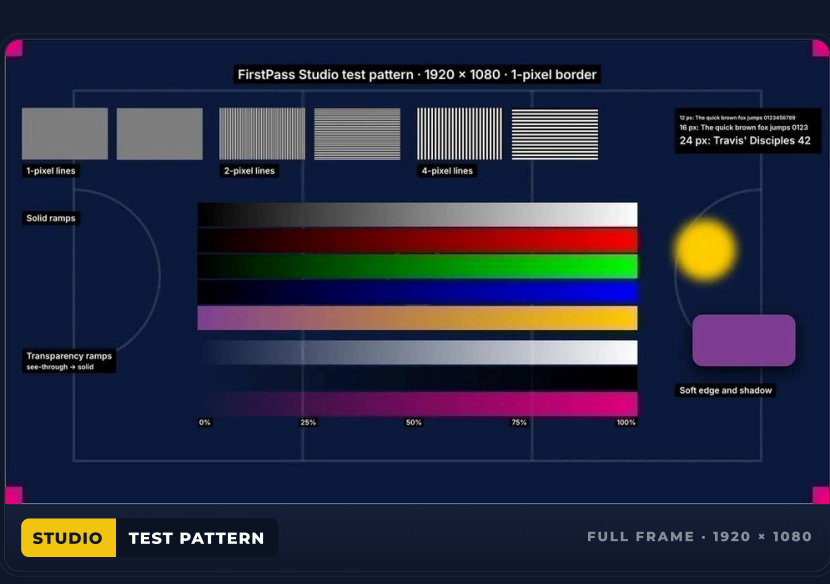


FIG 3.18 The **test pattern** (Add page) checks an output and the switcher's key end to end: a 1-pixel border, fine lines that shimmer if anything is resizing the picture, ramps for banding and soft shapes for the key. Not for air.

CHROMA KEY SAFE A chroma key cuts out one colour, so anything half see-through fringes. With a chroma output picked, **Chroma key safe** is on: every graphic goes out with solid panels, no glows or soft shadows, and fades become cuts. Pick another output and everything is back as it was. The Output page also lists any competition or club colour close to the key colour.

THE SWITCHER

Studio can also drive a Blackmagic ATEM switcher over the network, so the camera cuts can be pressed from the Output page or a Stream Deck beside the graphics. Open the **Switcher** card under Video output, press **Set up** and **Look for switchers**: Studio finds the ATEMs on the network the way ATEM Software Control does (macOS asks the first time to let it use the local network) and remembers the one you pick by name. Typing its address works too.

- **Each output's key, checked.** In each output's setup, **Into the switcher as** says which switcher key its graphics come into (usually downstream key 1 for Output 1, upstream keys 2 to 4 for the others, an upstream key for chroma) and which inputs its fill and key are plugged into. Studio reads that key's settings and lists anything that doesn't suit the output, such as the wrong fill source, Pre Multiplied Key off, or Straight ticked on a Blackmagic output. **Set up this key for me** fixes just those, after asking, and leaves the key on or off air as it was.
- **Graphics that go with a camera.** An output on an upstream key can tick the cameras it **goes with**, say a bench graphic with the bench camera. Studio's cuts, autos and stingers then bring its key on with those cameras and take it off with the others, on the same frame. With **Keys follow the camera** on, a cut on the switcher's own panel brings the key a frame or two later. A key turned off by hand stays off until the camera changes. The output's card says **On air with Bench** or **Waiting for Bench**.
- **Inputs in tally colours.** Each input shows by its name from ATEM Software Control, red on program and green on preview. Click one to put it on preview, then **Cut** or **Auto**. Fade to black and each upstream and downstream key are a press away too.
- **Only what your switcher has.** An ATEM Mini or Mini Pro has 4 inputs, one upstream key and one downstream key, so fill and key into the downstream key leaves two cameras; a Mini Extreme has 8 inputs, four upstream keys and two downstream keys. Studio offers what it finds, and says so when a key or link asks for something the switcher doesn't have.
- **Your hands stay on the switcher.** Its own buttons, ATEM Software Control and other controllers keep working, and the card follows them. Studio sends nothing by itself, not even on connecting.
- **If it drops off the network, or is off when Studio connects,** the card turns red within about five seconds and Studio keeps trying. Presses meanwhile are refused rather than saved up, so a cut never lands late.
- **Stinger cuts.** Add the Stinger page as a source and the card's **Stinger** button plays it and cuts the switcher underneath, right in the middle of the bands. It plays on the output on a downstream key, which is Output 1 unless another output that's switched on is. If the cut shows before the bands do (an output's picture can reach the switcher a few frames late, more over NDI), set the **line-up** in frames and press **Test the line-up** before you go live. Each output keeps its own line-up.
- **Camera cuts in sequences and scenes.** A sequence step's 🎥 button, or a scene's 🎥 **Switcher** under Edit scenes, adds switcher actions: cut, auto or stinger to a camera, put one on preview, a key on or off, fade to black, or run a macro, as the step starts or a set number of seconds in. So the walk-out sequence can cut to the bench camera as the line-ups come up and bring the scoreboard key back as it ends, with nobody on the switcher. They happen only on Output 1, and if the switcher isn't answering at that moment the action is skipped and logged rather than done late. You can always cut by hand on the switcher itself.
- **Stream Deck.** The plugin's **Switcher** key puts a camera on preview or program, plays a stinger, cuts, autos, turns a key on or off or runs a macro, and lights like the switcher's own buttons: red on program, green on preview. While the switcher isn't answering it goes dark with a yellow dot. Without the plugin, the links are `/deck/switcher/cut`, `auto` and `ftb`, `/deck/switcher/preview/2` (or `preview/Camera%202`), `/deck/switcher/usk/1/toggle`, `/deck/switcher/dsk/1/auto` and `/deck/switcher/macro/1`.

3.10 A night, start to finish

With the sample league's scenes and sequences, a solo operator's evening looks like this. Everything in the right-hand column happens by itself.

Arrive
Open the app. Studio goes on air, picks tonight's first game on Court 1 with Follow PlayHQ on. Turn the video output on and check the test pattern on the switcher.

15 min out
Run-up starts by itself: the Pre-game scene, then the sponsor wall, match up and ladder in turn over the warm-up playlist, repeating until 2 minutes before the start.

2 min out
Run-up hands over to **Walk out**: The Clash smashes the two logos together, the position match ups run through GS to GK, the countdown holds until the clock starts, then the Live game scene comes up and the music fades.

Q1 to Q4
The scoreboard follows the PlayHQ scorer's score, quarter and clock. Super shots, hot streaks and moments play by themselves. The super shot tab and song mark the last five minutes of each quarter. The operator watches the stream and, if they like, keys up a ladder lower third or a lower third for a commentator.

Quarter time
Quarter time starts a few seconds after the siren: shooting stats, the last 10 goals and the ladder, then two sponsor ads and the sponsor card, over the break music, round and round until play resumes, then back to Live game.

Half time
Half time does the same with Up next and the ladder lower third added, and the house ad if no sponsor has one due.

Full time
Full time starts after the final siren: the winner celebration, the full time card, game stats, the score worm, a sponsor break and Up next (which becomes the end-of-night card after the last game).

Next game
Automatic selection moves the board to the next game on Court 1 as its electronic scoring opens, with the quarter back to Q1 and a full clock. Run-up starts again 15 minutes out.

Close
Press **Stream close**: the Clear scene, then the outro with the night's teams. Turn the output off. The sponsor report has the night's exposure ready to send.

3.11 The Production Planner

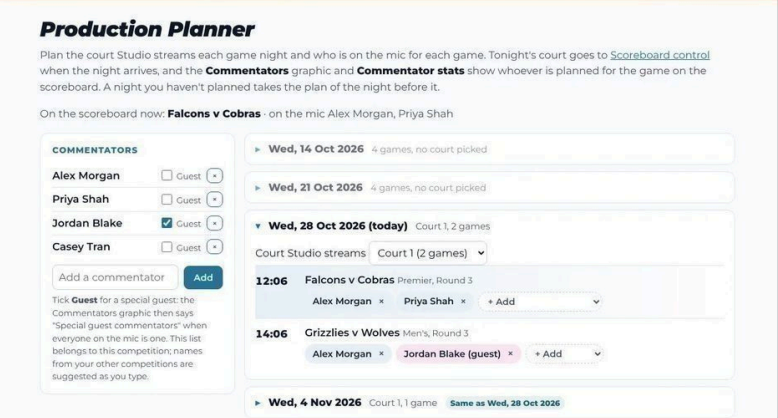


FIG 3.19 The Production Planner: the commentators down the side, and the season's game nights from PlayHQ, the next one open, with each game on that night's court and who calls it.

The **Production Planner** tab plans the season ahead: which court Studio streams each game night, and who is on the mic for each game. Studio streams one court a night; a second court is a second Mac running its own Studio, with its own plan.

COMMENTATORS

Add the people who call your games to the list on the left. Tick **Guest** for a special guest. The list belongs to the competition; as you type a name, people from your other competitions are suggested, so someone who calls in two leagues is added in a tap. Click the circle beside a name, or drop a photo on the row, to add their **headshot** for the Socials tab's Commentary team picture; the small × on it takes it off again.

A NIGHT

Every game night in the season is listed, read from PlayHQ's fixture, by date rather than round. Open one, pick the **Court Studio streams**, and its games on that court appear in time order. Tap **+ Add** on a game to put up to three people on the mic; tap × to take someone off. Everything saves as you go. A night you haven't planned takes the plan of the night before it (the same court, and the same people game by game), marked **Same as** that night, so a season with a regular team is planned once and only the nights that differ need changing. **Same as the night before** drops a night's own plan again.

WHEN THE FIXTURE CHANGES

PlayHQ's fixture changes through a season: a game moves court for availability, or is refixtured to another date. The commentators belong to the time on the court, because that is where the desk and cameras are set up, so they stay put and the game is flagged **Fixture changed** with what was there before. A game that has gone from the court keeps its row, so its commentators can be moved to wherever it went.

ON THE NIGHT

When the night arrives, the plan's court goes to Scoreboard control, which follows it game by game as usual. The **Commentators** graphic, with no names typed in its options, names whoever is planned for the game on the scoreboard, and says **Special guest commentators** when everyone on the mic is a guest. **Commentator stats** shows them under the clock, and the **Fixture board** full page puts the whole night's fixture on a screen in the competition room or backstage, with a camera on the streamed court's games and who is on the mic for each. Manual control always wins: if you change the court or pick a game by hand on Scoreboard control, the Planner says so at the top, so the commentary team can be moved to match. Swapping someone in the Planner takes effect the next time the graphic comes on.

3.12 Socials

The **Socials** tab turns a game night into pictures for Instagram and the league's other channels, from the same fixture the stream uses, so the posts and the broadcast always agree. Pick the **Game night** (by date, so every division playing that night is in it), **Upcoming** or **Results** (a night with results opens on Results), and the sizes: **Square** 1080 × 1080, **Portrait** 1080 × 1350 (the most room in the feed) and **Story** 1080 × 1920. Every picture shows as a preview in each size, in the competition's colours, font and logos.

UPCOMING PICTURES

The **Game night fixture** lists every game of the night on every court, by time. A **Match up** for each game puts the two logos side by side with the date, time, court and venue, one for each club to share. **On the stream** shows the court Studio streams, with each game's division and who calls it, from the Production Planner. **Commentary team** is a picture for each game someone calls, with that game's commentators side by side with their headshots and the game's time and teams once underneath, to share before the livestream. Someone who calls more than one game is on each of their games' pictures, so the social team can post them one at a time or grouped as they like. Long names shrink together so the row stays balanced, breaking only between words or after a hyphen; anyone without a headshot shows their initials, and a guest gets a **Special guest** pill. The **Polaroid** is a tilted white print for a match photo: drop a photo on it, drag the preview to frame it and set the zoom. With no photo, its window is see-through, so the social team can put their own photo under it in Canva or a phone app.

RESULTS PICTURES

After the night, **Results** shows every finished game's final score, division by division, with the winner in the highlight colour. Each game gets a **Full time** card, with the logos and score big and who won by how much, and a **Polaroid** with the score as its caption and the division and round under it. Each division that played gets its **Ladder**, with the finals line, labelled "after" the round when it is that division's latest night with results. Each division also gets its own **Top shooters**, naming its best six that night from each game's line ups (with super shots only where that division plays them), and its **Tournament top shooters**: the best eight over every finished game so far, after this night's round. It's called **Season top shooters** when the fixture runs 15 days or more; pick either name under Game day on Competitions setup to override. On a carnival day with several rounds, set **As at** to an earlier round to count only the games up to it. Tick **Highlight movers** and each shooter shows the places they climbed or fell since the round before, even when both were played that day, with NEW for anyone new to the top eight; the tick is remembered for the competition. A game appears once PlayHQ has its final score.

Finals get their own pictures. After a division's last home and away round, Results adds **Finals bound**, the teams that made the finals with what each one's ladder place earned (a double chance, a week off, an elimination final), and the **Finals draw**, the finals week by week. On each finals night the draw is there again, under Upcoming to post before the games and under Results with the winners filled in, and the Match up, Full time and Results pictures name each final, such as Major semi final or Grand final. A division gets these when it has a finals format on Competitions setup.

SPONSORS, THE LOGOS AND SAVING

Each picture has a **Sponsor strip** tick, which puts your naming, elite and major sponsors along the bottom and is remembered for next week. Under **Options**, **FirstPass Studio mark** (bottom right) and **PIVT logo** (bottom left) turn those on or off for every picture. **Template** sets the look of every picture: **Default** is the stream's own, and Dynamic, Junior, Retro, Professional, Thunder, Gradient, High contrast, Neon, Poster, Splash, Game Day, Hero and Final each bring their own shapes and lettering, and three netball sets draw on the game itself: Court (the court's lines and circles), Bib (a bib with its position letters) and Goal (the ring, net and post), and five sporting sets bring the kit around it: Bottle, Cones, Sneakers, Stadium and Locker room. Star (☆) the templates you like and tick **Favourites only** to list just those, and the one in use; it's remembered for the competition. Any template works with any colours, and every title stays on one line beside the competition logo, shrinking a little where it would otherwise wrap. A short ladder or a couple of games grow to fill the picture, and a Story keeps its words and logos out of the top and bottom, where Instagram puts its own bar and reply box. A long team name shrinks to stay on one line in its row. Athletes' photos come from the Athletes page, and only for athletes ticked **Approved for photo use**. Top shooters shows each shooter's photo automatically. On a Match up, Full time or Polaroid card, pick an athlete by name, narrowing the list by division and team. Hero and Final stand the picked athletes tall, in front of the title; a cut-out PNG with a transparent background looks best. **Colours** gives every picture a different look: **Competition colours** is the stream's own, and the others swap the competition's main, highlight and accent colours around (Accent first, Highlight ground, Accent ground), or go darker (Night), paler (Day) or down to two colours. **New colour set** makes your own from three colours, for a finals night or a sponsor round. A set marked **low contrast** has titles that may be hard to read. Your choice is remembered for the competition. Untick anything you don't want (**Select all** and **Select none** tick or untick every picture at once), then **Save pictures**: they are saved as PNGs in a folder for the night in **Pictures › FirstPass Studio**, which opens in Finder. **Share…** saves them too and opens the Mac's Share menu, to AirDrop them to your phone or send them by Messages or Mail. Tick **As a zip** to also save them as one .zip beside the folder; Share then sends that one file, which is easier to email or upload.

3.13 Recordings

For each PlayHQ game on the scoreboard while on air, Studio keeps a recording of what PlayHQ said and what went to air, timed: the live score, clock and quarter, the line ups, every scoreboard and Stream Deck press, and every change to the scene and the graphics. There's no video or sound in it, only what the graphics need, so a game is a megabyte or so. A recording starts when its game is due on the scoreboard, 10 minutes before the scheduled start unless automatic game selection on the competition setup page says otherwise, and stops 10 minutes after PlayHQ says the game is finished (change it with **Keep recording for … minutes after full time**), when the scoreboard moves on to the next game, or when Studio goes off air. Being on air between games, or while you set things up, records nothing. The **Recordings** tab lists each game's recording, with **Download** to keep a copy and **Remove**, and says when the next one starts. Recording is on unless you untick **Record each PlayHQ game on air**. A recording older than 60 days is tidied away. Recordings stay on this Mac and hold nothing secret.

PRACTICE GAMES

Press **Practise** beside a game to play it back as if it were live. Studio doesn't ask PlayHQ anything while it runs: the recording answers, with the game's date and times moved to now, so the game starts a few minutes from now (5 to begin with: set **A practice game starts in** above the games, from 0 to 60 minutes, for time to run a pre-game sequence) and everything works as on a real night. The scoreboard follows the score and the clock, sequences and quarter-break triggers fire, and the outputs can be on. At the top of the Recordings tab, **Jump to** goes to the start, any quarter or full time, **Pause** stops the clock where it is, and **Stop** ends it and puts the scoreboard back exactly as it was. A red strip on every Studio window says a practice game is running. Nothing in it counts for your sponsors, and it isn't recorded as a game of its own. Use it to rehearse before the season, try a new sequence, or show Studio to someone.

4K REPLAY

Streamed a night at 1080p and recorded the camera in 4K? Press **Replay** beside the game and Studio puts that game's graphics back on every output, exactly as they went to air. Turn on **4K** on the Output page and they go out at 2160p, sharp, to key over the 4K recording in a 4K switcher or recorder, ready for a YouTube upload. Studio doesn't play the video. The replay starts paused at the game: pause the video on a moment you can see, such as the first centre pass, pick the same moment from the list of what went on air and press **Jump there**, then press Play on both together. **Nudge** moves the graphics a frame or a second if they drift. The live scene and scoreboard are kept as they were while it runs, and **Stop** puts them back. Sound comes from the HyperDeck's 1080p programme recording.

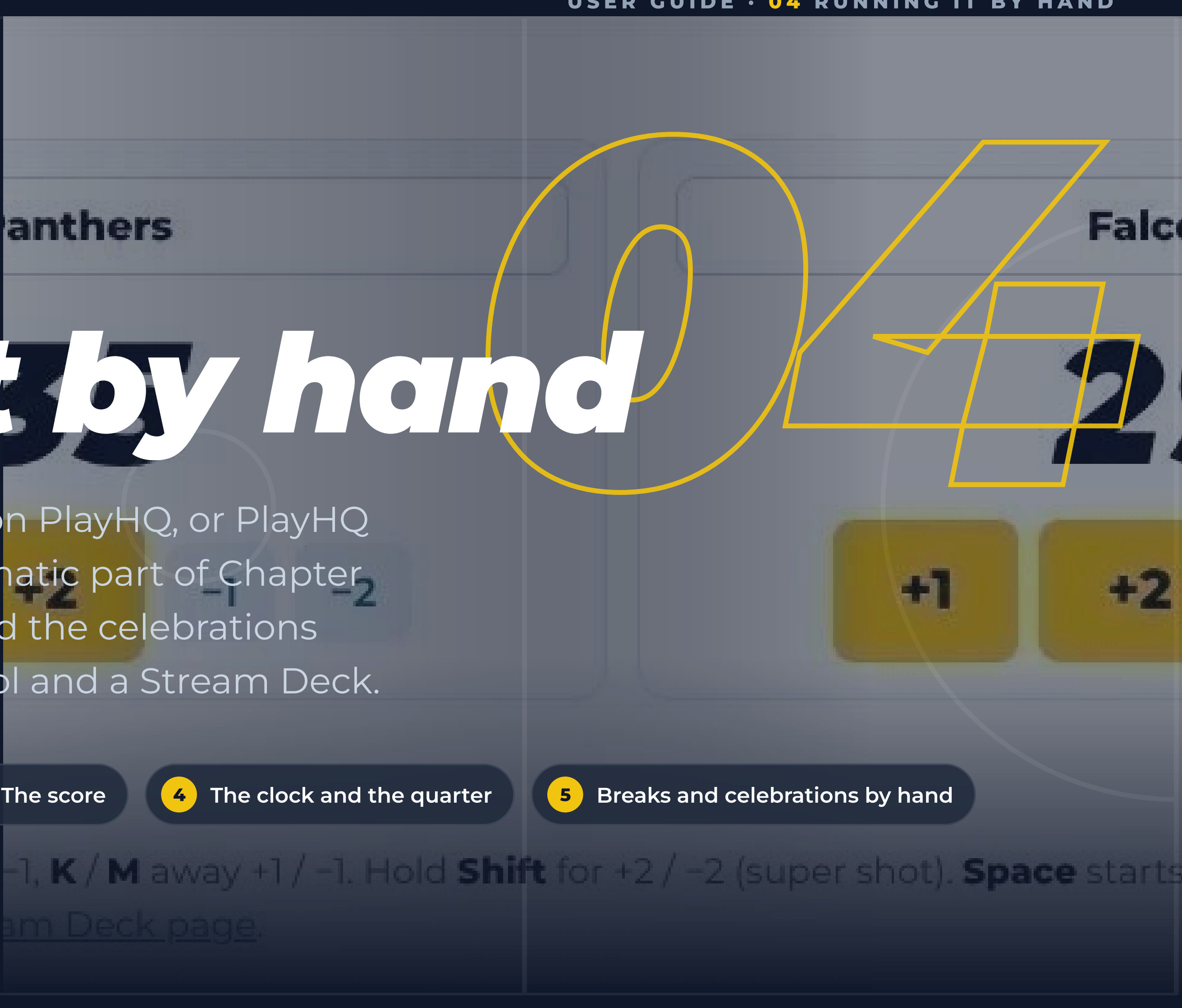
With the 4K video on a HyperDeck, set **Follow** to **A HyperDeck's timecode** and type its IP address. The graphics then go wherever the HyperDeck is: play, stop or scrub it and they follow, frame for frame, with nothing to line up by hand. If its clock was a little off on the night, stop it on a moment you can see, pick the same moment and press **Jump there** once. Studio remembers that sync for the recording. It only reads the HyperDeck's timecode and never controls it.

Any other player works through the UltraStudio: send its SDI into the UltraStudio's SDI input and set **Follow** to **Timecode into the UltraStudio (SDI)**. Studio reads the timecode that comes with the picture and follows it the same way. HDMI doesn't carry timecode, so it has to be SDI.

On a Stream Deck, the plugin's **Replay** key plays or pauses, jumps to the next moment, nudges or stops the replay, or a practice game, whichever is running. It lights while one runs.

DEMO GAMES

Showing Studio to another league? Set up their competition, then press **Demo** beside one of your recorded games. Pick one of their divisions and a finished game: Studio plays your recorded game as theirs, with their competition, team names, logos, colours and athletes, while the score, clock, quarters, moments and timing come from your recording. Athletes are matched by position, so your GS's goals become their GS's; anyone without a match is left unnamed and their goals go to the team. If their game has no line up, Studio uses each team's season squad, by the position each athlete played most. While the demo runs, the **Athletes** table shows who each of your athletes is shown as; change any of them there. **Save demo** keeps the whole setup by name, so before the meeting it's one click under **Saved demos**. **Swap home and away** lets either team win. The other teams in the division take the places of yours on the ladder and fixture. Play the recorded game's video wherever you like and lay Studio's output over it: the pictures won't show their teams, but the graphics will. Their competition's setup is read from PlayHQ, so a demo needs the internet.



CHAPTER 4 OF 11

Running it by hand

When nobody is live-scoring the game on PlayHQ, or PlayHQ stops answering mid-game, every automatic part of Chapter 3 has a button. The board, the breaks and the celebrations can all be driven from Scoreboard control and a Stream Deck.

1 Two people, not one
2 Switching to manual
3 The score
4 The clock and the quarter
5 Breaks and celebrations by hand
6 Finals

Studio still needs the competition on PlayHQ for its fixture, ladder, squads and results; what changes in manual mode is who runs the score and the clock. The same competition can be followed one week and run by hand the next, and nothing on the Output page changes between the two.

4.1 Two people, not one

A followed game can be run by one person because the score, the clock and the breaks take care of themselves. A game run by hand is a different job: someone has to press a key for every goal within a second or two of it, start and stop the clock with the umpire's whistle, and still call the breaks and the graphics. We recommend two people share it:

Camera operator

- Frames and follows play on the cameras, and cuts between them on the switcher.
- Runs the game clock from the whistle, with **Start / Stop** and **Reset clock** keys beside the camera controls.
- Scores the goals as they go in, **+1** or **+2**, because they are already watching the ring.
- Nothing else: no graphics, no breaks, no music.

Production manager

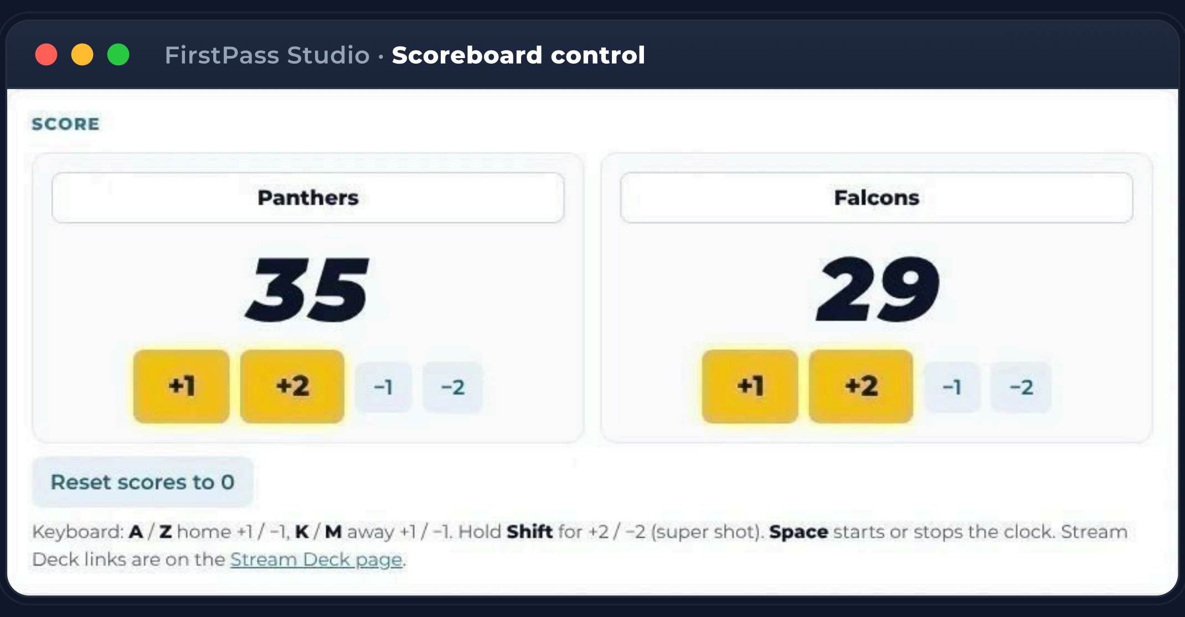
- Keeps the Output page open and watches the stream itself, not the court.
- Corrects the score (**−1**, **−2**) and the quarter, and sets the clock when it drifts from the venue's.
- Plays the run-up, break and full time sequences from their keys, and the lower thirds, ladders and stats during play.
- Runs the music, the sponsor ads and the end of night, and picks the finals winners.

If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks, exactly as they do in automated mode: the game triggers follow the scoreboard's clock whoever runs it.

> **If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks.**
>
> CHAPTER 4 · RUNNING IT BY HAND

4.2 Switching to manual

Press **Manual** on Scoreboard control, or just press any score key: either switches the board to manual scoring and stops following the PlayHQ clock as well. Press **Follow PlayHQ** to go back; the score catches up with PlayHQ on the next check and the clock follows the scorer's again.



Keyboard: **A** / **Z** home +1 / −1, **K** / **M** away +1 / −1. Hold **Shift** for +2 / −2 (super shot). **Space** starts or stops the clock. Stream Deck links are on the Stream Deck page.

FIG 4.1 Score and clock on Scoreboard control. Keyboard: A and Z for the home team, K and M for away, Shift for a super shot, Space for the clock.

WHEN PLAYHQ DROPS OUT If PlayHQ stops answering during a followed game, the scoreboard keeps its last score, the ladder, fixtures and game list keep what they last read, and the debug log records each failed read. Switch to Manual and run the board by hand; Follow PlayHQ picks it up again when PlayHQ is back, and corrects the score to the scorer's.

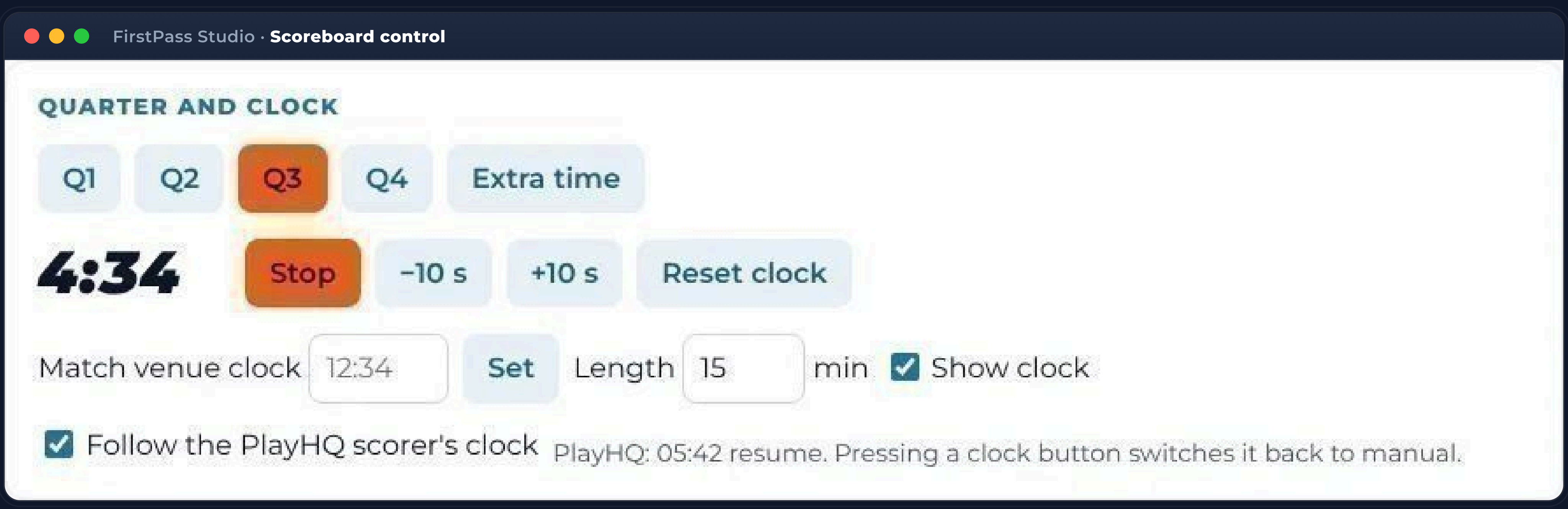
4.3 The score

Each team has **+1**, **+2**, **−1** and **−2**. A +2 is a super shot: it scores two and plays the super shot celebration, so a +2 pressed by mistake is fixed with −2 (which plays nothing). **Reset scores to 0** clears both. The hot streaks and game moments still play in manual mode, because Studio works out the goal order from the presses just as it does from PlayHQ's score.

KEY	WHAT IT DOES
Home +1 / Away +1	A goal.
Home +2 / Away +2	A super shot: two points and the celebration.
−1, −2	Take a goal or a super shot back, quietly.
Reset scores to 0	Both scores to zero, for the next game.
Super shot (home / away)	The celebration only, score unchanged: for a test, or a super shot that was scored as +1 +1.

4.4 The clock and the quarter

Start and **Stop** are one key on the Stream Deck. **Reset clock** puts a full quarter back (the competition's quarter length; a key can set any time). **+10 s** and **−10 s** nudge it, and a key can add or take off any amount. Type the venue's clock in **Match venue clock** to line the two up, and untick **Show clock** for a game with no reliable clock at all. Quarters are **Q1** to **Q4** plus extra time, or H1 and H2 for a competition played in halves; **Next quarter** steps through them. Changing the quarter doesn't touch the clock, so make the quarter key a Stream Deck Multi Action that also resets the clock.



Match venue clock 12:34
Length 15 min
☑ Show clock
☑ Follow the PlayHQ scorer's clock

PlayHQ: 05:42 resume. Pressing a clock button switches it back to manual.

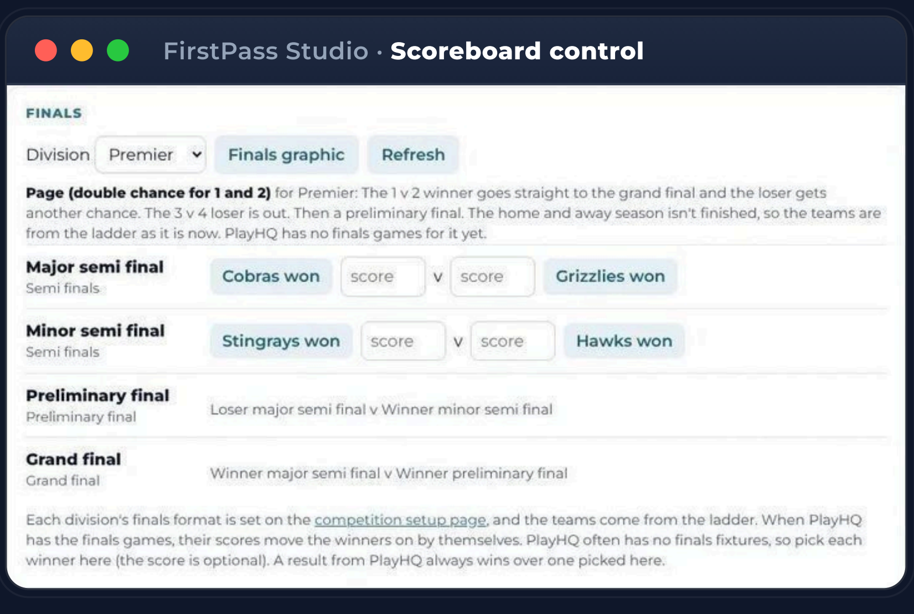
FIG 4.2 Quarter and clock. The super shot tab, the super shot song and the break sequences all read this clock, so they work by hand as they do when following PlayHQ.

4.5 Breaks and celebrations by hand

A sequence that starts by itself at quarter time is started by its key too, so the production manager can run **Quarter time** the moment the siren goes instead of waiting for the clock, or run **Half time** early when a game is called. **Next**, **Pause** and **Stop** work on whatever is playing, and **Automatic sequences on** on the Output page can be unticked for a night where every break is called by hand.

- **Winner.** One key. The celebration reads the board's score when it comes on, so it covers a home win, an away win and a draw. Check the score first. A source with **Result** set to home, away or draw forces it.
- **Super shot and streaks, once.** Besides the automatic sources, a Super shot source set to **Plays: home** or **away**, or a Hot streaks source set to one streak, plays once each time it is turned on. Don't keep a manual source and the automatic one on together, or it plays twice.
- **The Clash.** A pre-game animation, played from its key or as a step in the walk-out sequence, with the two teams, their logos and club colours.
- **Lower thirds.** The person lower third's name box on the Output page suggests the competition's athletes; type a guest's name and title for a commentator or an official. Turn it on when they appear and off a few seconds later.
- **Stats during play.** The ladder lower third, shooting stats, last 10 goals and match up are all Source keys. Each loads fresh when shown, so it has the latest numbers.

4.6 Finals



Division: Premier

Page (double chance for 1 and 2) for Premier: The 1 v 2 winner goes straight to the grand final and the loser gets another chance. The 3 v 4 loser is out. Then a preliminary final. The home and away season isn't finished, so the teams are from the ladder as it is now. PlayHQ has no finals games for it yet.

Major semi final Semi finals	Cobras won score v score Grizzlies won
Minor semi final Semi finals	Stingrays won score v score Hawks won
Preliminary final Preliminary final	Loser major semi final v Winner minor semi final
Grand final Grand final	Winner major semi final v Winner preliminary final

Each division's finals format is set on the competition setup page, and the teams come from the ladder. When PlayHQ has the finals games, their scores move the winners on by themselves. PlayHQ often has no finals fixtures, so pick each winner here (the score is optional). A result from PlayHQ always wins over one picked here.

FIG 4.3 Finals on Scoreboard control. The format for each division is set on the Competitions setup page, and the teams come from the ladder.

PlayHQ often has no finals fixtures until the day. In **Finals** on Scoreboard control, pick the division, then pick each game's winner (the score is optional) as the finals are played; the finals graphic and the ladder's finals key move on with each pick. A result from PlayHQ, when it has one, always wins over one picked here.

CHAPTER 5 OF 11

The graphics

Every graphic Studio draws, grouped by when it is used. Each one is added to the scene with **Add page** on the Output page, and its options appear as fields under its address; the options named here are the ones worth knowing.

1 Always on · 2 Before the game · 3 During play · 4 After the game · 5 The competition · 6 The breaks · 7 Checking the outputs

All of them follow the game night, game and court picked on Scoreboard control, take the competition's colours, font and logos from its setup, and fit long team names. Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits. Every page, with its link and a Preview button, is also listed on the Docs tab.

5.1 Always on

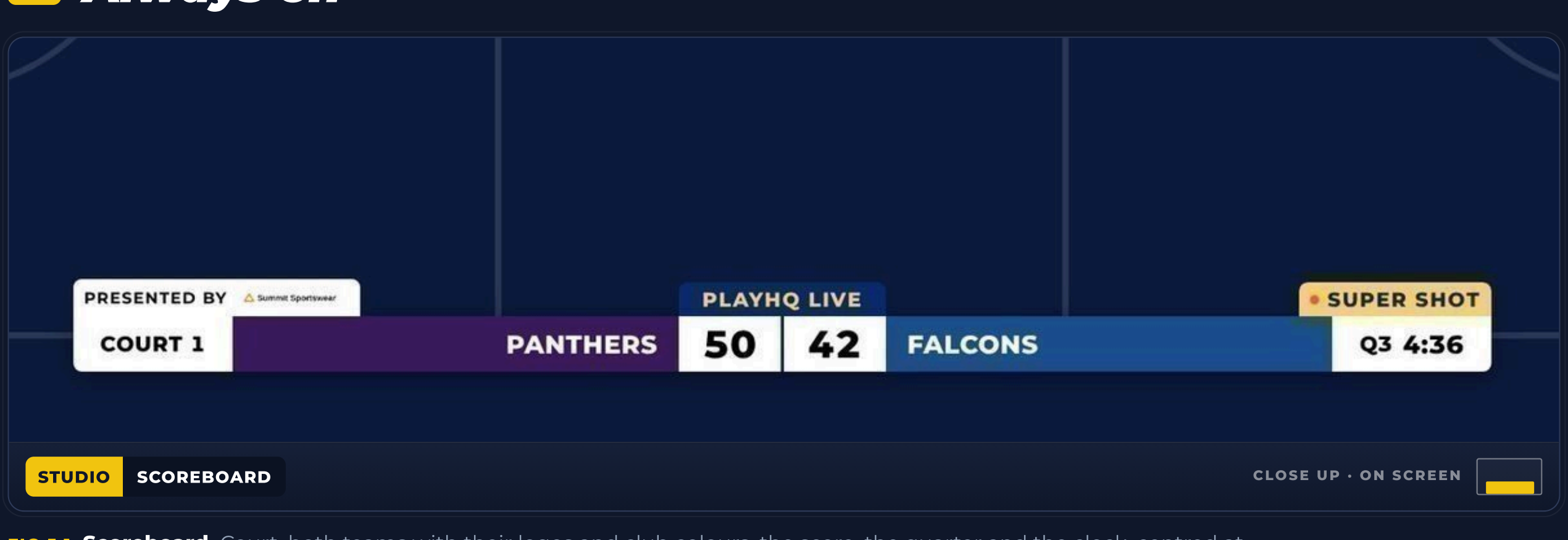


FIG 5.1 Scoreboard. Court, both teams with their logos and club colours, the score, the quarter and the clock, centred at the bottom above the ticker. Options: position (top centre, bottom left and so on), super shot tab on or off, and the naming-rights sponsor's tab.

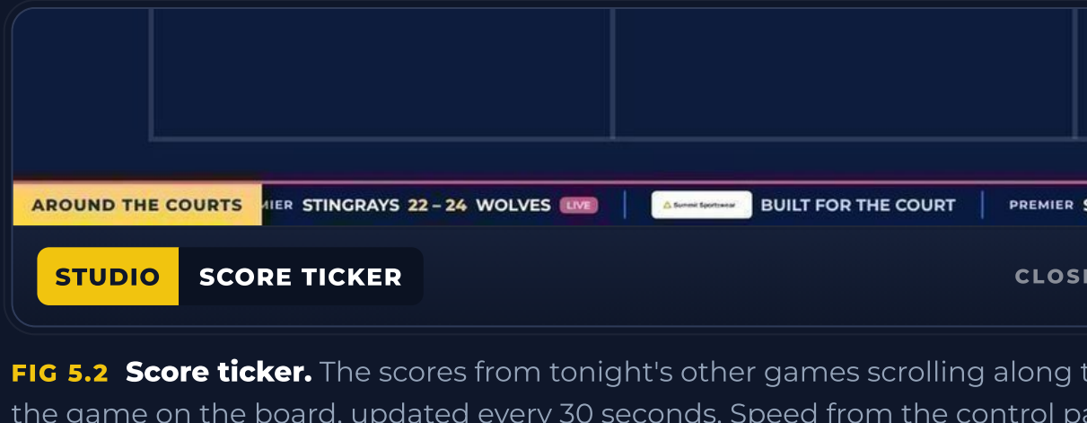


FIG 5.2 Score ticker. The scores from tonight's other games scrolling along the bottom, leaving out the game on the board, updated every 30 seconds. Speed from the control page or a Stream Deck key; a Partner sponsor's logo takes a slot in the line.

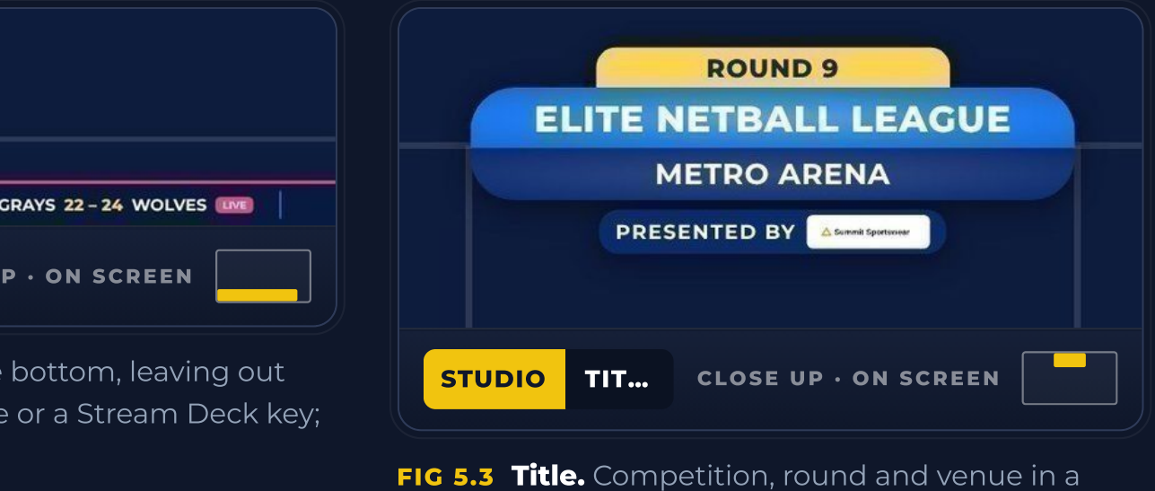


FIG 5.3 Title. Competition, round and venue in a banner at the top. Options: an emoji either side, the competition logo above it, top or bottom, and the naming-rights sponsor under it.

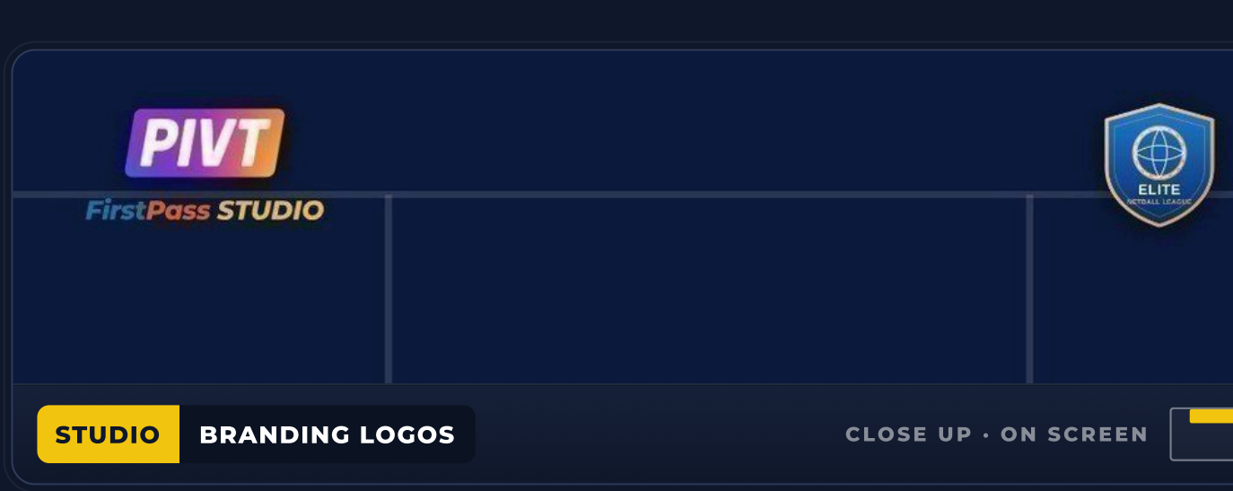


FIG 5.4 Branding logos. PIVT and FirstPass Studio left of the title, the competition logo right. Each can be hidden; gap, height and size adjust the placement.

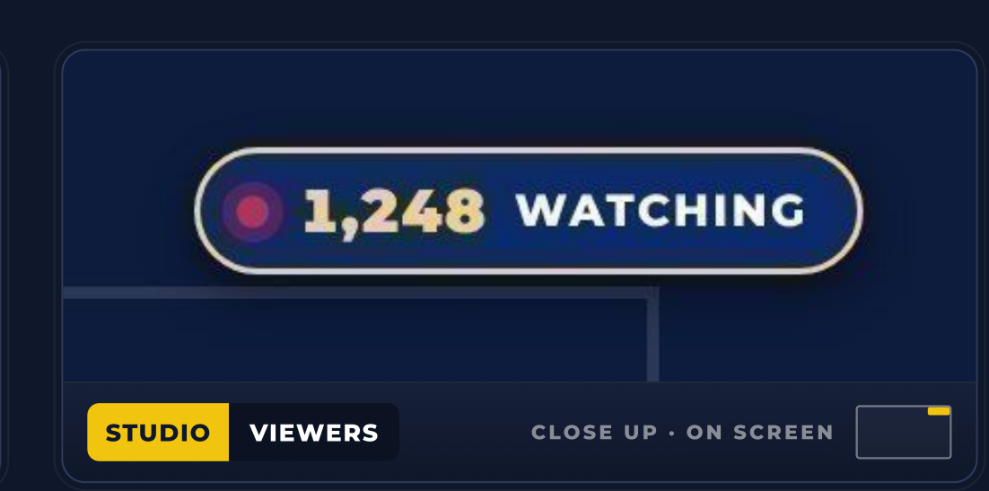


FIG 5.5 Viewers. "N watching" from the competition's YouTube and Facebook streams, top right, shown only once the count passes the competition's minimum. **Shows: peak** makes it a card for the end of the night.

5.2 Before the game

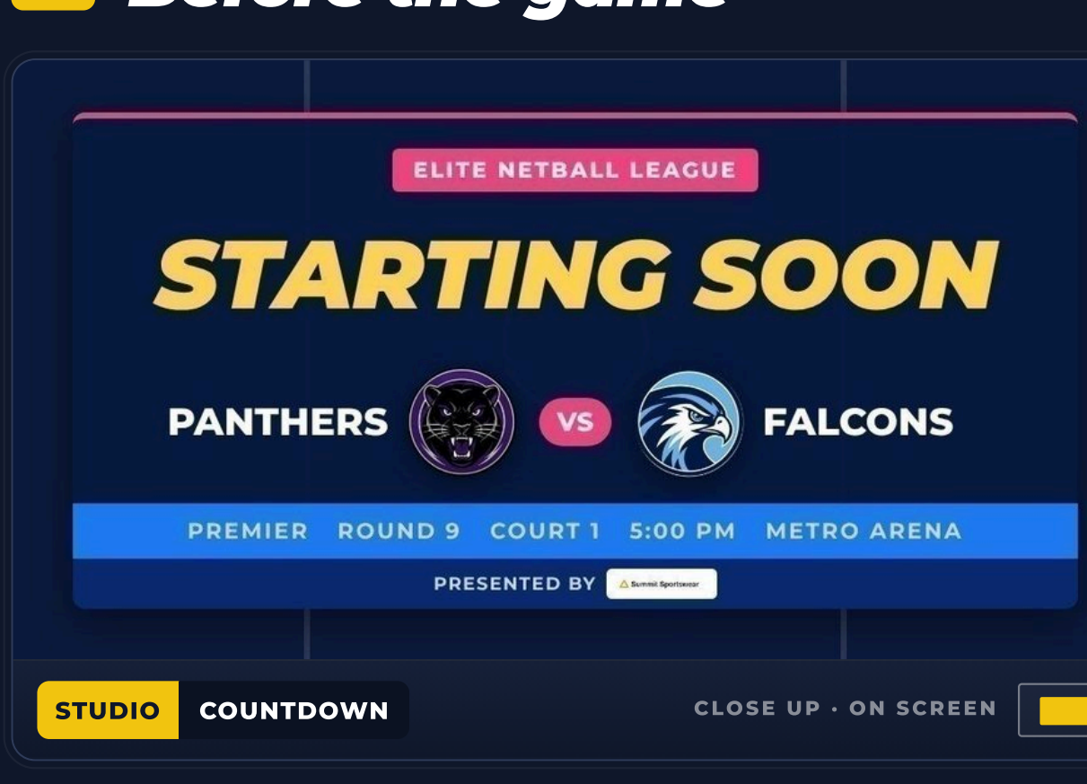


FIG 5.6 Countdown. Counts down to the picked game's scheduled start, with the two teams. Options: heading, the words at zero, a fade over the last seconds, position and a sponsor.

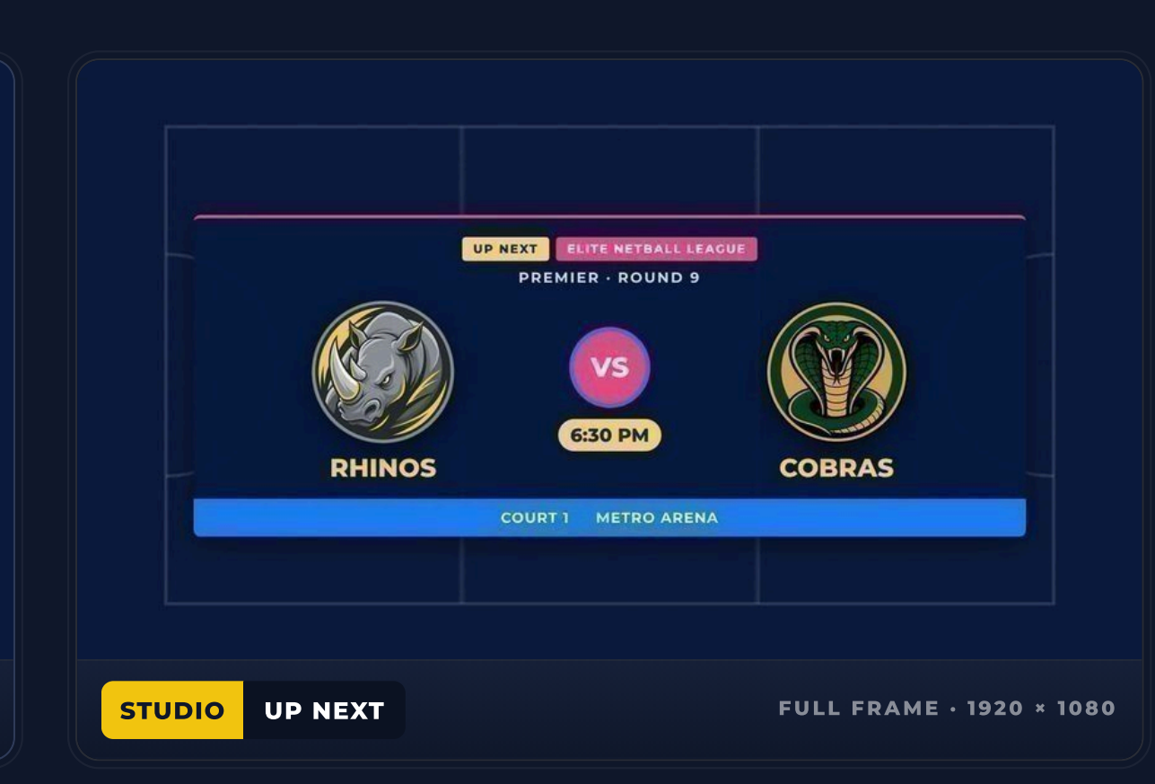


FIG 5.7 Up next. The next game on this court (or any court), with its time, teams and logos. After the last game it becomes the end-of-night card.

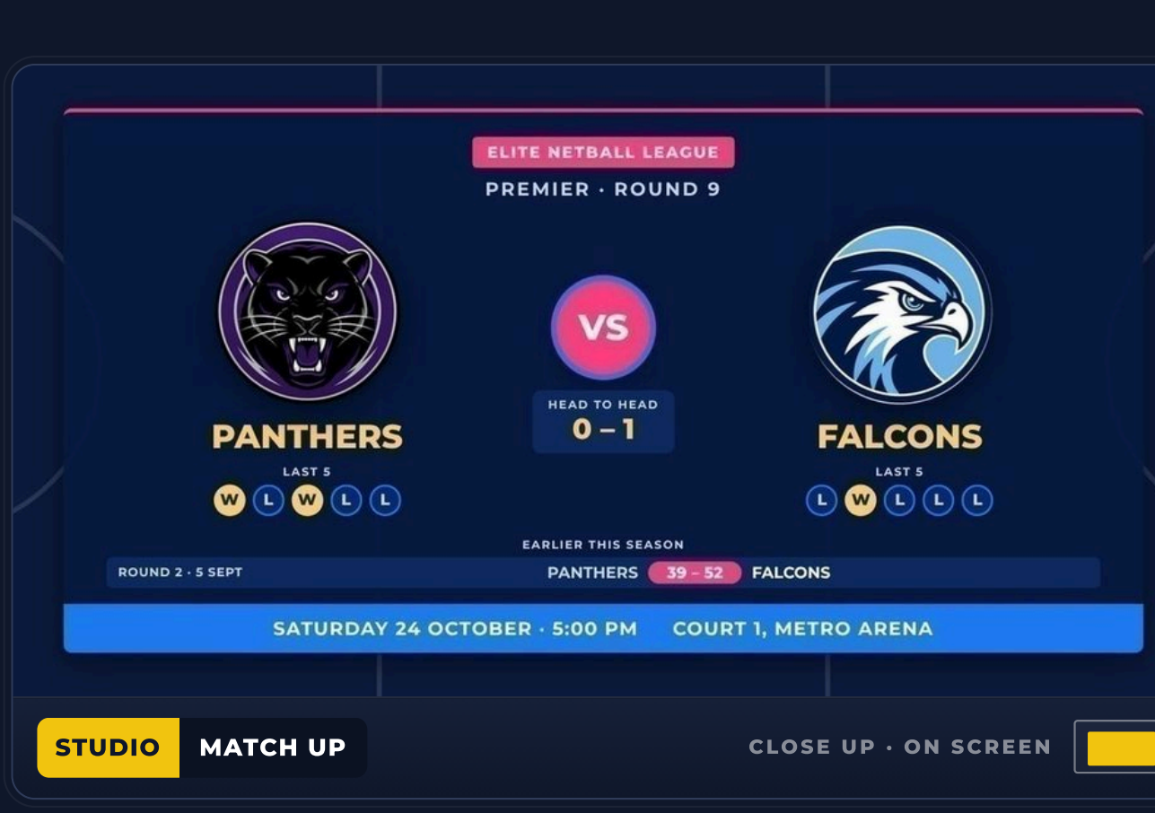


FIG 5.8 Match up. Both teams side by side with their last five results as form dots and this season's head to head. **Form** and **Head to head** can each be hidden.

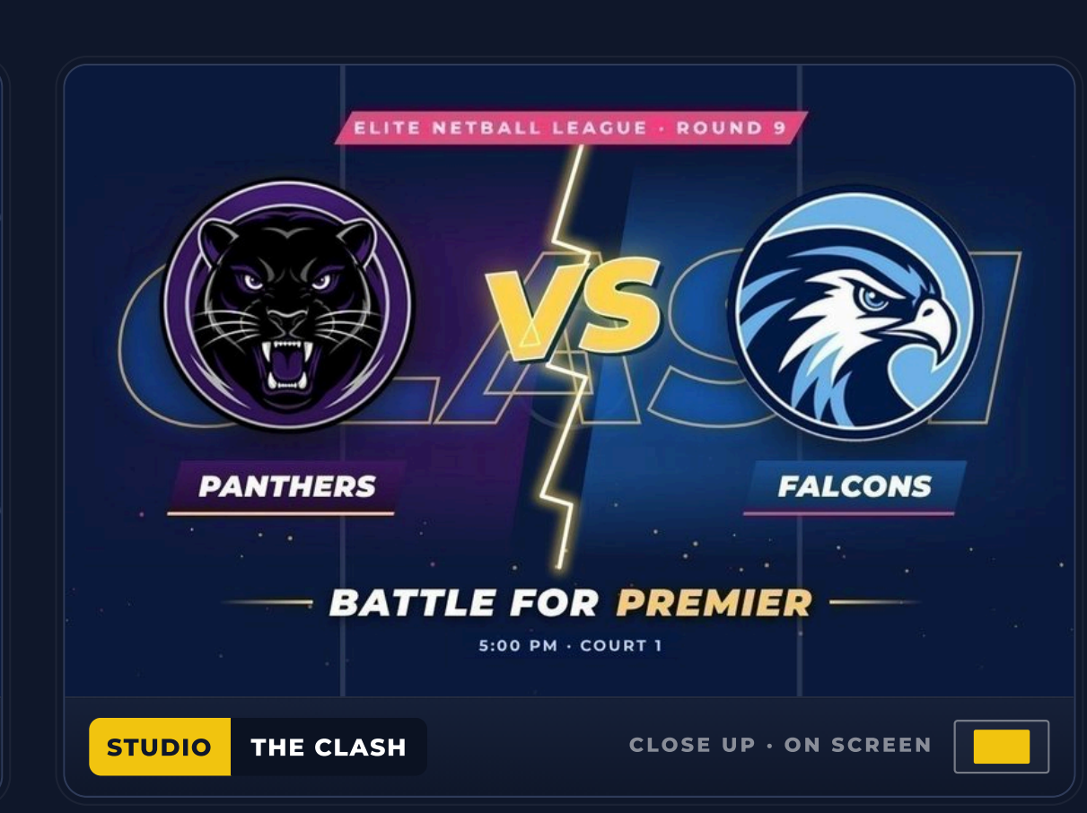


FIG 5.9 The Clash. The two teams slam together behind a big word (IMPACT, CLASH, SHOWDOWN, your own) in their club colours. Options: the word, the bottom line, animate away after N seconds.

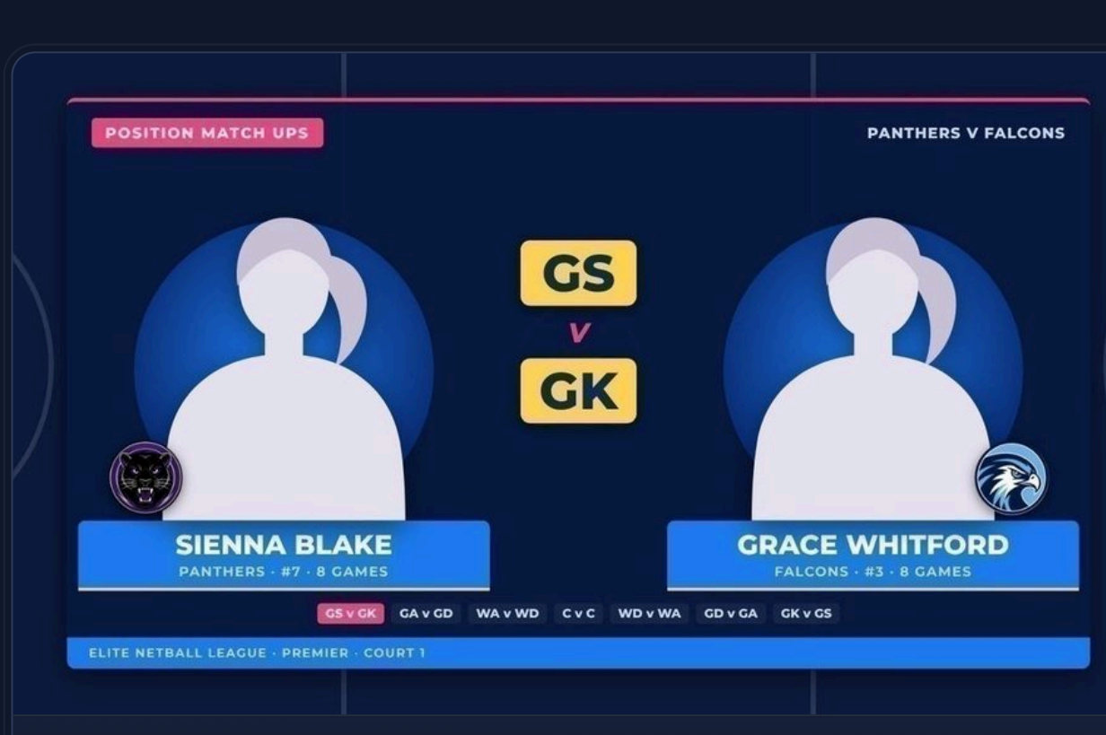
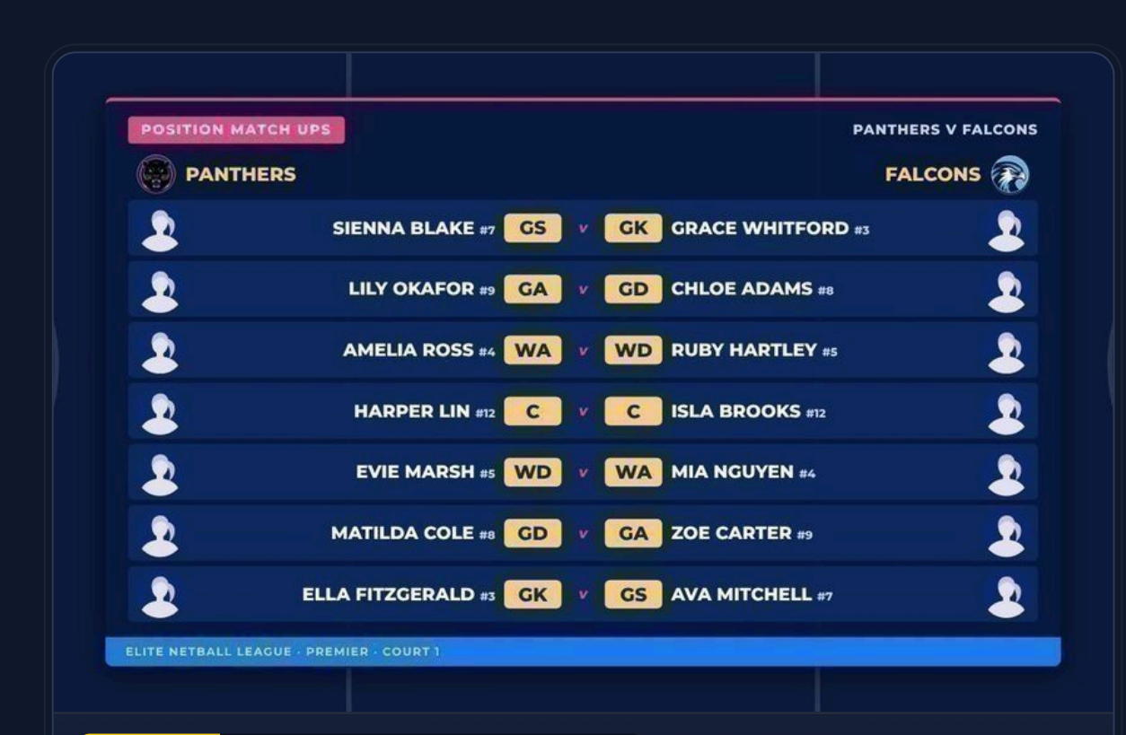
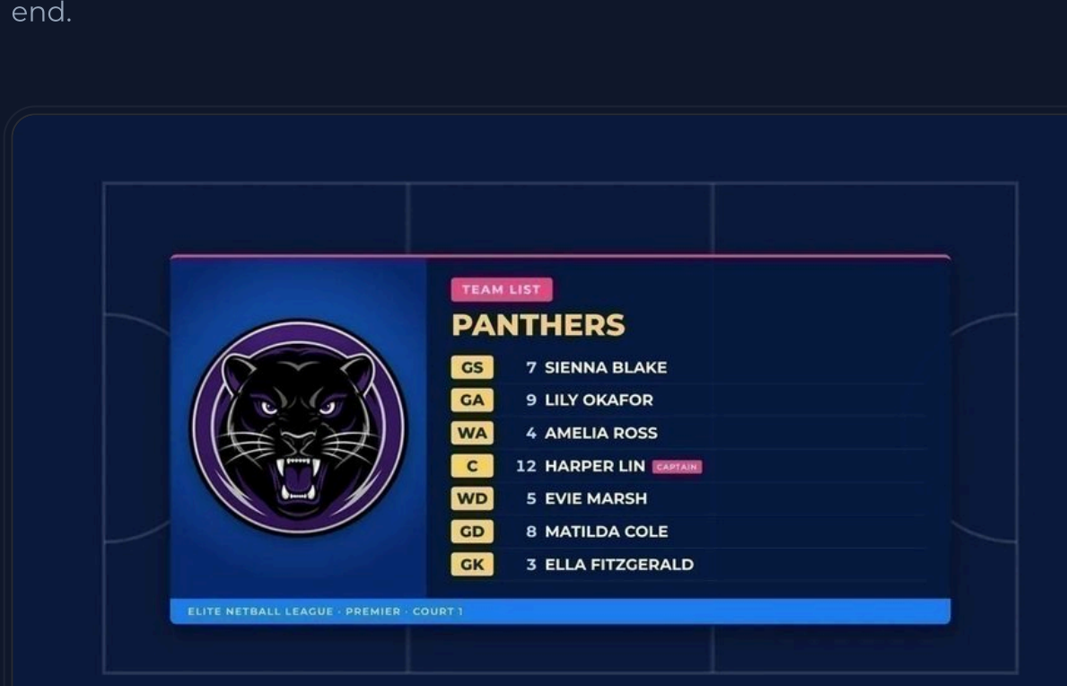


FIG 5.10 Position match ups. Who marks whom, position by position, with the athletes' photos from the Athletes page or a silhouette. Options: seconds per pairing, loop or stop, start with GS or GK, one pairing only, animate away at the end.



PANTHERS				FALCONS
SIENNA BLAKE #7	GS	v	GK	GRACE WHITFORD #3
LILY OKAFOR #9	GA	v	GD	CHLOE ADAMS #8
AMELIA ROSS #4	WA	v	WD	RUBY HARTLEY #5
HARPER LIN #12	C	v	C	ISLA BROOKS #12
EVIE MARSH #5	WD	v	WA	MIA NGUYEN #4
MATILDA COLE #8	GD	v	GA	ZOE CARTER #9
ELLA FITZGERALD #3	GK	v	GS	AVA MITCHELL #7

FIG 5.11 Position match ups, all seven. The same page can show every pairing at once as one card.



GS	7	SIENNA BLAKE
GA	9	LILY OKAFOR
WA	4	AMELIA ROSS
C	12	HARPER LIN CAPTAIN
WD	5	EVIE MARSH
GD	8	MATILDA COLE
GK	3	ELLA FITZGERALD

FIG 5.12 Team list. One team's line-up from PlayHQ with positions, captain and bench, warmed up before the game so it shows at once. Add one source per side.

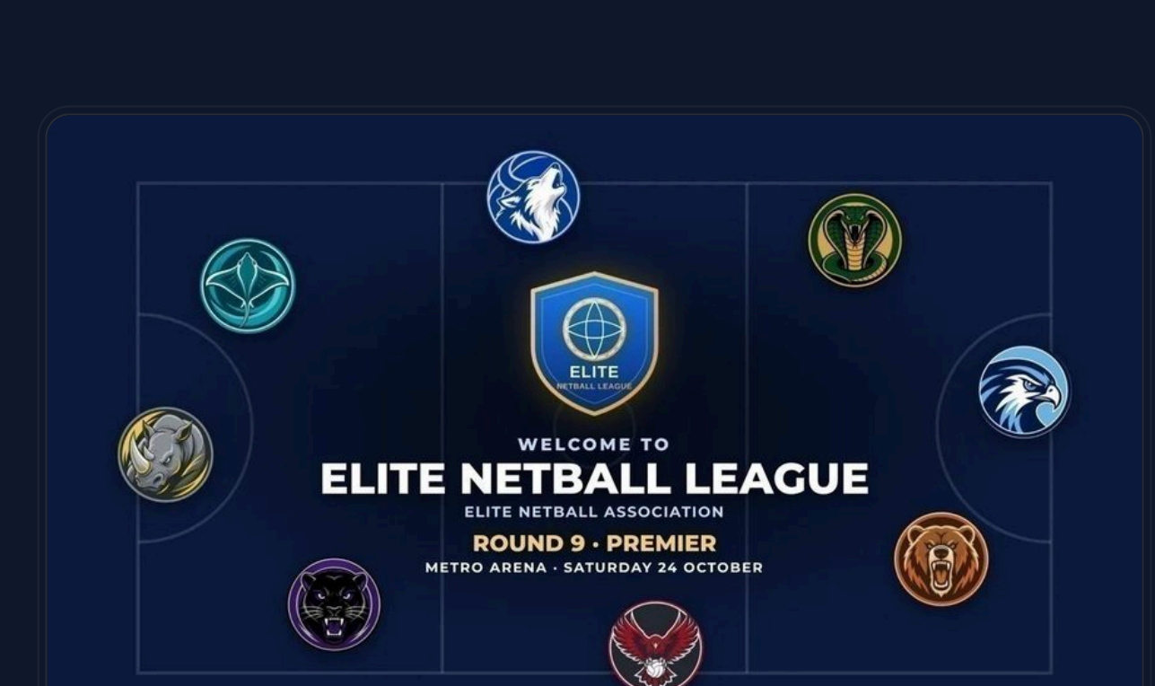


FIG 5.13 Intro and outro. Six full-screen styles, about 8 seconds, with the round's teams and a yellow message line. Options: style, teams from a division or round, time on screen, a theme song with its timing and volume, sound effects, a background.

5.3 During play

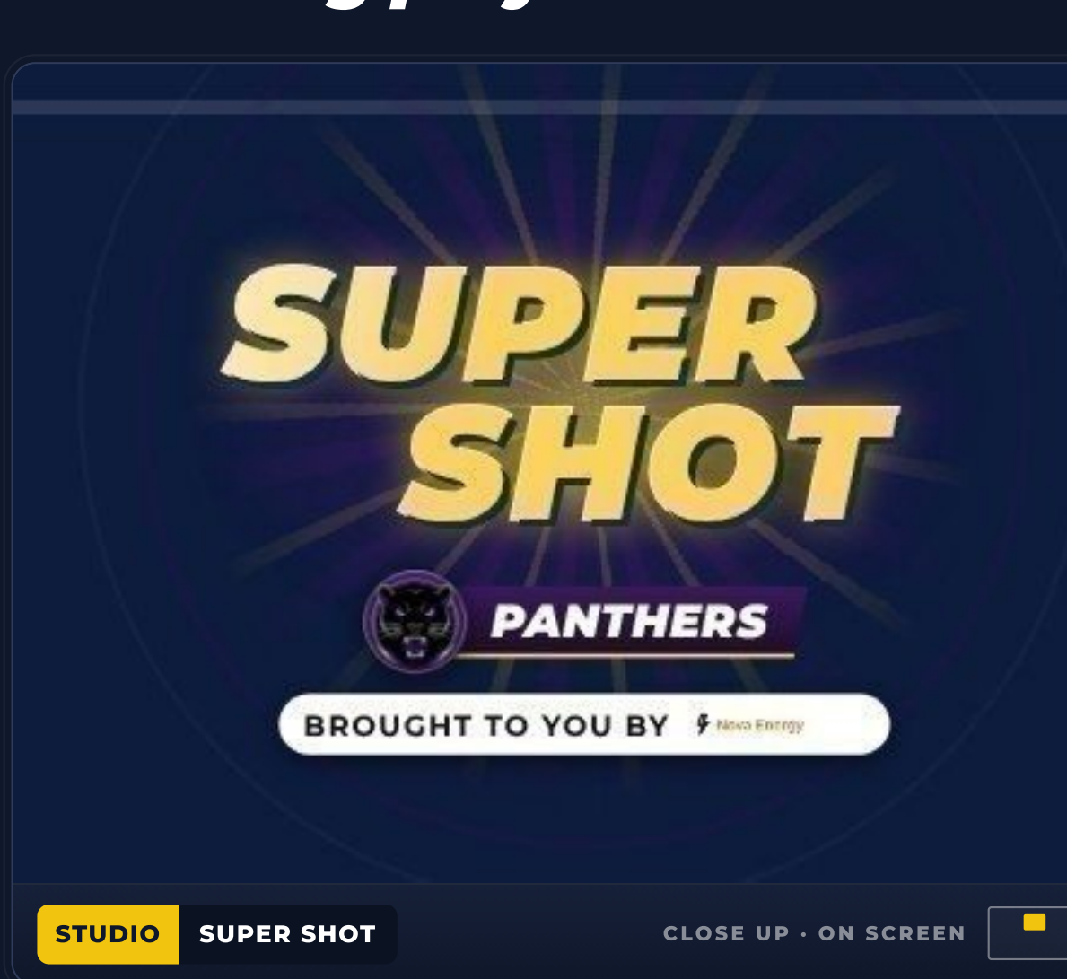


FIG 5.14 Super shot. Automatic, 3 seconds, the scoring team's logo and colour; the sponsored-moment tag names who brought it. **Plays: home / away / no team** makes a manual one.

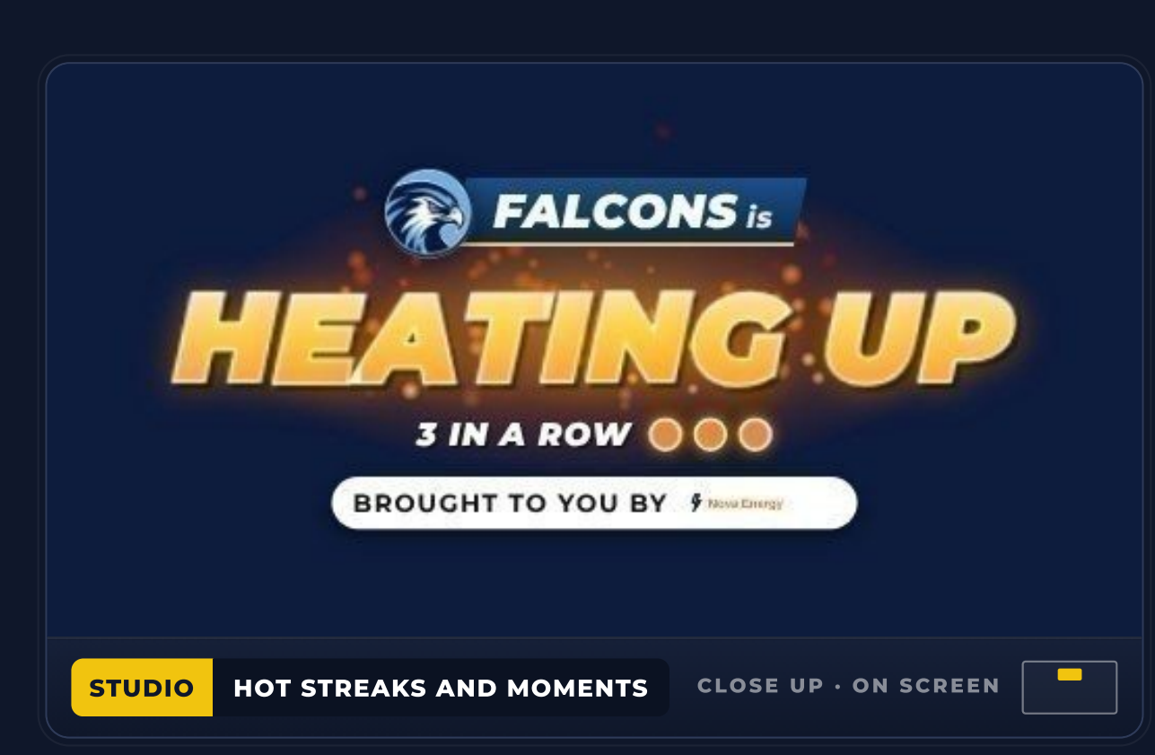


FIG 5.15 Hot streaks and moments. Heating up, on fire, goal for goal, hits the front, all square, coming back, on the siren, back to back. Automatic from the goal order, or **Plays** picks one to play once.

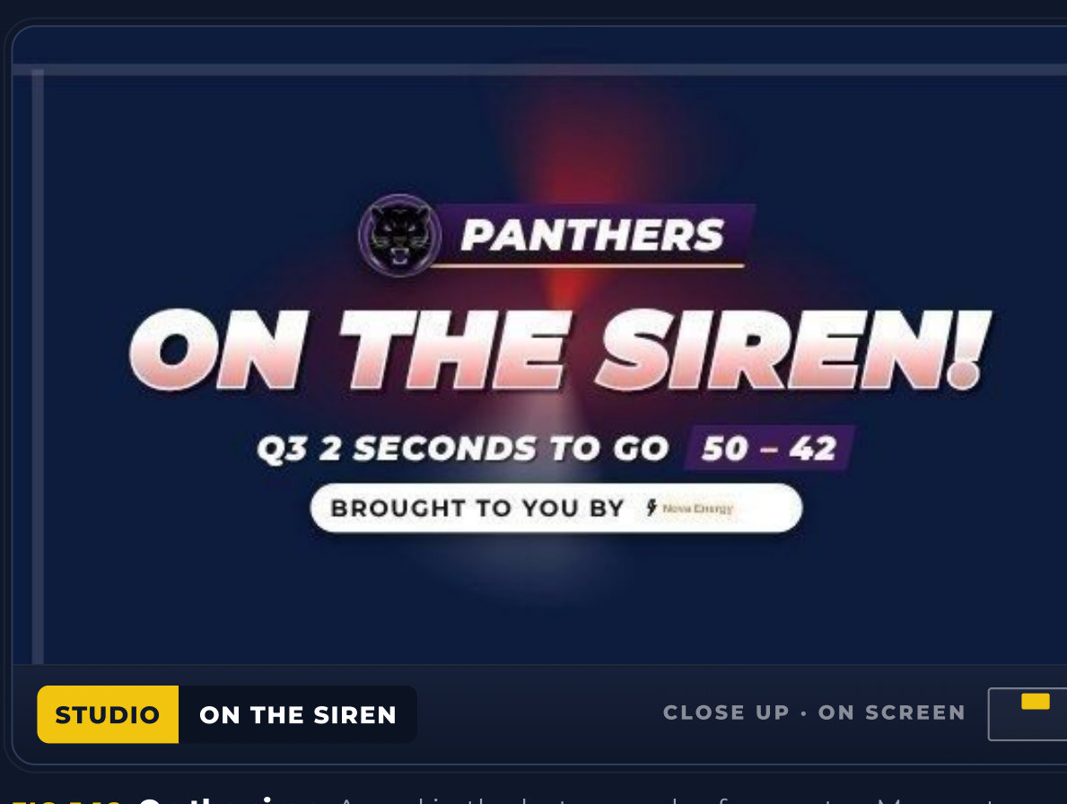


FIG 5.16 On the siren. A goal in the last seconds of a quarter. Moments are the only animations that need nothing from you; they also take the sponsor's "brought to you by" tag.

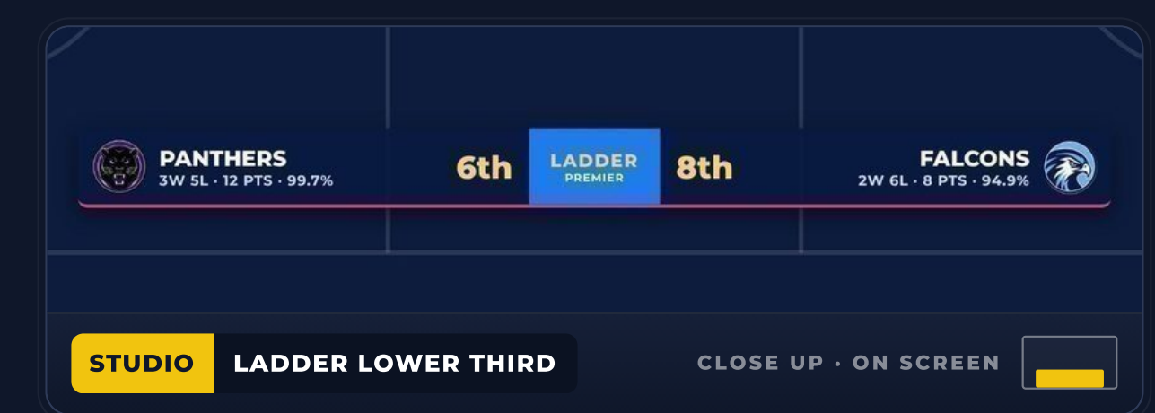


FIG 5.17 Ladder lower third. Both teams' ladder position and record as one strip right above the scoreboard, the same width as it, for a glance during play without covering the game.

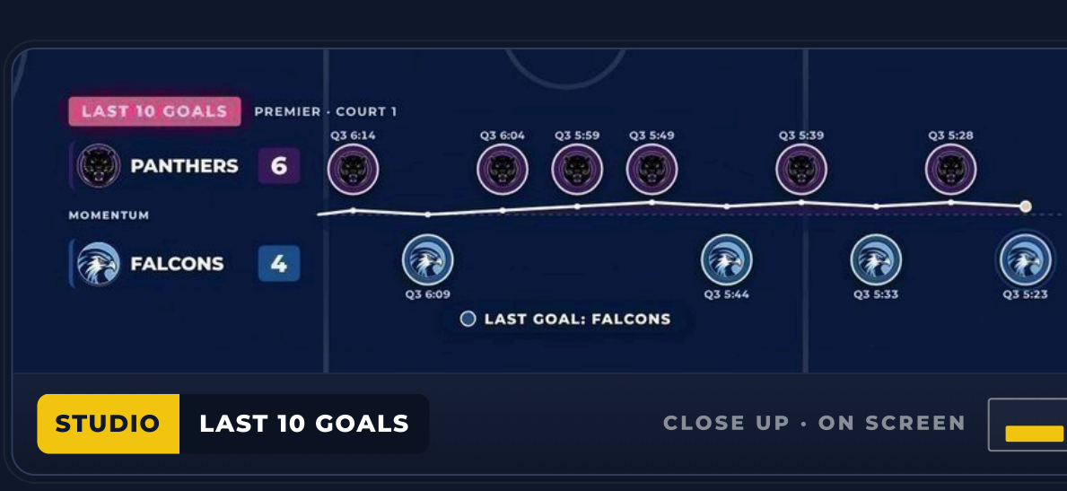
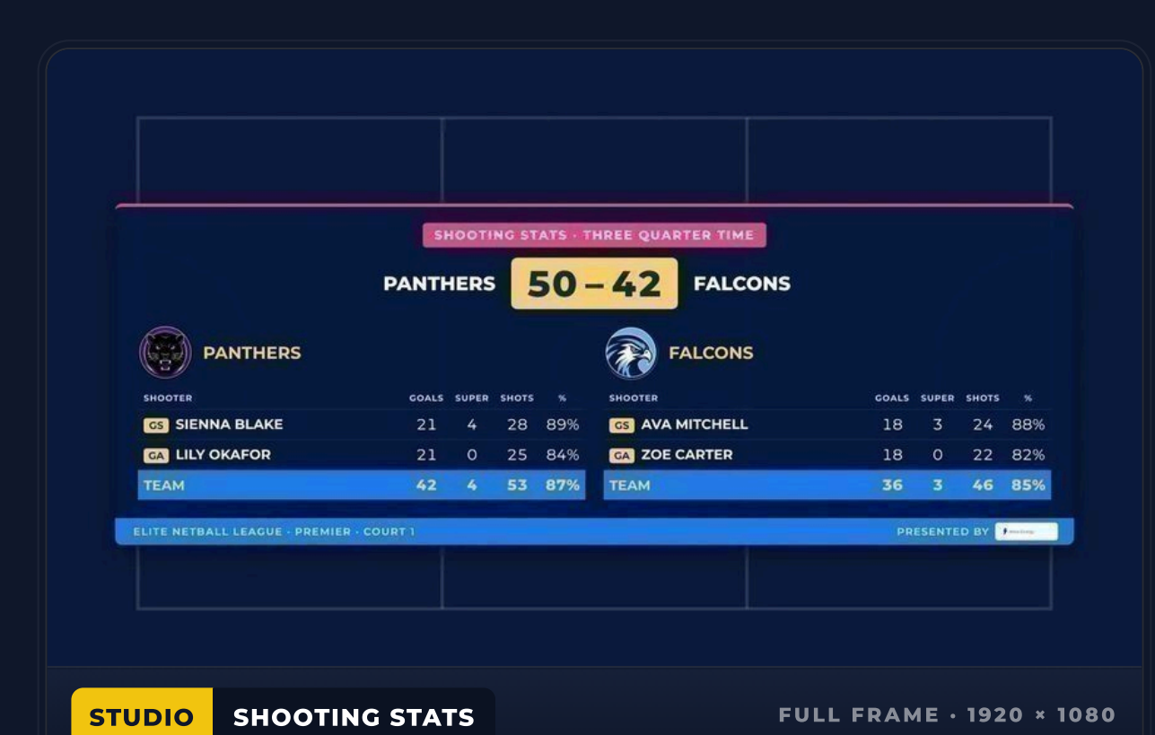


FIG 5.18 Last 10 goals. The game's recent goals oldest to newest with a momentum line and the current run, worked out by Studio from the score, updating while it stays on. Options: across the top or down the side, how many goals, position and heading.



PANTHERS SHOOTER	GOALS	SUPER	SHOTS	%	FALCONS SHOOTER	GOALS	SUPER	SHOTS	%
GS SIENNA BLAKE	21	4	28	89%	GS AVA MITCHELL	18	3	24	88%
GA LILY OKAFOR	21	0	25	84%	GA ZOE CARTER	18	0	22	82%
TEAM	42	4	53	87%	TEAM	36	3	46	85%

FIG 5.19 Shooting stats. Goals, super shots and attempts per shooter from PlayHQ's game page, for a break or a quarter-time sequence. Super columns are left off for a division without super shots.

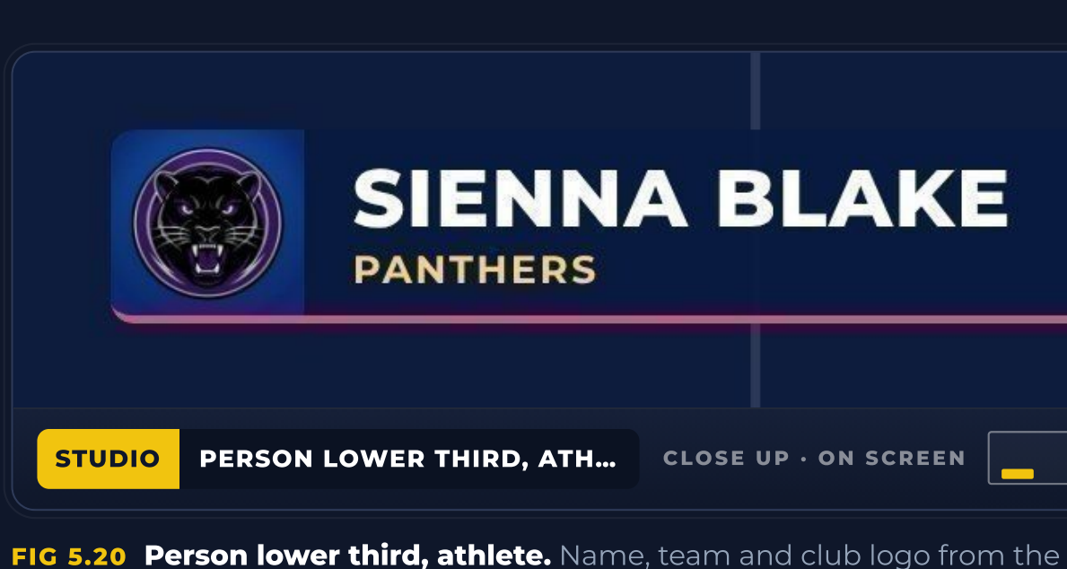


FIG 5.20 Person lower third, athlete. Name, team and club logo from the athlete register, with their photo if there is one. The Output page suggests names as you type.

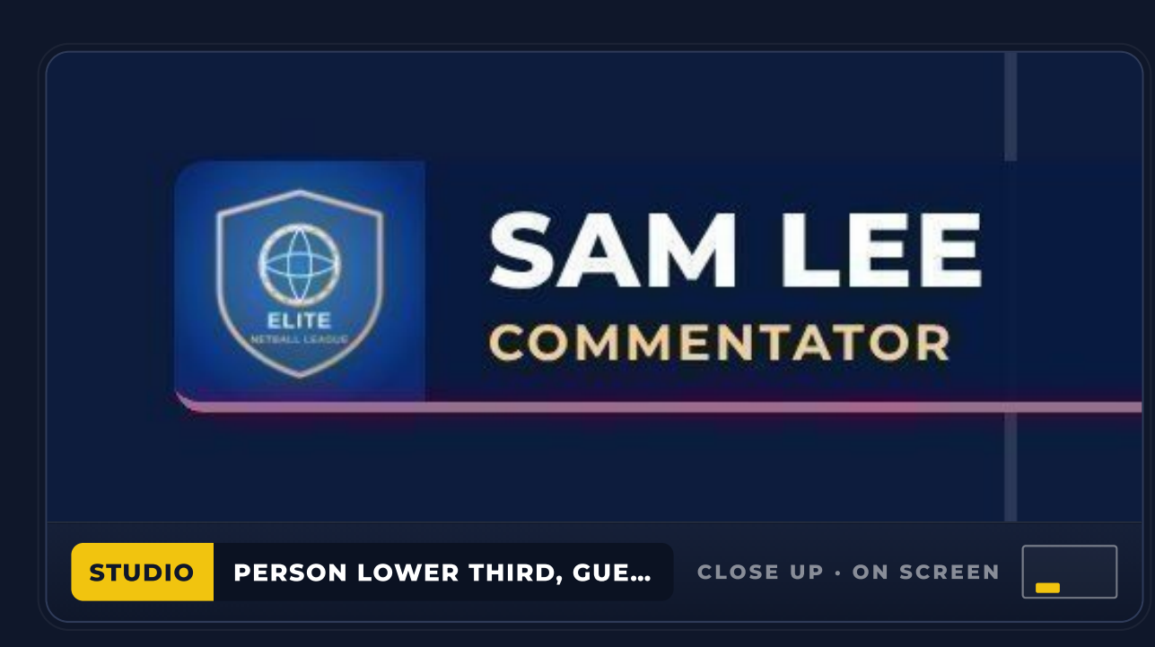


FIG 5.21 Person lower third, guest. Name, title and the competition logo, for a commentator, an official or a sponsor's guest. Options: a tab above, position left or right, animate away.

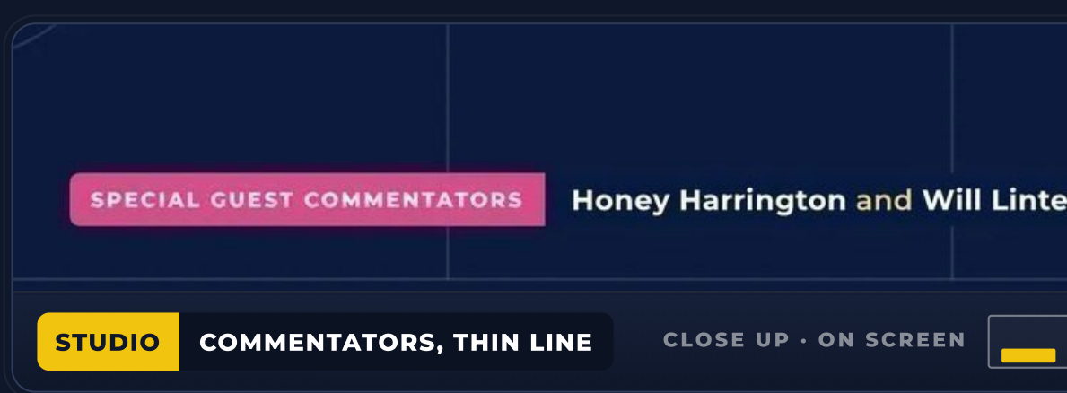


FIG 5.22 Commentators, thin line. The commentary team in one slim strip above the scoreboard: a pink tab, like Special guest commentators, and up to four names joined with "and". With no names typed, whoever the Production Planner has on the mic.

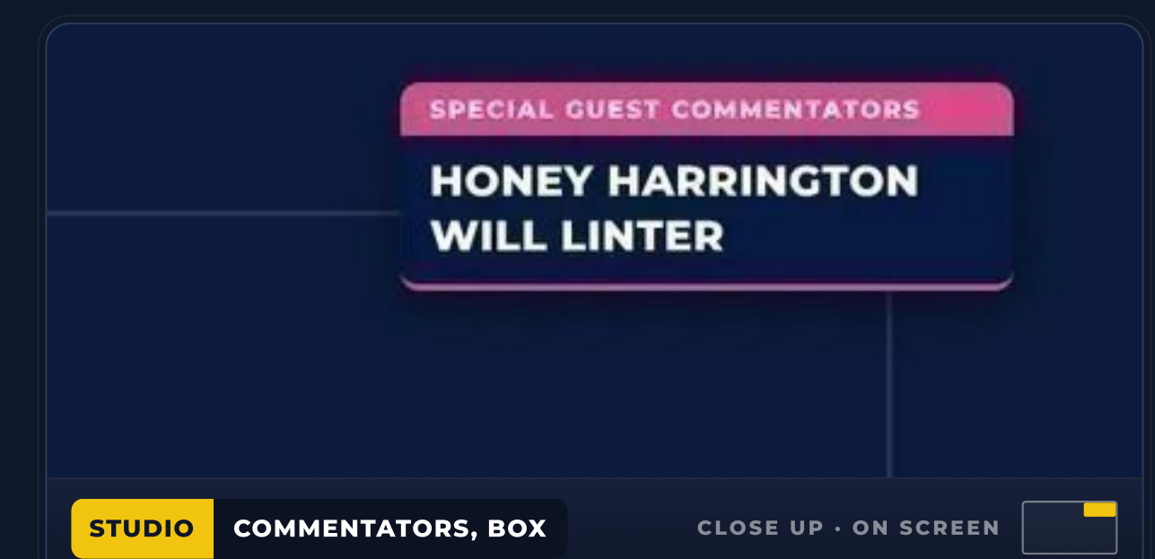


FIG 5.23 Commentators, box. The same names as an upper third in the top right corner beside the competition logo, a line per name. Options: under the title instead, the tab's words, animate away.

5.4 After the game

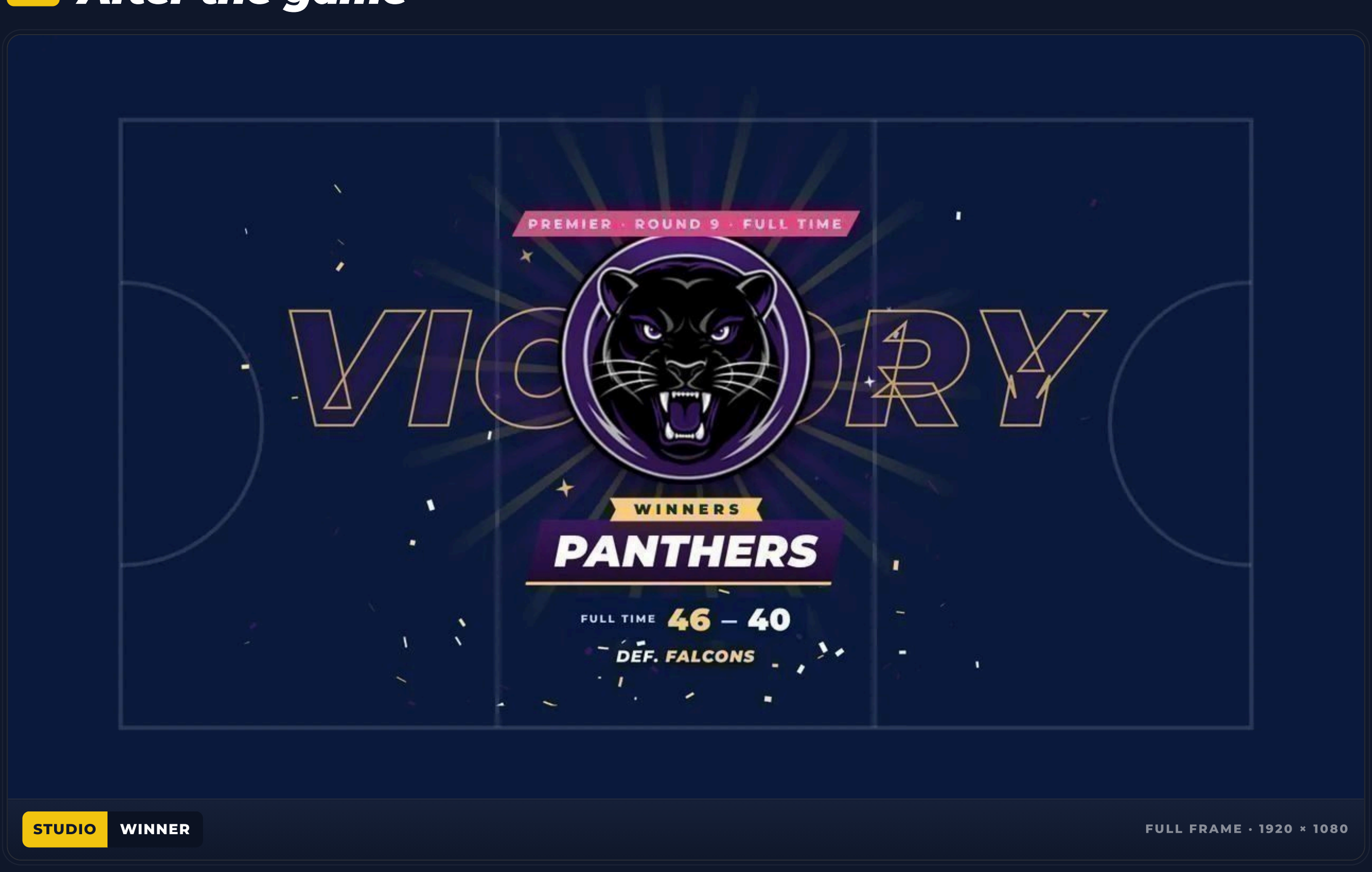


FIG 5.24 Winner. The winning team, logo, final score and a big word, worked out from the board each time it comes on. Options: force a result, the word, the line under the score, animate away.

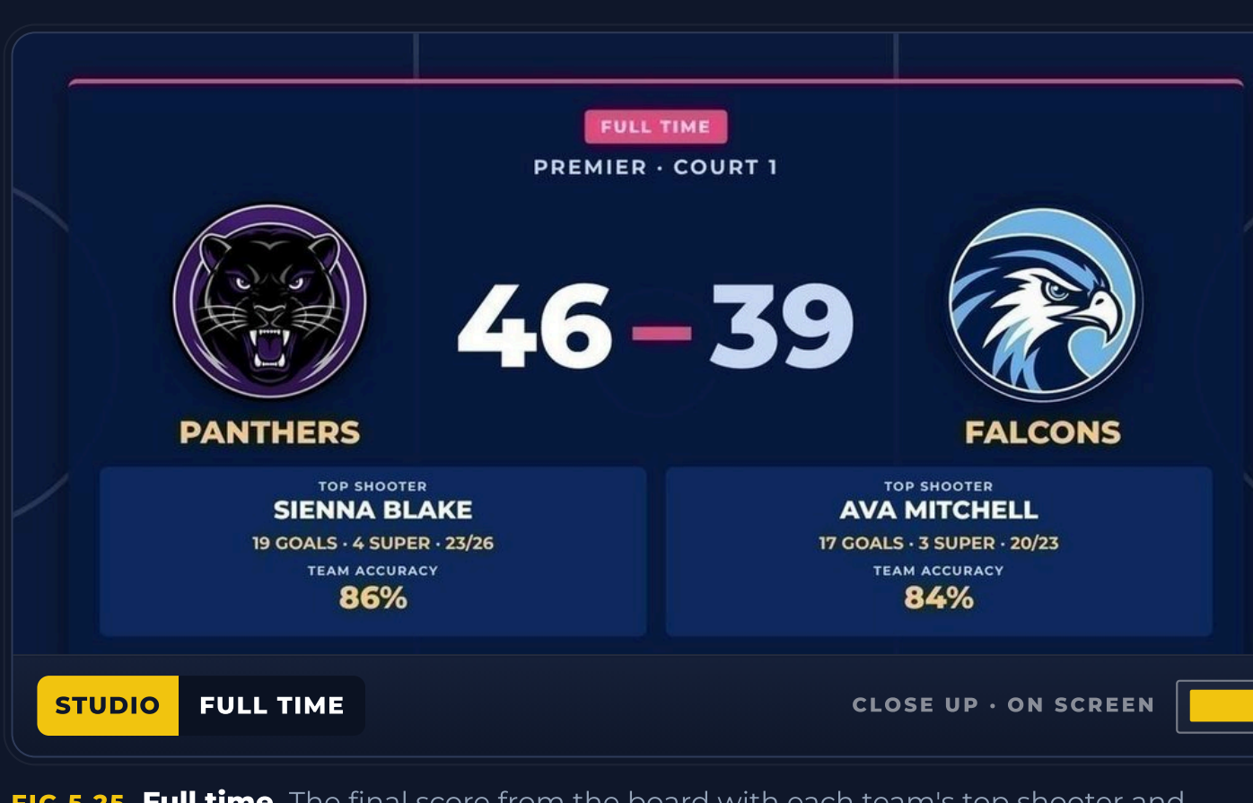
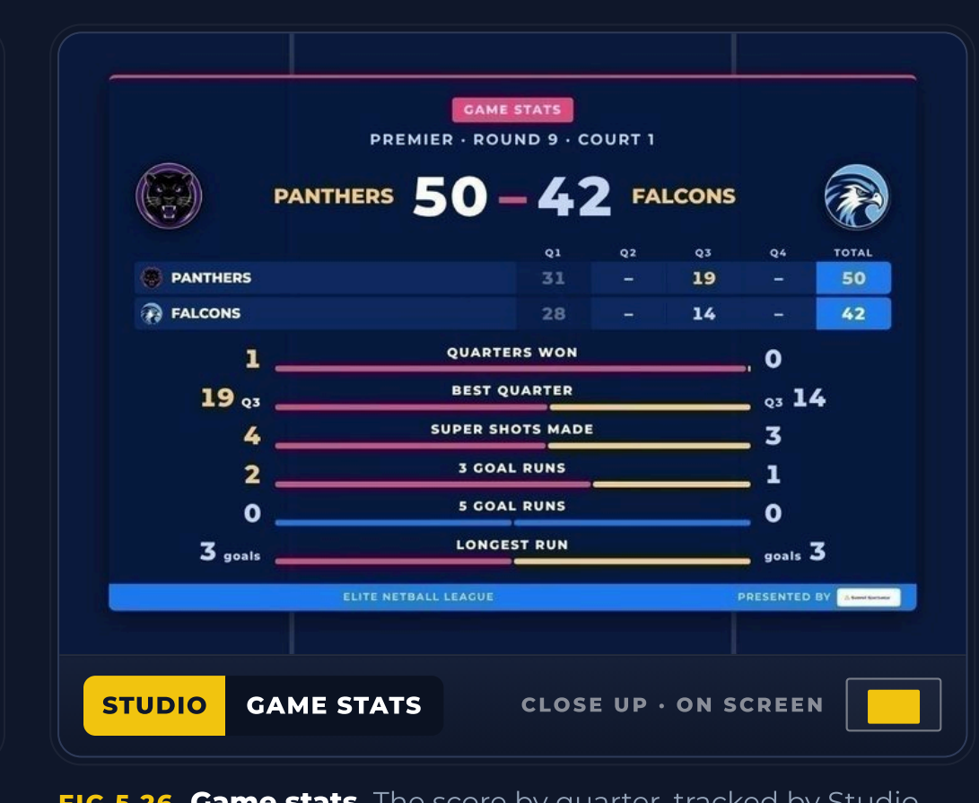


FIG 5.25 Full time. The final score from the board with each team's top shooter and shooting accuracy from PlayHQ's game stats. **Heading** makes it a half time card.



	Q1	Q2	Q3	Q4	TOTAL
PANTHERS	31	–	19	–	50
FALCONS	28	–	14	–	42

PANTHERS		FALCONS
1	QUARTERS WON	0
19 Q3	BEST QUARTER	Q3 14
4	SUPER SHOTS MADE	3
2	3 GOAL RUNS	1
0	5 GOAL RUNS	0
3 goals	LONGEST RUN	goals 3

FIG 5.26 Game stats. The score by quarter, tracked by Studio from the board, then team against team: quarters won, best quarter, super shots made, 3 and 5 goal runs and the longest run, with a sponsor's footer.

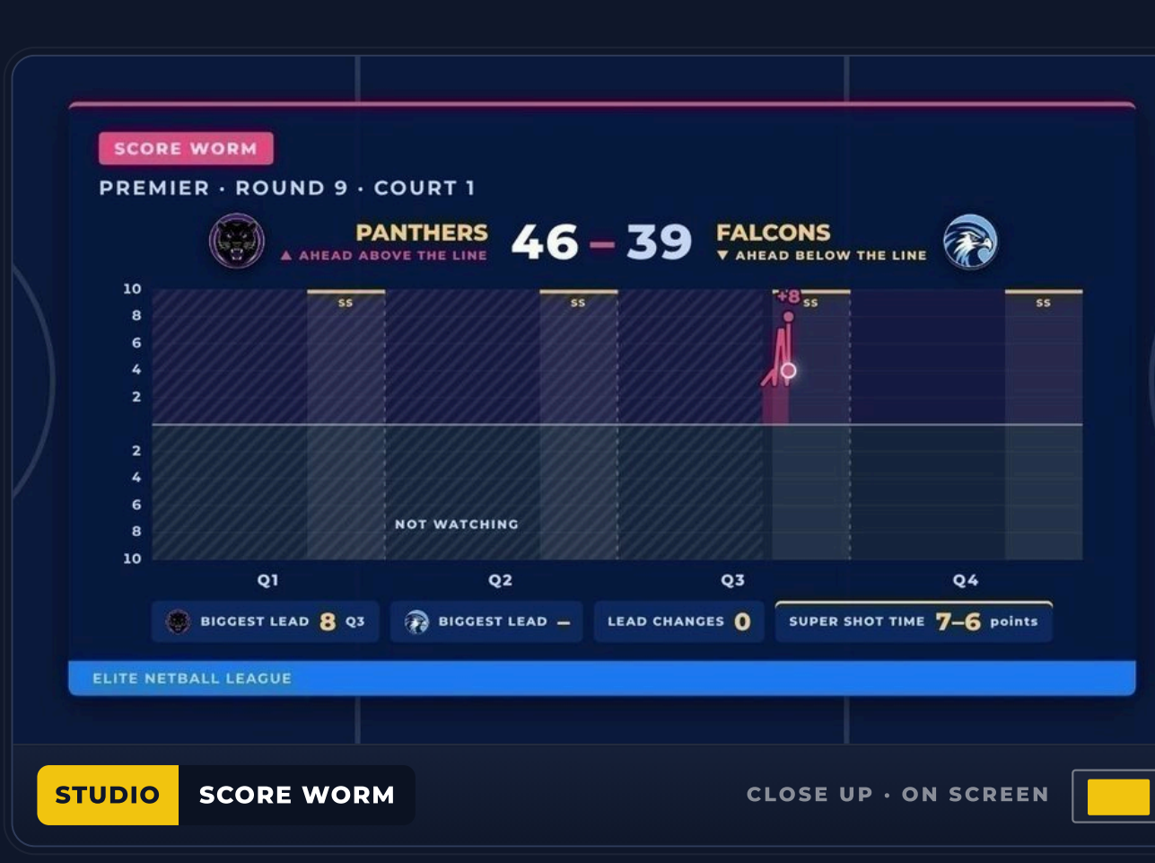


FIG 5.27 Score worm. The margin through the game as a line, drawn in over a few seconds. Options: drawing time, club colours or the competition's.

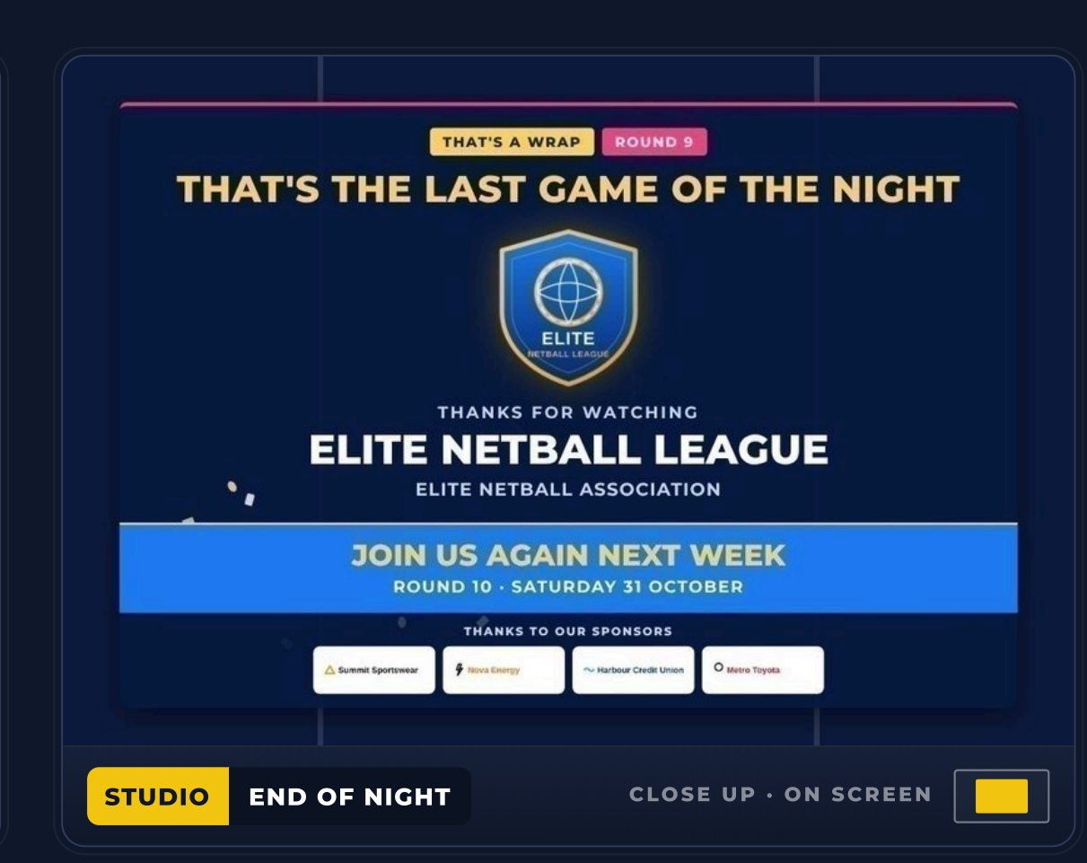
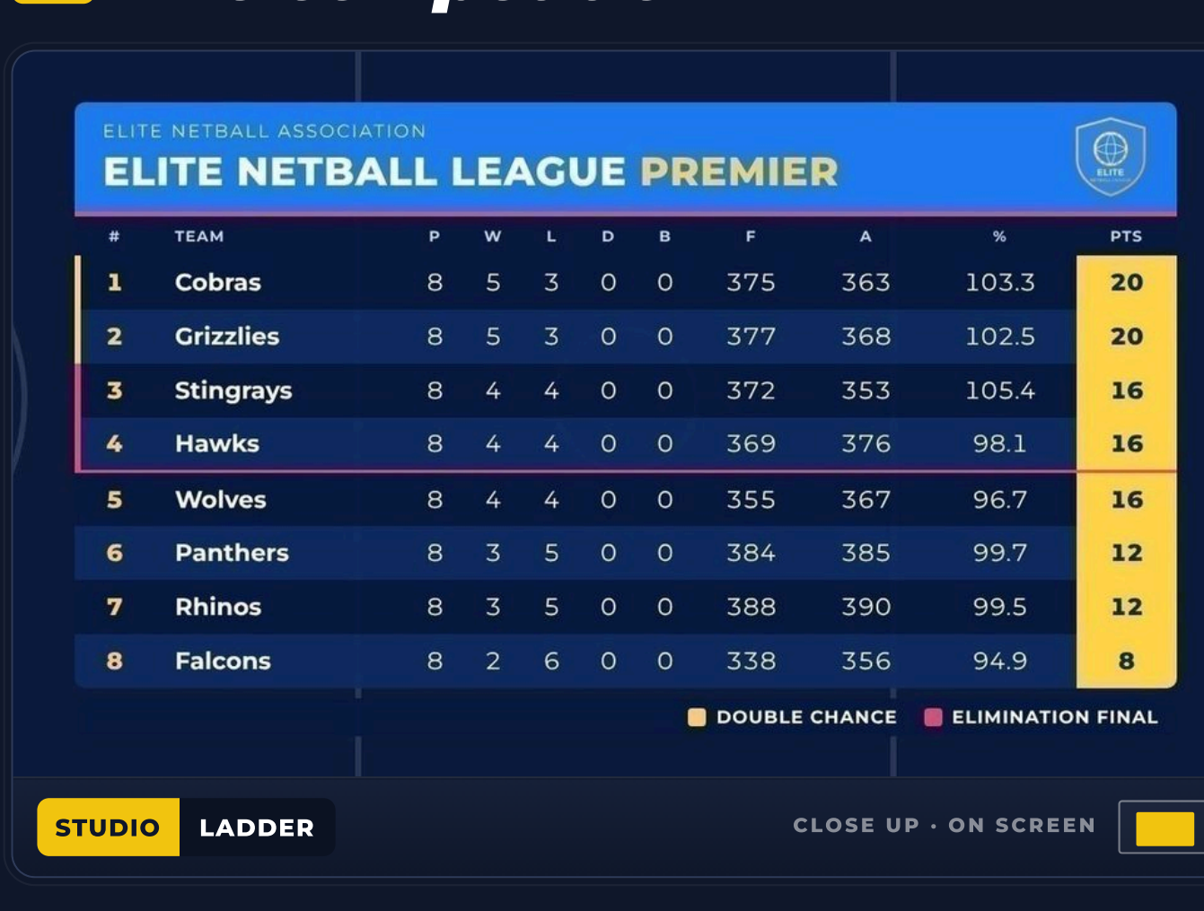


FIG 5.28 End of night. Up next after the last game: thanks for watching, next week's date or the finals, and the Supporter sponsors in a strip. Options: the invite line, sponsors on or off.

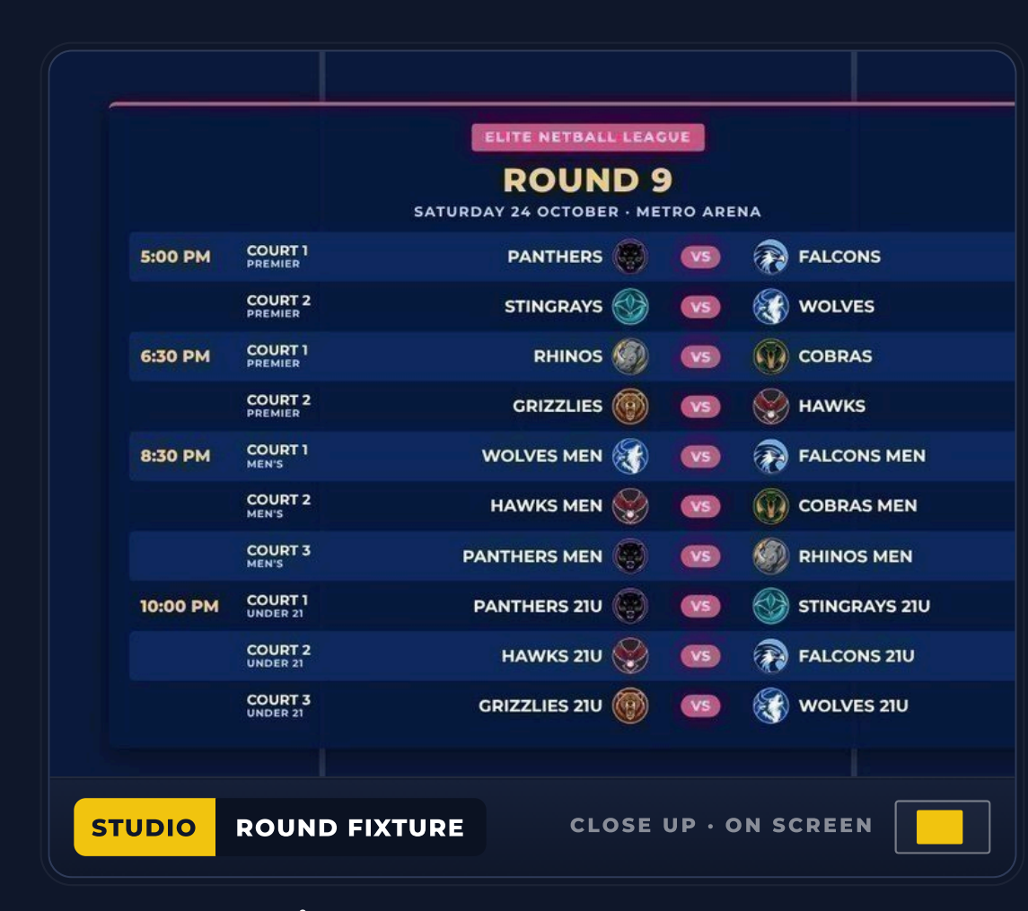
> **Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits.**
>
> CHAPTER 5 · THE GRAPHICS

5.5 The competition



#	TEAM	P	W	L	D	B	F	A	%	PTS
1	Cobras	8	5	3	0	0	375	363	103.3	20
2	Grizzlies	8	5	3	0	0	377	368	102.5	20
3	Stingrays	8	4	4	0	0	372	353	105.4	16
4	Hawks	8	4	4	0	0	369	376	98.1	16
5	Wolves	8	4	4	0	0	355	367	96.7	16
6	Panthers	8	3	5	0	0	384	385	99.7	12
7	Rhinos	8	3	5	0	0	388	390	99.5	12
8	Falcons	8	2	6	0	0	338	356	94.9	8

FIG 5.29 Ladder. The division's ladder with the finals places marked and a key under it. Options: division, rows, logo on or off, light or dark mode logo.



5:00 PM	COURT 1 PREMIER	PANTHERS	VS	FALCONS
	COURT 2 PREMIER	STINGRAYS	VS	WOLVES
6:30 PM	COURT 1 PREMIER	RHINOS	VS	COBRAS
	COURT 2 PREMIER	GRIZZLIES	VS	HAWKS
8:30 PM	COURT 1 MEN'S	WOLVES MEN	VS	FALCONS MEN
	COURT 2 MEN'S	HAWKS MEN	VS	COBRAS MEN
	COURT 3 MEN'S	PANTHERS MEN	VS	RHINOS MEN
10:00 PM	COURT 1 UNDER 21	PANTHERS 21U	VS	STINGRAYS 21U
	COURT 2 UNDER 21	HAWKS 21U	VS	FALCONS 21U
	COURT 3 UNDER 21	GRIZZLIES 21U	VS	WOLVES 21U

FIG 5.30 Round fixture. This round's games across all divisions with day, time and court. **Round** takes next, prev, +2, a number or a name like Final. When divisions are on different rounds, each game shows its own.

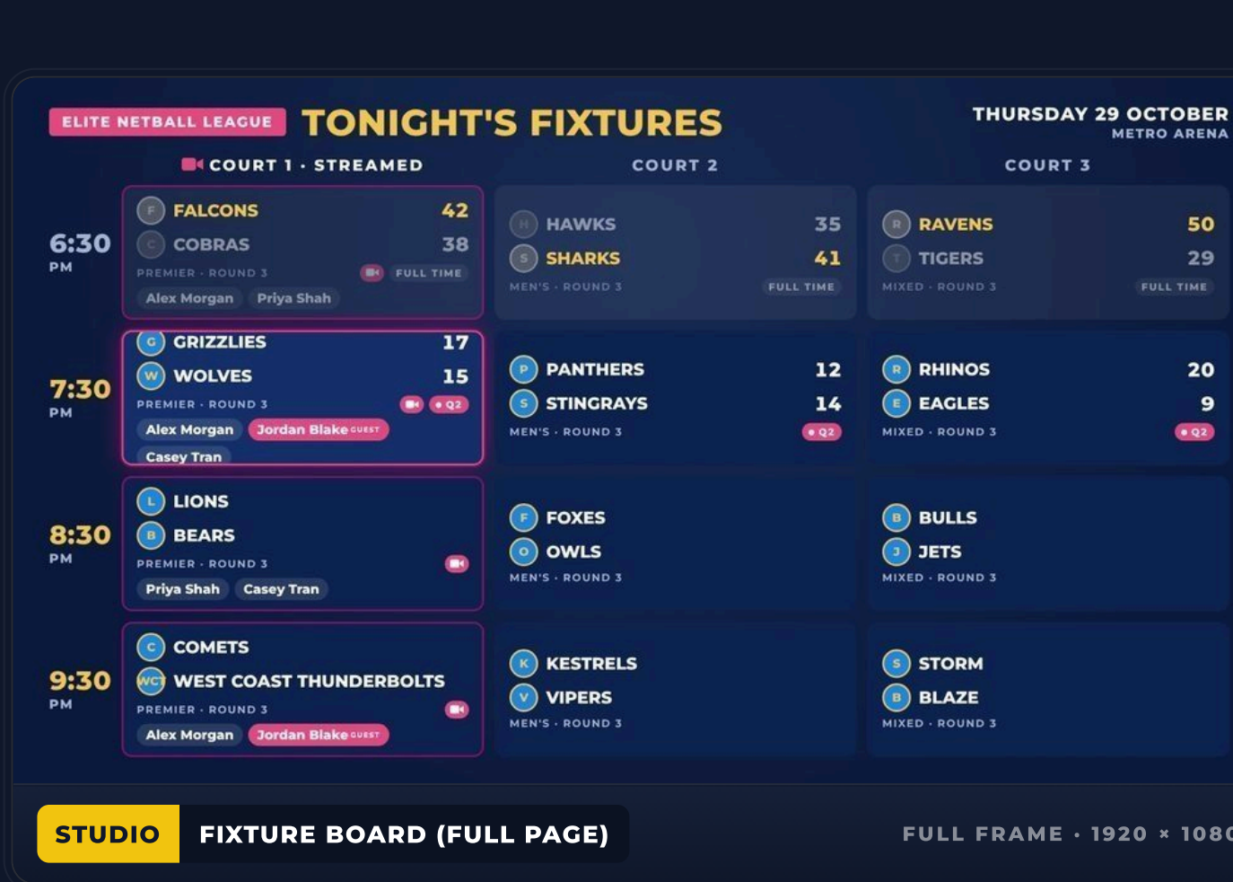
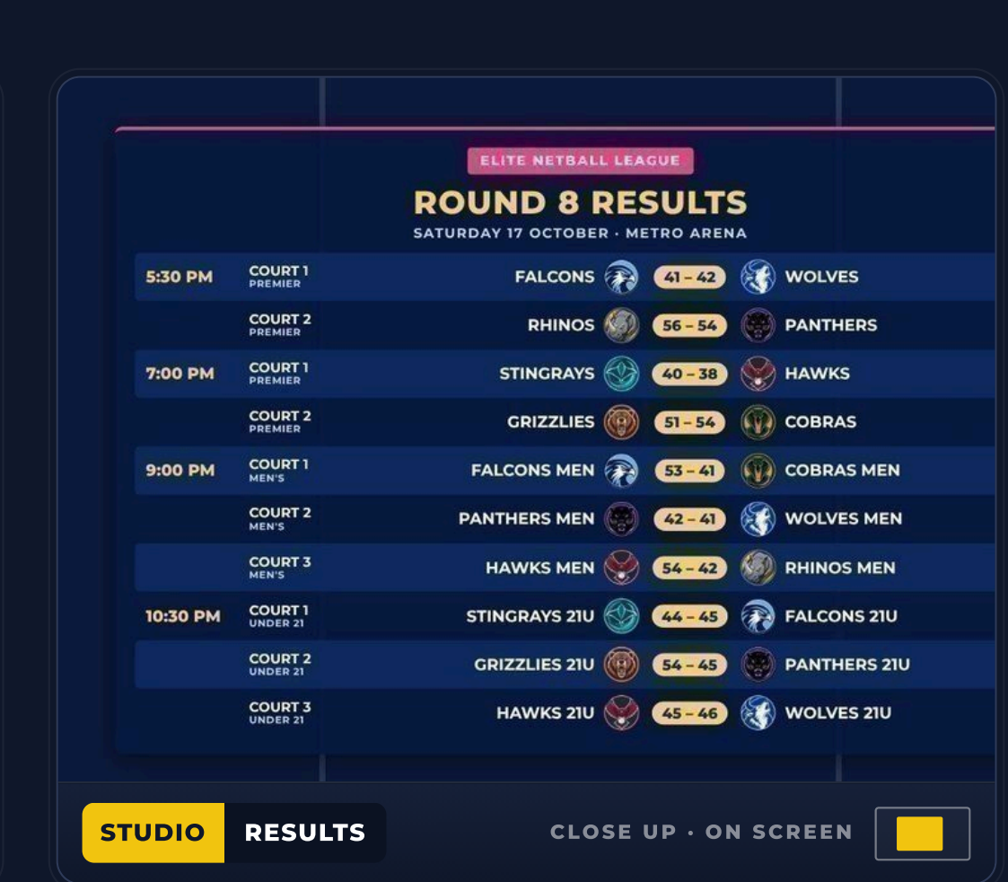
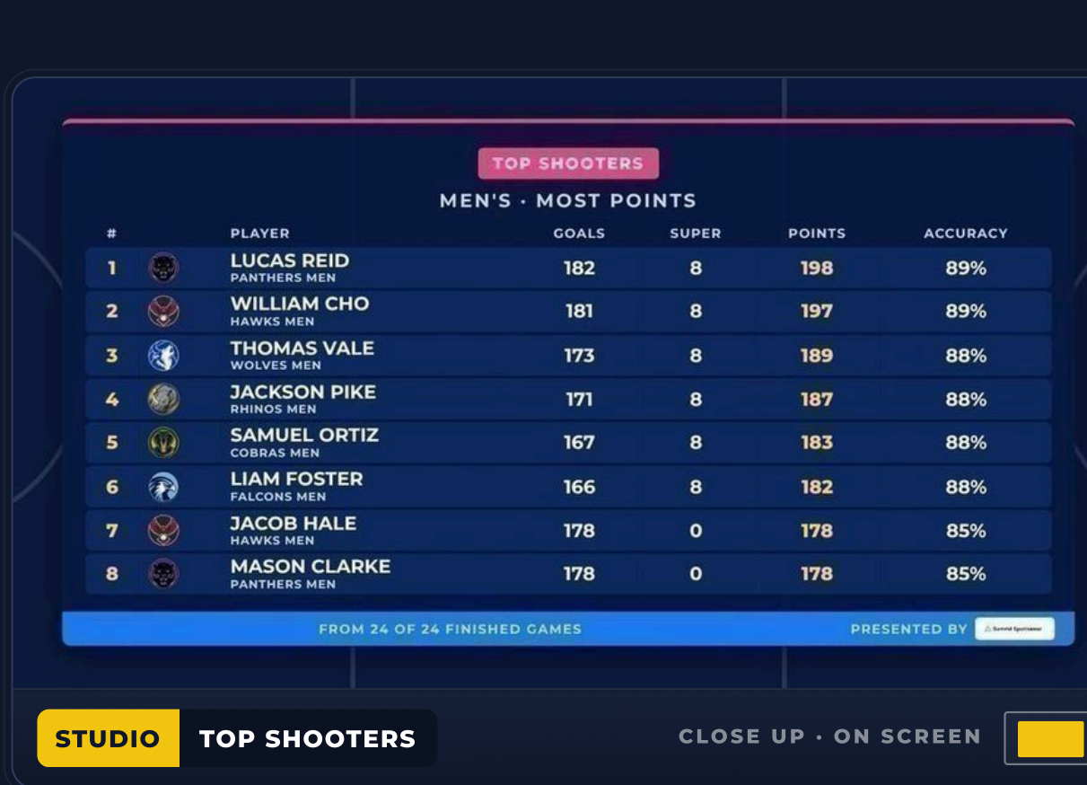


FIG 5.31 Fixture board (Full Page). The night's whole fixture, a card per court with its games by time, for a screen in the competition room or backstage, or a picture to share before the night (**Tuesday Night's Fixtures**). The streamed court has a camera and its games name the commentators from the Production Planner; games being played show their live score, and finished ones grey out with the winner in yellow. Not for the stream.



5:30 PM	COURT 1 PREMIER	FALCONS	41 – 42	WOLVES
	COURT 2 PREMIER	RHINOS	56 – 54	PANTHERS
7:00 PM	COURT 1 PREMIER	STINGRAYS	40 – 38	HAWKS
	COURT 2 PREMIER	GRIZZLIES	51 – 54	COBRAS
9:00 PM	COURT 1 MEN'S	FALCONS MEN	53 – 41	COBRAS MEN
	COURT 2 MEN'S	PANTHERS MEN	42 – 41	WOLVES MEN
	COURT 3 MEN'S	HAWKS MEN	54 – 42	RHINOS MEN
10:30 PM	COURT 1 UNDER 21	STINGRAYS 21U	44 – 45	FALCONS 21U
	COURT 2 UNDER 21	GRIZZLIES 21U	54 – 45	PANTHERS 21U
	COURT 3 UNDER 21	HAWKS 21U	45 – 46	WOLVES 21U

FIG 5.32 Results. The same round once its games are final, with scores. The same round option applies.



#	PLAYER	GOALS	SUPER	POINTS	ACCURACY
1	LUCAS REID PANTHERS MEN	182	8	198	89%
2	WILLIAM CHO HAWKS MEN	181	8	197	89%
3	THOMAS VALE WOLVES MEN	173	8	189	88%
4	JACKSON PIKE RHINOS MEN	171	8	187	88%
5	SAMUEL ORTIZ COBRAS MEN	167	8	183	88%
6	LIAM FOSTER FALCONS MEN	166	8	182	88%
7	JACOB HALE HAWKS MEN	178	0	178	85%
8	MASON CLARKE PANTHERS MEN	178	0	178	85%

FIG 5.33 Top shooters. The season's leading shooters for a division. Options: order by goals or average, how many shooters, a sponsor footer.

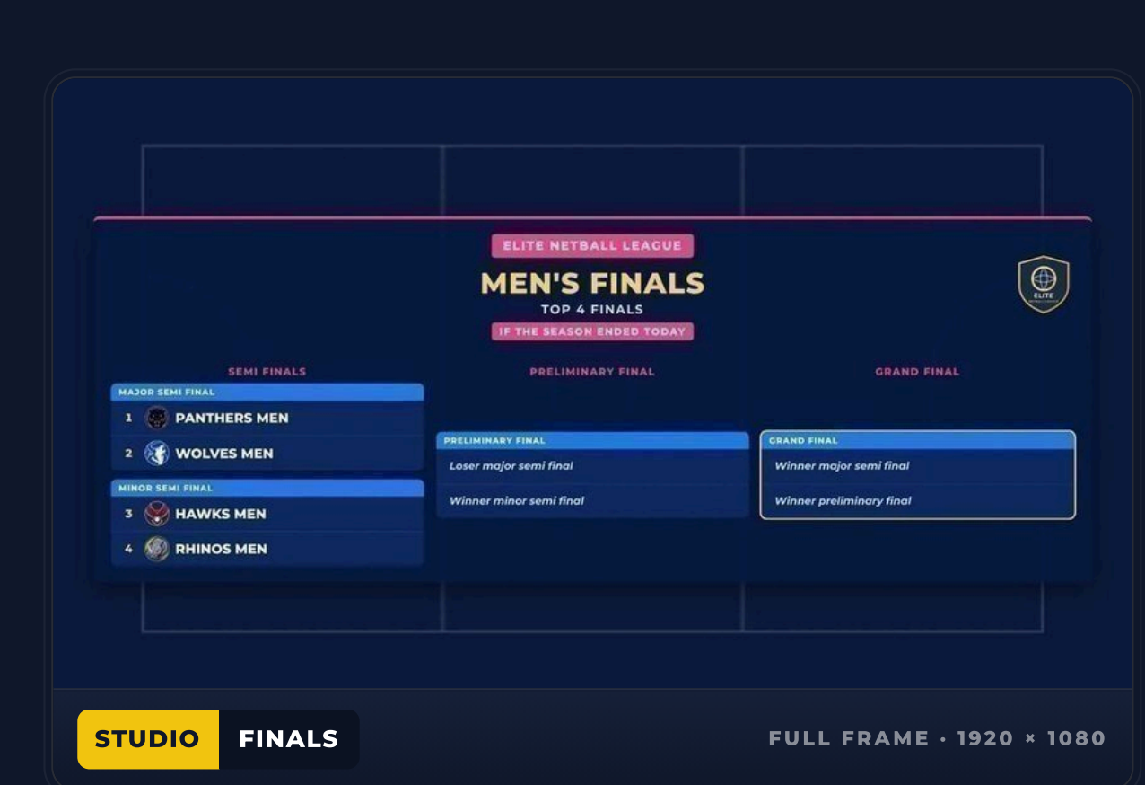


FIG 5.34 Finals. Who plays whom week by week in the division's finals format, with ladder positions and scores; "If the season ended today" until the home-and-away rounds finish. Options: one week only, the heading and the line under it.

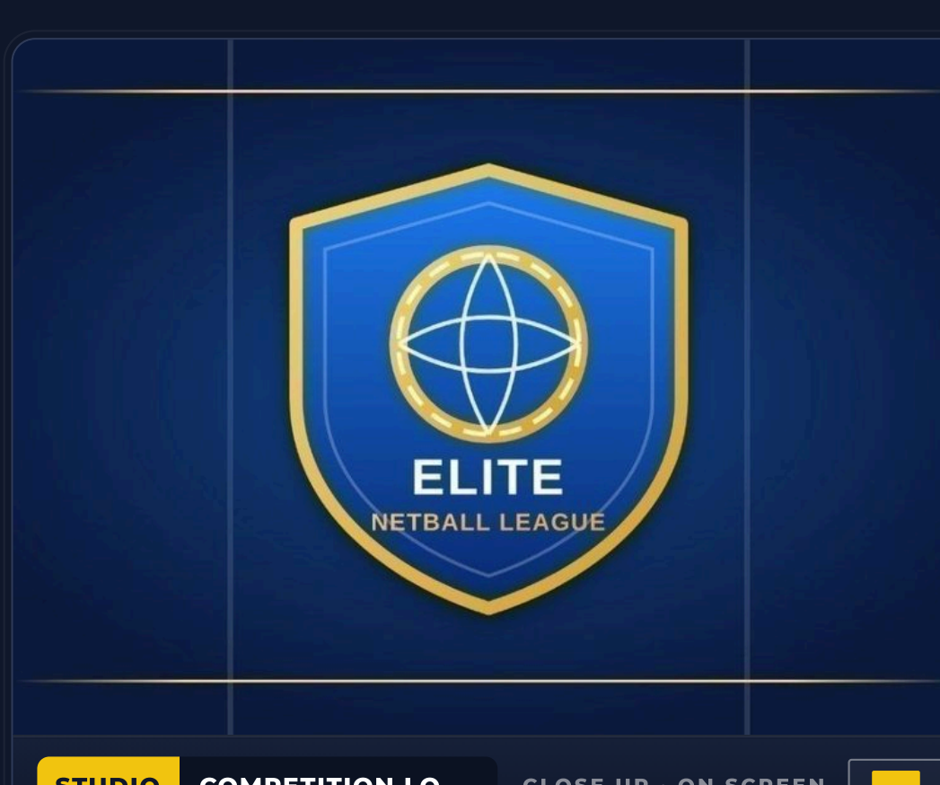


FIG 5.35 Competition logo animation. Seventeen templates (pro, coin, tiles, glitch, badge, lightning, court, bibs, net and more), about 5 seconds, with optional words and a smaller line.

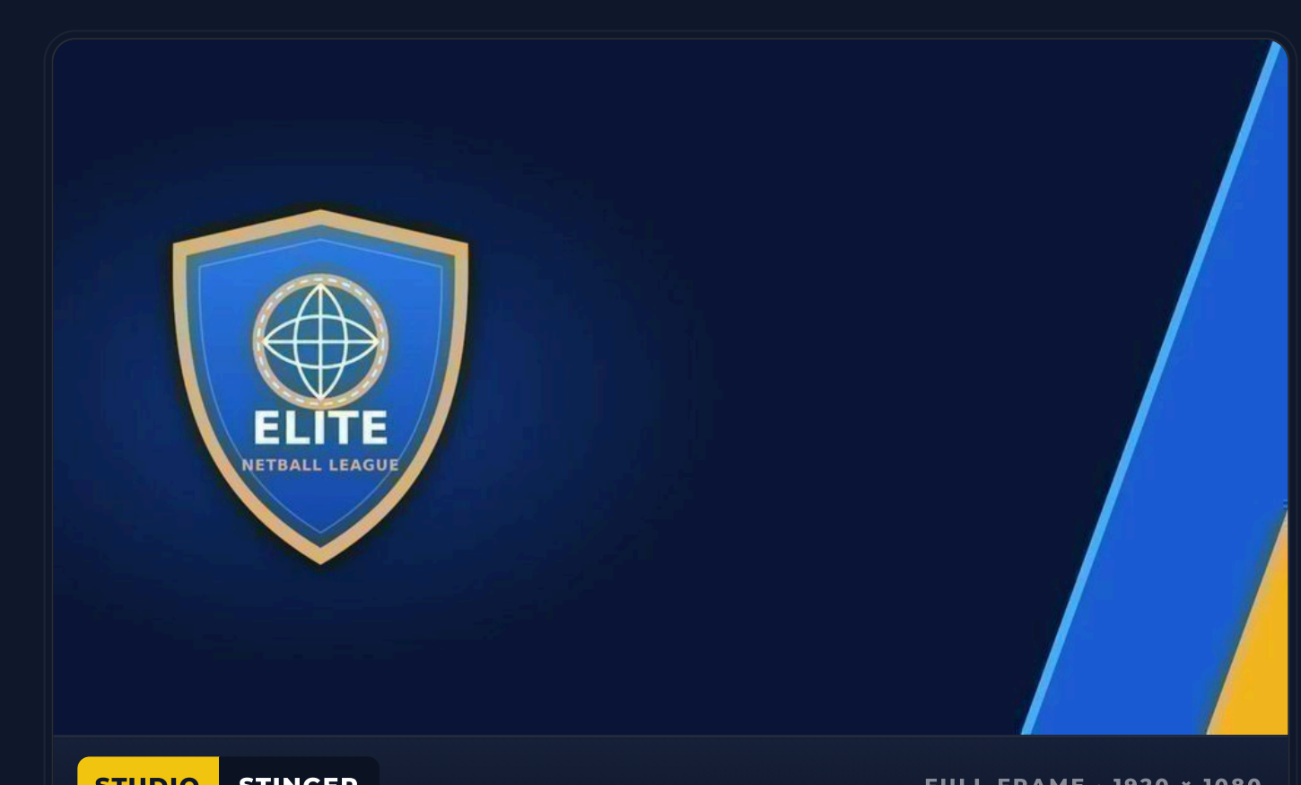


FIG 5.36 Stinger. A 1.5 second transition: bands in the competition's colours sweep across and cover the whole picture with the competition logo for half a second, so a cut can happen underneath, then carry on off the other side. Options: hold, length, logo on or off, light or dark mode logo, whoosh volume, and the music quieted for the next scene or dipped while it plays.

5.6 The breaks

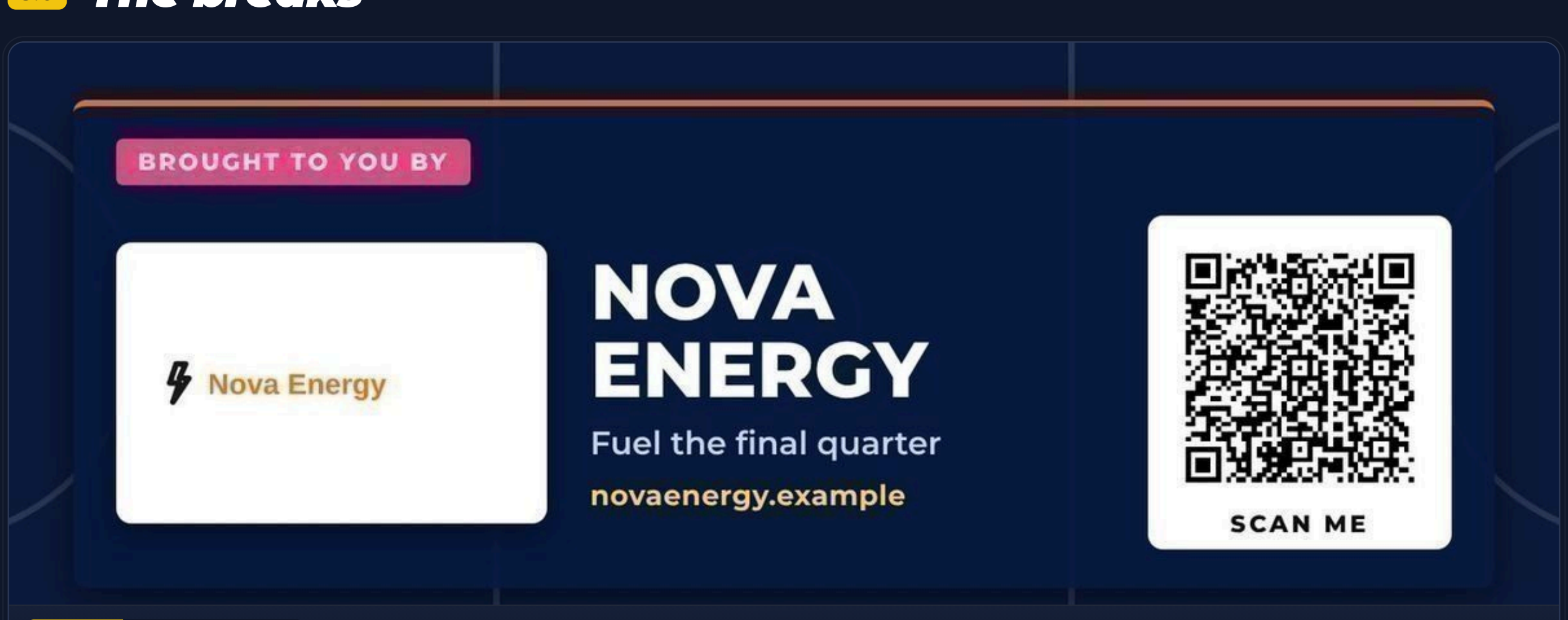


FIG 5.37 Sponsor card. One sponsor's logo, tagline, offer and a QR code to their website, taking turns by tier. Options: one sponsor always, which tiers rotate, animate away.

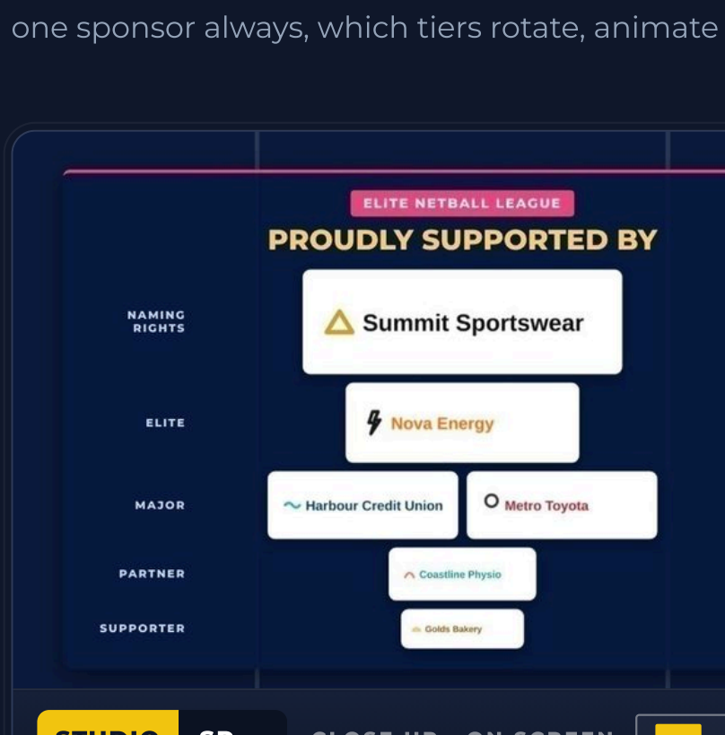


FIG 5.38 Sponsor wall. Every sponsor by tier on one card, for the run-up and the close. Options: which tiers, names under the logos.

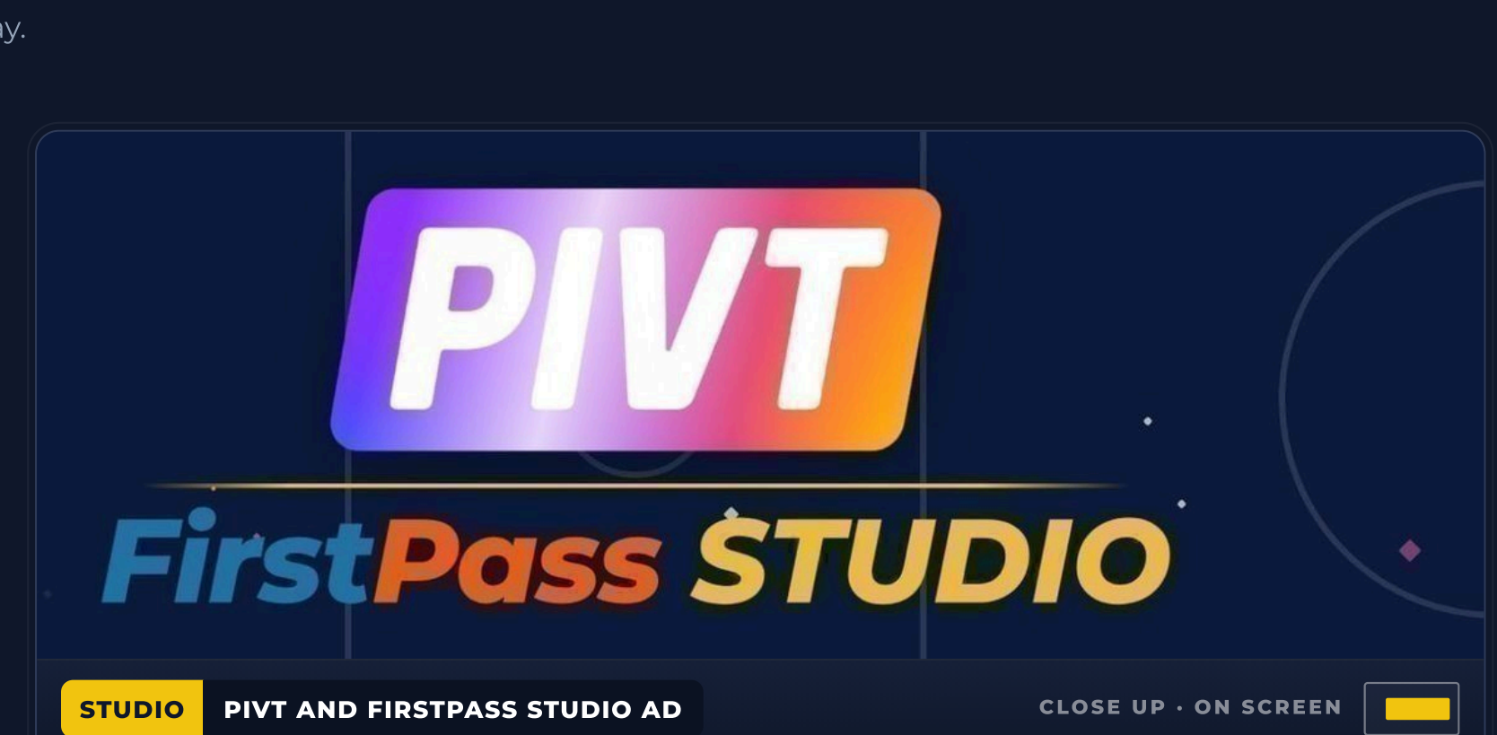


FIG 5.39 PIVT and FirstPass Studio ad. A 10-second house ad for a break with nothing sold. A Sponsor ads step plays it when no sponsor's ad is due, if **House ad** is ticked.

5.7 Checking the outputs

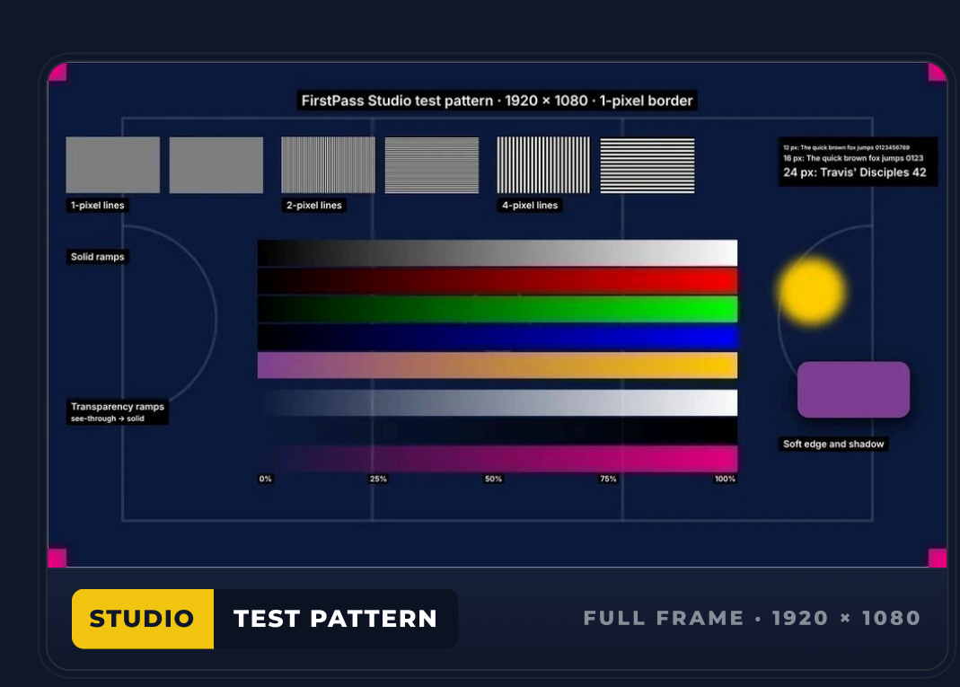


FIG 5.40 Test pattern. Borders, fine lines, ramps and soft shapes to check an output, the switcher's scaling and its key end to end before the stream. Never for air.

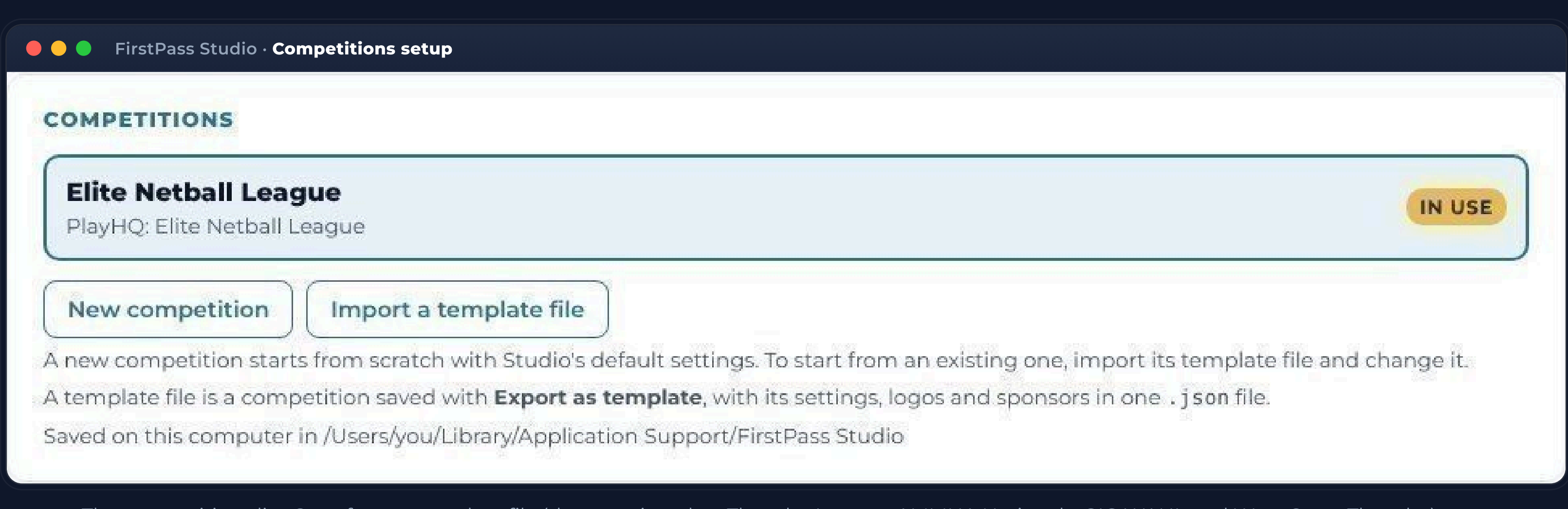
CHAPTER 6 OF 11

Competitions setup

Everything about a competition that doesn't come from PlayHQ lives on one page: its PlayHQ link, the names on the graphics, the game rules, finals, game day, colours, font and logos. Set it up once a season and every graphic follows.

1. PlayHQ
2. Names on the graphics
3. Game rules
4. Finals
5. Game night
6. Colours and font
7. Logos
8. Live viewers
9. Templates, export and more

Studio keeps a setup per competition. The one marked **In use** is what every graphic, Scoreboard control, the Sponsors page and the Stream Deck use. Changes aren't kept until you press **Save** (or **Save and use this competition**), and leaving the page with unsaved changes asks first.


FirstPass Studio · **Competitions setup**

COMPETITIONS

Elite Netball League — IN USE
PlayHQ: Elite Netball League

A new competition starts from scratch with Studio's default settings. To start from an existing one, import its template file and change it.

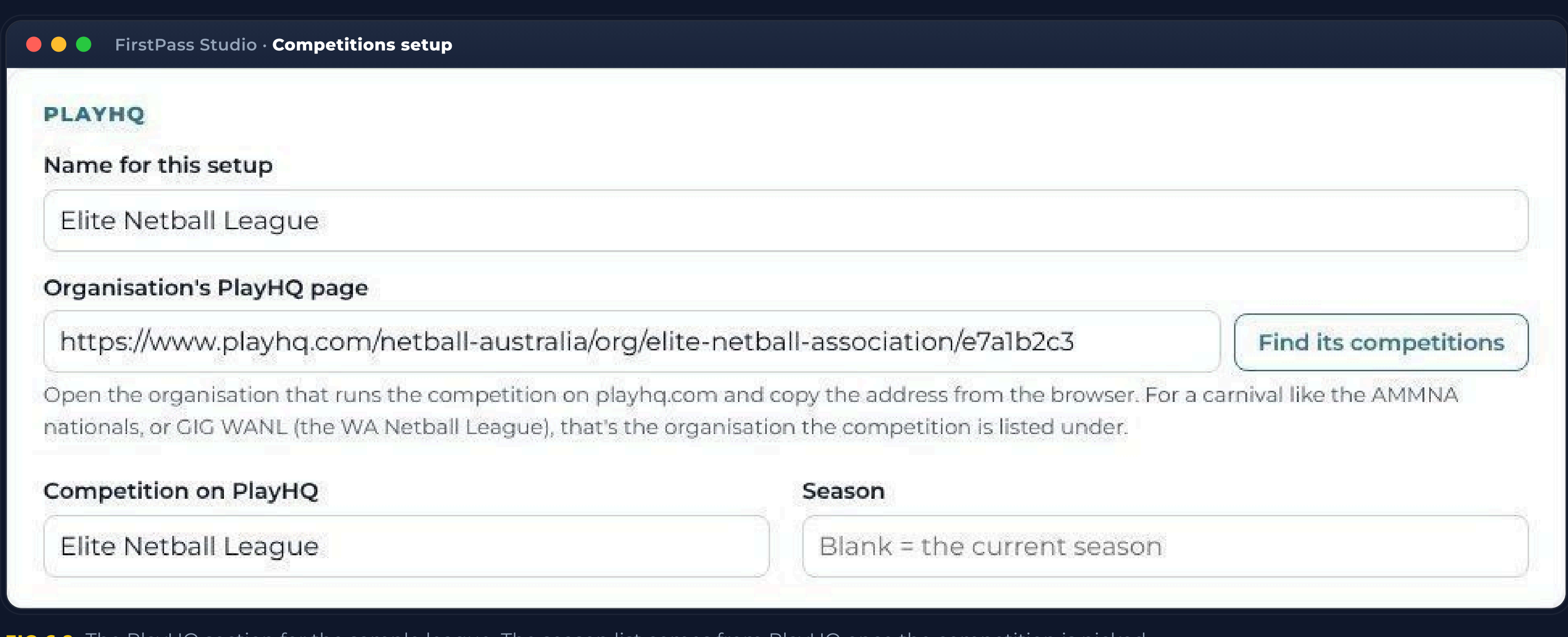
A template file is a competition saved with **Export as template**, with its settings, logos and sponsors in one `.json` file.

Saved on this computer in /Users/you/Library/Application Support/FirstPass Studio

FIG 6.1 The competitions list. Start from a template file (the repository has Thunder League, AMMNA Nationals, GIG WANL and West Coast Thunder) or press New competition for Studio's defaults.

6.1 PlayHQ

Paste the address of the organisation's page on playhq.com and press **Find its competitions**; then pick the competition and the season. For a carnival or a league run under another body, use the organisation the competition is listed under. Nothing else about PlayHQ is needed: no account, no key.


FirstPass Studio · **Competitions setup**

PLAYHQ

Name for this setup: Elite Netball League

Organisation's PlayHQ page: https://www.playhq.com/netball-australia/org/elite-netball-association/e7a1b2c3

Open the organisation that runs the competition on playhq.com and copy the address from the browser. For a carnival like the AMMNA nationals, or GIG WANL (the WA Netball League), that's the organisation the competition is listed under.

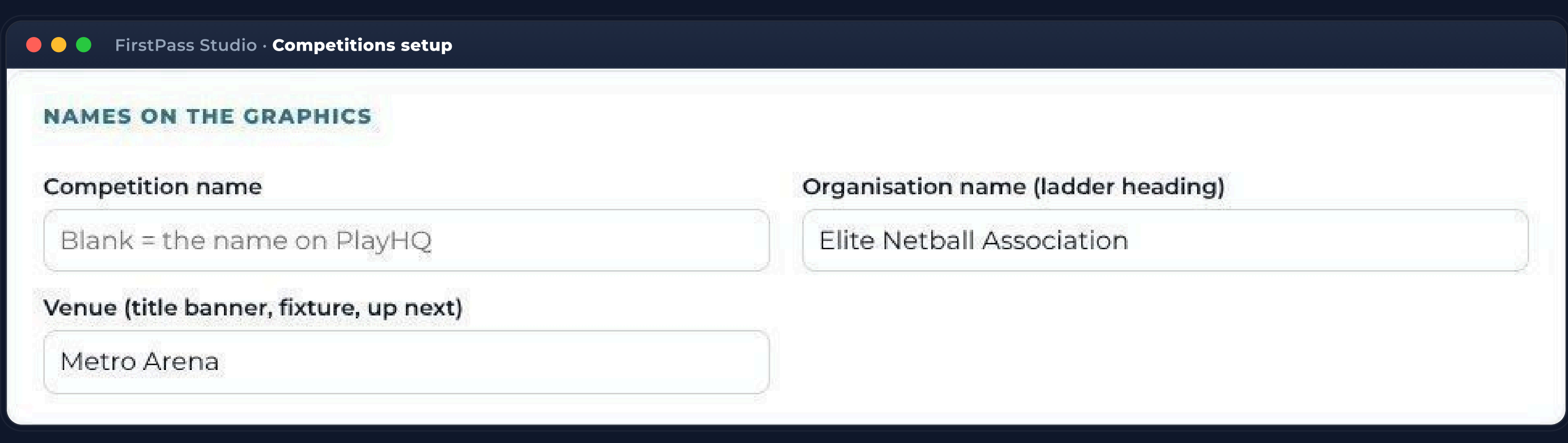
Competition on PlayHQ: Elite Netball League

Season: Blank = the current season

FIG 6.2 The PlayHQ section for the sample league. The season list comes from PlayHQ once the competition is picked.

6.2 Names on the graphics

The competition name on the title banner, the organisation name as the ladder's heading, and the venue for the title, fixture and Up next. PlayHQ's names are often long or formal; these are the ones viewers see.


FirstPass Studio · **Competitions setup**

NAMES ON THE GRAPHICS

Competition name: Blank = the name on PlayHQ

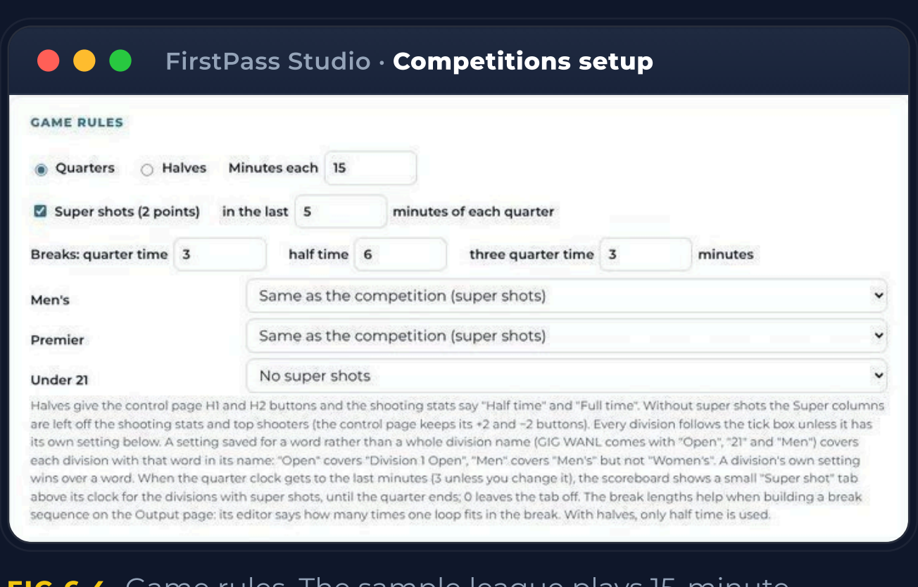
Organisation name (ladder heading): Elite Netball Association

Venue (title banner, fixture, up next): Metro Arena

FIG 6.3 Names on the graphics.

6.3 Game rules

Quarters or halves and their length set the clock, the quarter buttons and the wording of the break cards. **Super shots** turns on the +2 scoring, the celebration, the super shot tab in the last minutes of each quarter and the Super columns on the shooting graphics; a division can have its own setting, by its name or by a word in it ("Open", "Men"). The break lengths let the sequence editor say how many loops of a break sequence fit in the break.


FirstPass Studio · **Competitions setup**

GAME RULES

● Quarters ○ Halves Minutes each 15

☑ Super shots (2 points) in the last 5 minutes of each quarter

Breaks: quarter time 3 half time 6 three quarter time 3 minutes

Men's: Same as the competition (super shots)

Premier: Same as the competition (super shots)

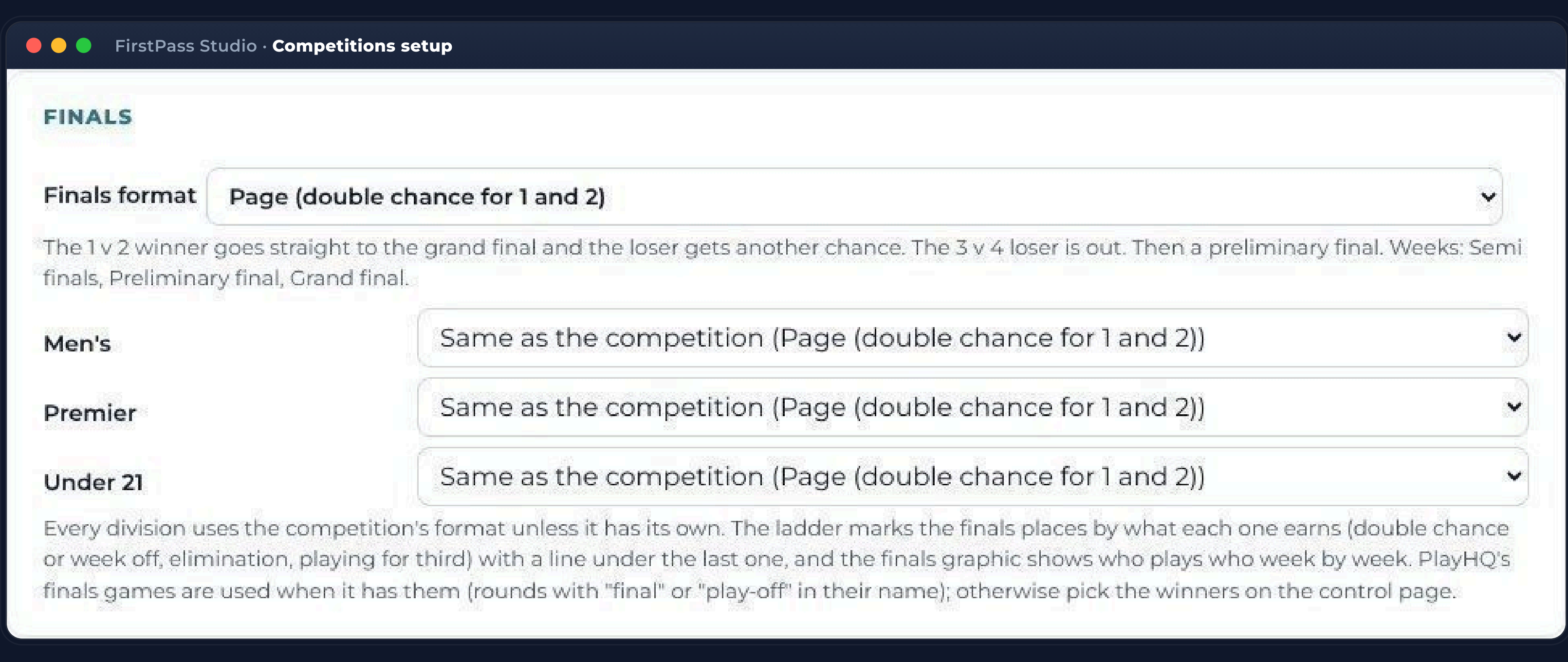
Under 21: No super shots

Halves give the control page H1 and H2 buttons and the shooting stats say "Half time" and "Full time". Without super shots the Super columns are left off the shooting stats and top shooters (the control page keeps its +2 and −2 buttons). Every division follows the tick box unless it has its own setting below. A setting saved for a word rather than a whole division name (GIG WANL comes with "Open", "21" and "Men") covers each division with that word in its name: "Open" covers "Division 1 Open", "Men" covers "Men's" but not "Women's". A division's own setting wins over a word. When the quarter clock gets to the last minutes (3 unless you change it), the scoreboard shows a small "Super shot" tab above its clock for the divisions with super shots, until the quarter ends; 0 leaves the tab off. The break lengths help when building a break sequence on the Output page: its editor says how many times one loop fits in the break. With halves, only half time is used.

FIG 6.4 Game rules. The sample league plays 15-minute quarters with super shots in the last 5 minutes, except in its Under 21 division.

6.4 Finals

Pick the finals format for the competition, or per division: it decides the finals places the ladder marks (double chance or a week off, elimination, playing for third) and the bracket the finals graphic draws week by week. PlayHQ's finals games are used when it has them; otherwise the winners are picked on Scoreboard control.


FirstPass Studio · **Competitions setup**

FINALS

Finals format: Page (double chance for 1 and 2)

The 1 v 2 winner goes straight to the grand final and the loser gets another chance. The 3 v 4 loser is out. Then a preliminary final. Weeks: Semi finals, Preliminary final, Grand final.

Men's: Same as the competition (Page (double chance for 1 and 2))

Premier: Same as the competition (Page (double chance for 1 and 2))

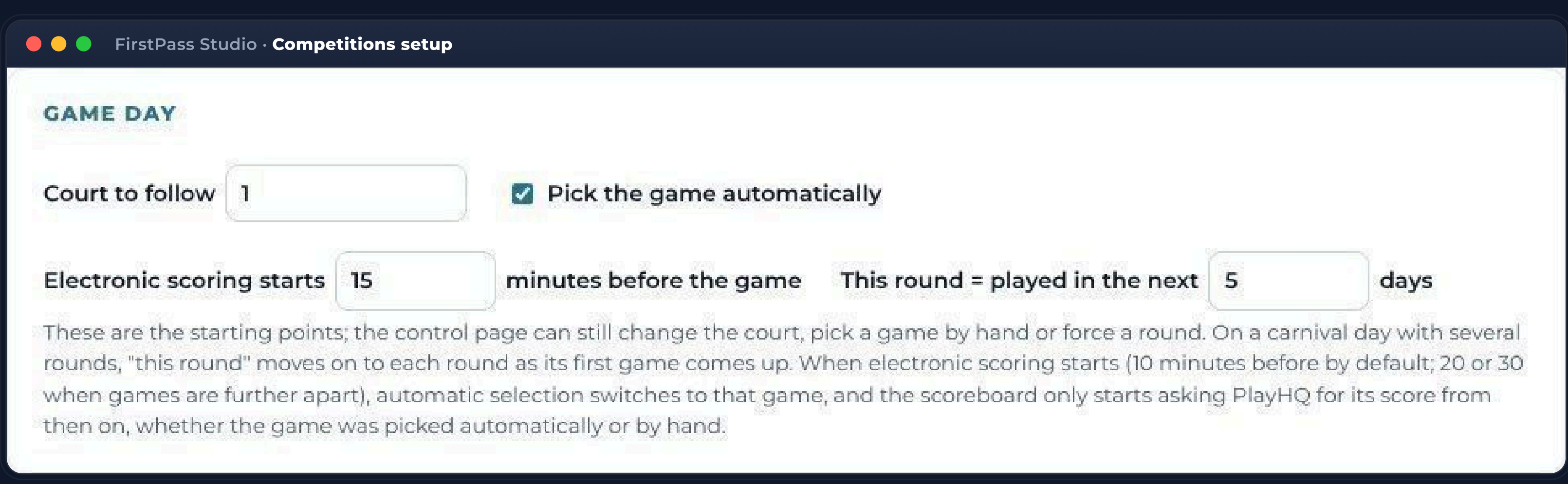
Under 21: Same as the competition (Page (double chance for 1 and 2))

Every division uses the competition's format unless it has its own. The ladder marks the finals places by what each one earns (double chance or week off, elimination, playing for third) with a line under the last one, and the finals graphic shows who plays who week by week. PlayHQ's finals games are used when it has them (rounds with "final" or "play-off" in their name); otherwise pick the winners on the control page.

FIG 6.5 Finals formats, per division where they differ.

6.5 Game night

The court Studio follows, whether it picks the game automatically, when electronic scoring starts before each game (10 minutes by default; 20 or 30 when games are further apart) and how many days ahead tonight's games are found. These are starting points: Scoreboard control can change the court, pick a game by hand or pick another game night. On a carnival day with several rounds, this round moves on as each round's first game comes up. **Running top shooters are called** sets the Socials picture's name: Automatic calls it Tournament top shooters when the fixture runs under 15 days and Season top shooters otherwise.


FirstPass Studio · **Competitions setup**

GAME DAY

Court to follow: 1

☑ Pick the game automatically

Electronic scoring starts 15 minutes before the game

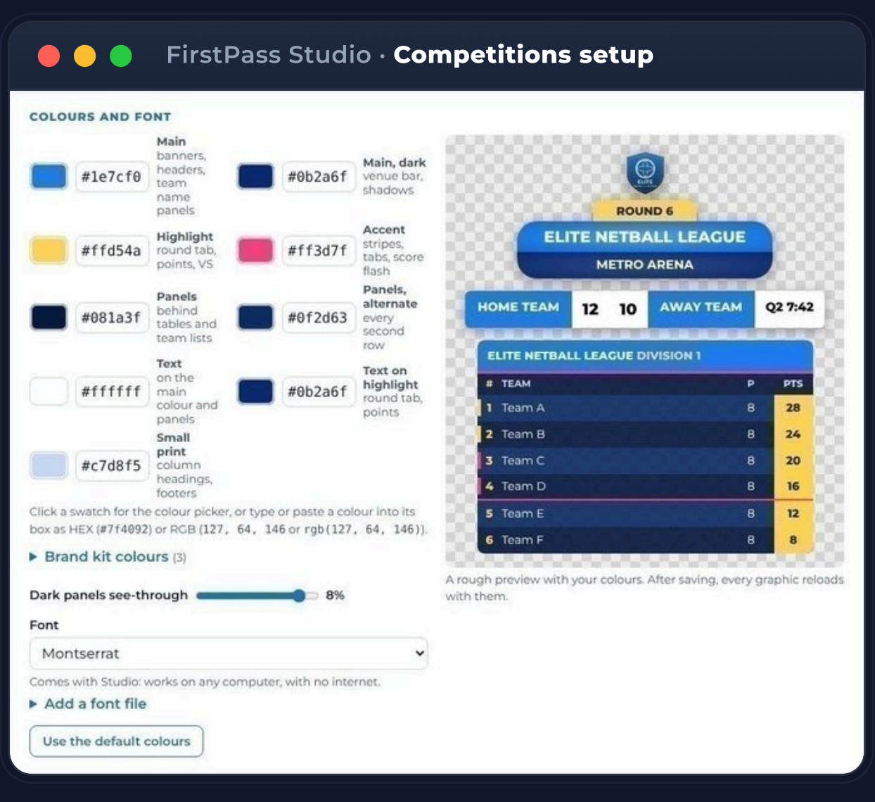
This round = played in the next 5 days

These are the starting points; the control page can still change the court, pick a game by hand or force a round. On a carnival day with several rounds, "this round" moves on to each round as its first game comes up. When electronic scoring starts (10 minutes before by default; 20 or 30 when games are further apart), automatic selection switches to that game, and the scoreboard only starts asking PlayHQ for its score from then on, whether the game was picked automatically or by hand.

FIG 6.6 Game night.

6.6 Colours and font

The competition's colours, see-through dark panels, and the font. Paste a brand kit's text into **Brand kit colours** and every HEX or RGB value in it becomes a swatch in each colour picker. Add a font file (TTF, OTF, WOFF or WOFF2, one per weight or a variable font) and it travels with the competition and its template, so the graphics look the same on any Mac; every font is sized to match the capitals of Montserrat, which the graphics were drawn with. The preview under the pickers shows a rough title and scoreboard in the colours before you save.


FirstPass Studio · **Competitions setup**

COLOURS AND FONT

Colour	Used for	Colour	Used for
#1e7cf0	Main banners, headers, team name panels	#0b2a6f	Main, dark venue bar, shadows
#ffd54a	Highlight round tab, points, VS	#ff3d7f	Accent stripes, tabs, score flash
#081a3f	Panels behind tables and team lists	#0f2d63	Panels, alternate every second row
#ffffff	Text on the main colour and panels	#0b2a6f	Text on highlight round tab, points
#c7d8f5	Small print column headings, footers		

Click a swatch for the colour picker, or type or paste a colour into its box as HEX (#7f4092) or RGB (127, 64, 146 or rgb(127, 64, 146)).

▸ Brand kit colours (3)

Dark panels see-through 8%

Font: Montserrat

Comes with Studio: works on any computer, with no internet.

▸ Add a font file

Use the default colours
ROUND 6
ELITE NETBALL LEAGUE
METRO ARENA
HOME TEAM 12 10 AWAY TEAM Q2 7:42
ELITE NETBALL LEAGUE DIVISION 1

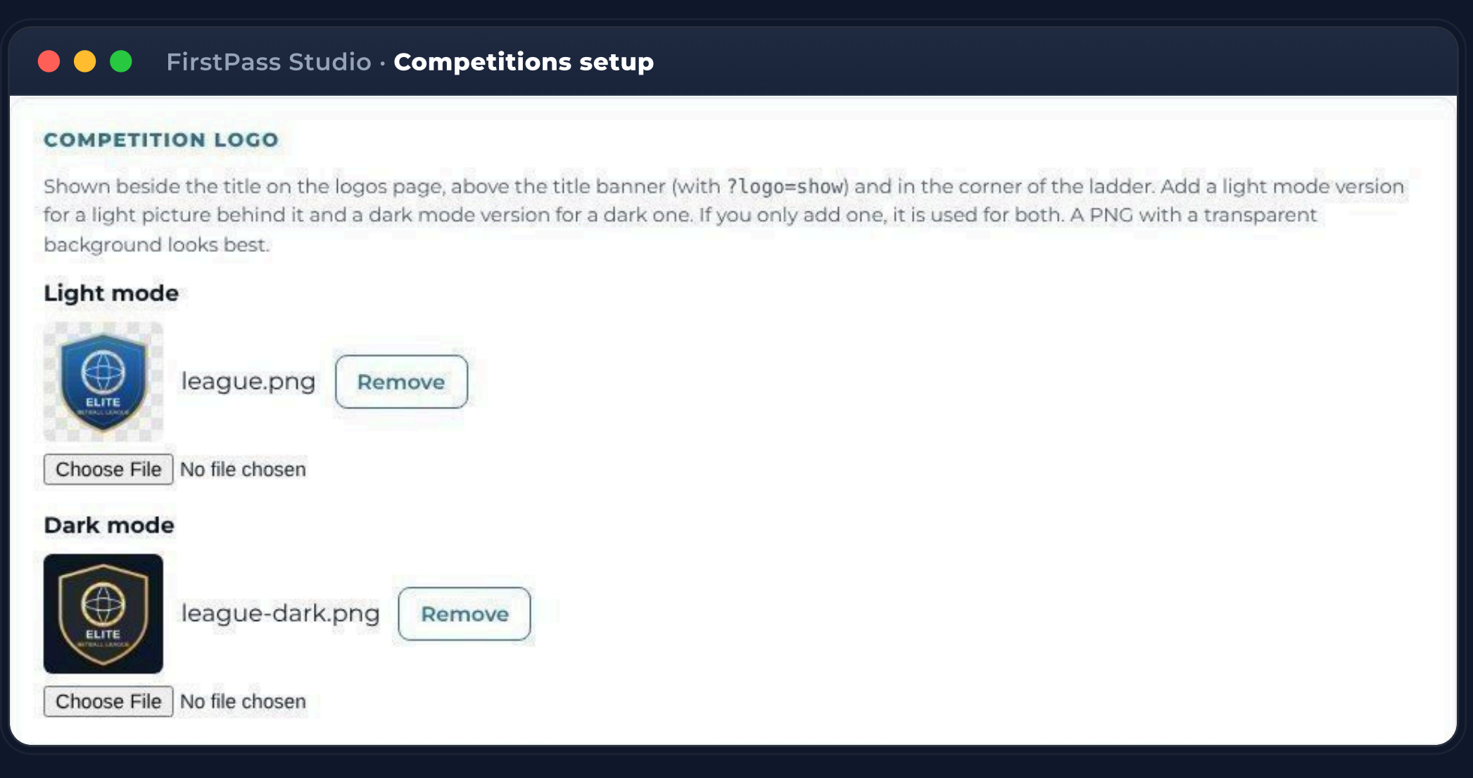
#	TEAM	P	PTS
1	Team A	8	28
2	Team B	8	24
3	Team C	8	20
4	Team D	8	16
5	Team E	8	12
6	Team F	8	8

A rough preview with your colours. After saving, every graphic reloads with them.

FIG 6.7 Colours and font, with the preview.

6.7 Logos

The **competition logo** has a light mode and a dark mode version (one is used for both if that's all you add); it shows beside the title in the branding logos, above the title with its option on, in the corner of the ladder and in the logo animations. **Team logos** are matched to PlayHQ team names by file name, so drop in a folder of files named after the teams. Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, worked out from the team name without its division ("Falcons 21U" is Falcons). Fix a wrong match in the **Club** box. A club colour goes behind its teams' names on the scoreboard and behind their logo on The Clash; without one the competition's colours are used.


FirstPass Studio · **Competitions setup**

COMPETITION LOGO

Shown beside the title on the logos page, above the title banner (with `?logo=show`) and in the corner of the ladder. Add a light mode version for a light picture behind it and a dark mode version for a dark one. If you only add one, it is used for both. A PNG with a transparent background looks best.

Light mode

league.png — Remove

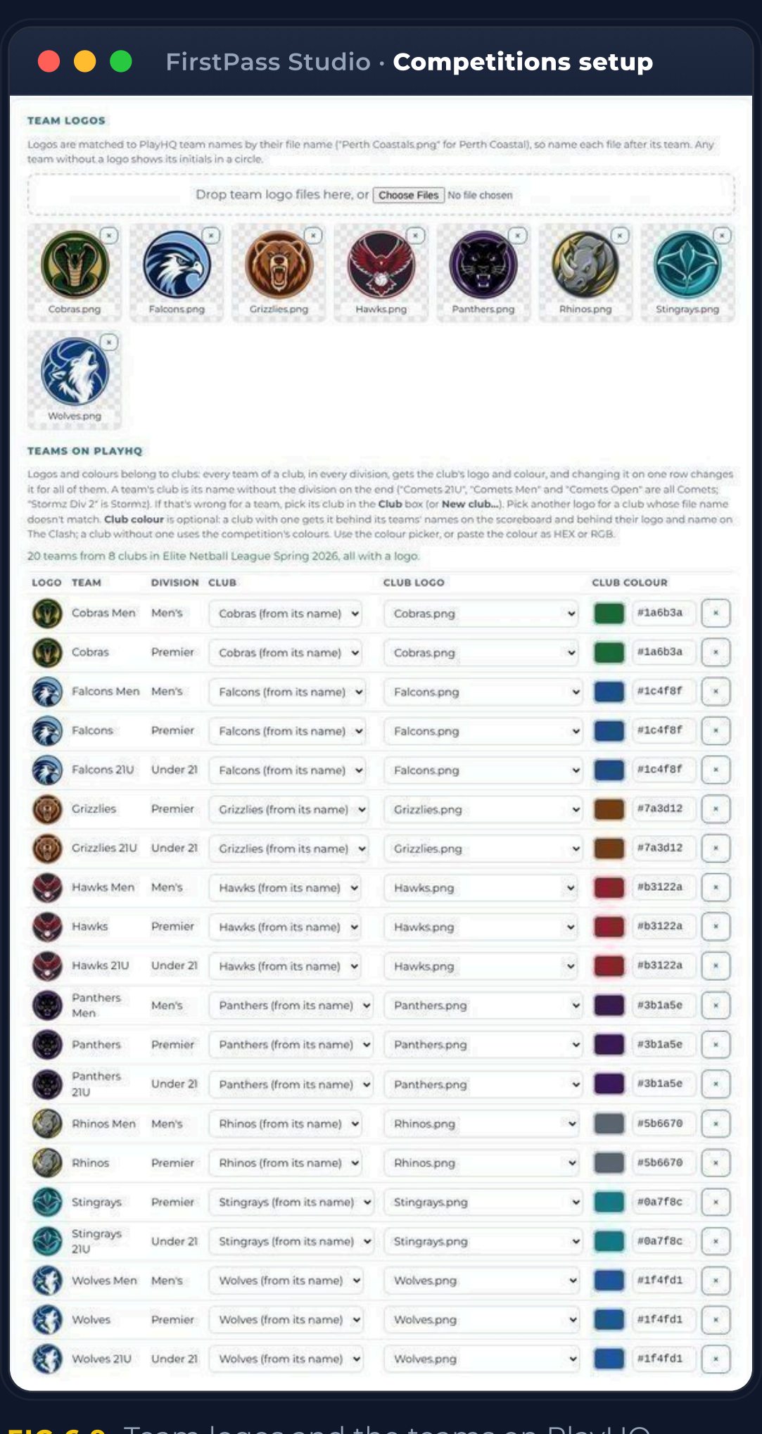
Choose File No file chosen

Dark mode

league-dark.png — Remove

Choose File No file chosen

FIG 6.8 Competition logo, light and dark mode.


FirstPass Studio · **Competitions setup**

TEAM LOGOS

Logos are matched to PlayHQ team names by their file name ("Perth Coastals.png" for Perth Coastal), so name each file after its team. Any team without a logo shows its initials in a circle.

Drop team logo files here, or Choose Files No file chosen

TEAMS ON PLAYHQ

Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, and changing it on one row changes it for all of them. A team's club is its name without the division on the end ("Comets 21U", "Comets Men" and "Comets Open" are all Comets; "Stormz Div 2" is Stormz). If that's wrong for a team, pick its club in the **Club** box (or **New club…**). Pick another logo for a club whose file name doesn't match. **Club colour** is optional: a club with one gets it behind its teams' names on the scoreboard and behind their logo and name on The Clash; a club without one uses the competition's colours. Use the colour picker, or paste the colour as HEX or RGB.

20 teams from 8 clubs in Elite Netball League Spring 2026, all with a logo.

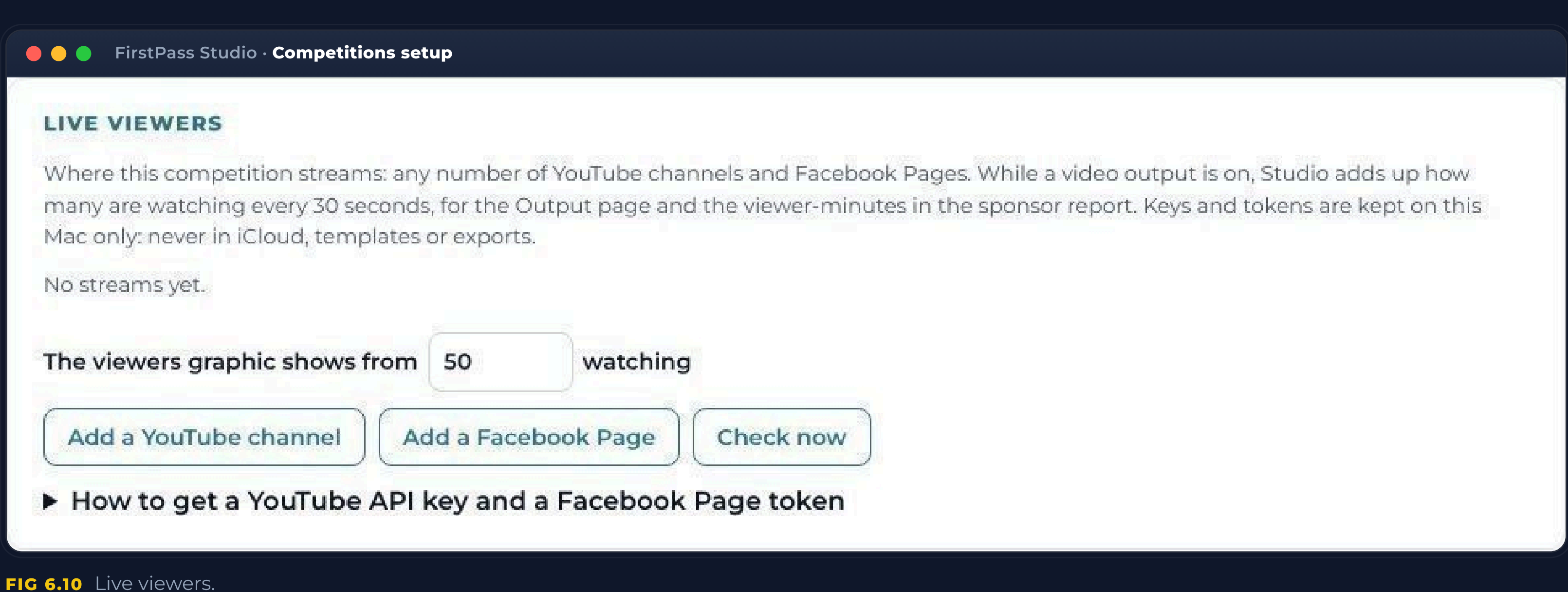
LOGO	TEAM	DIVISION	CLUB	CLUB LOGO	CLUB COLOUR
	Cobras Men	Men's	Cobras (from its name)	Cobras.png	#1a6b3a
	Cobras	Premier	Cobras (from its name)	Cobras.png	#1a6b3a
	Falcons Men	Men's	Falcons (from its name)	Falcons.png	#1c4f8f
	Falcons	Premier	Falcons (from its name)	Falcons.png	#1c4f8f
	Falcons 21U	Under 21	Falcons (from its name)	Falcons.png	#1c4f8f
	Grizzlies	Premier	Grizzlies (from its name)	Grizzlies.png	#7a3d12
	Grizzlies 21U	Under 21	Grizzlies (from its name)	Grizzlies.png	#7a3d12
	Hawks Men	Men's	Hawks (from its name)	Hawks.png	#b3122a
	Hawks	Premier	Hawks (from its name)	Hawks.png	#b3122a
	Hawks 21U	Under 21	Hawks (from its name)	Hawks.png	#b3122a
	Panthers Men	Men's	Panthers (from its name)	Panthers.png	#3b1a5e
	Panthers	Premier	Panthers (from its name)	Panthers.png	#3b1a5e
	Panthers 21U	Under 21	Panthers (from its name)	Panthers.png	#3b1a5e
	Rhinos Men	Men's	Rhinos (from its name)	Rhinos.png	#5b6670
	Rhinos	Premier	Rhinos (from its name)	Rhinos.png	#5b6670
	Stingrays	Premier	Stingrays (from its name)	Stingrays.png	#0a7f8c
	Stingrays 21U	Under 21	Stingrays (from its name)	Stingrays.png	#0a7f8c
	Wolves Men	Men's	Wolves (from its name)	Wolves.png	#1f4fd1
	Wolves	Premier	Wolves (from its name)	Wolves.png	#1f4fd1
	Wolves 21U	Under 21	Wolves (from its name)	Wolves.png	#1f4fd1

FIG 6.9 Team logos and the teams on PlayHQ, grouped by club with a club colour each.

> **Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour.**
>
> CHAPTER 6 · COMPETITIONS SETUP

6.8 Live viewers

Add the YouTube channels and Facebook Pages the competition streams to. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page, the viewers graphic (which shows only from the minimum set here) and the viewer-minutes in the sponsor report. The page explains how to get a YouTube API key and a Facebook Page token; both are kept on this Mac only, never in iCloud, templates or exports.


FirstPass Studio · **Competitions setup**

LIVE VIEWERS

Where this competition streams: any number of YouTube channels and Facebook Pages. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page and the viewer-minutes in the sponsor report. Keys and tokens are kept on this Mac only: never in iCloud, templates or exports.

No streams yet.

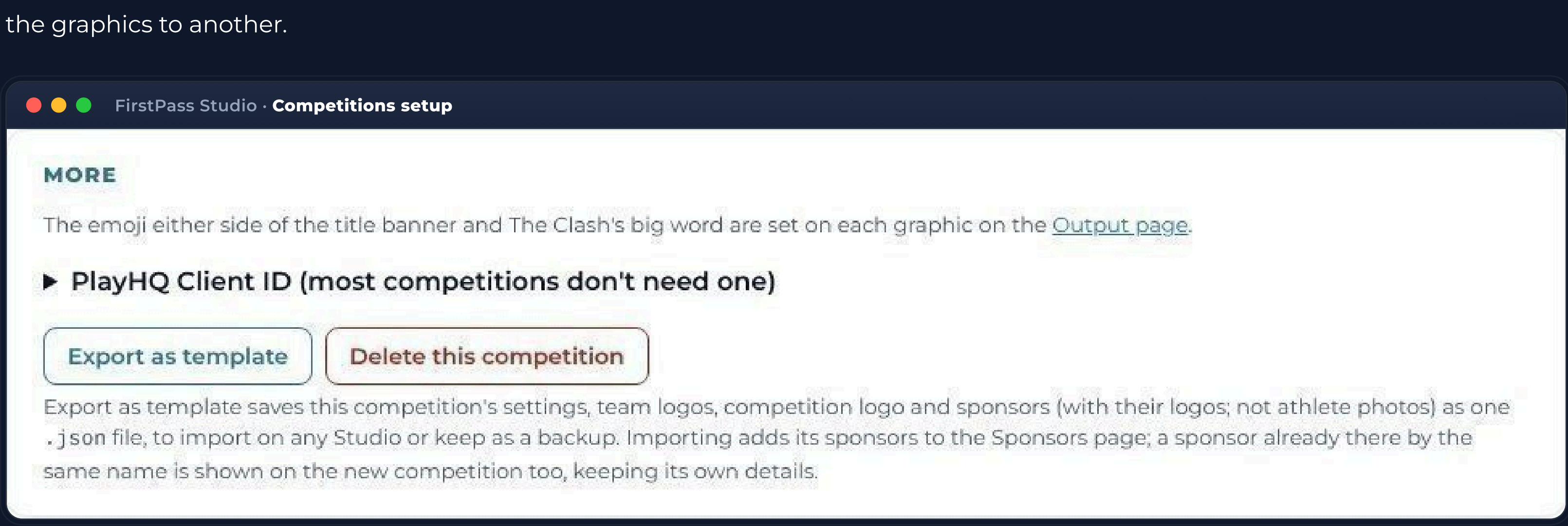
The viewers graphic shows from 50 watching

▸ How to get a YouTube API key and a Facebook Page token

FIG 6.10 Live viewers.

6.9 Templates, export and more

Export as template saves the competition's settings, team logos, competition logo, font and sponsors (with their logos) as one .json file to import on any Studio or keep as a backup; **Import a template file** adds it. A PlayHQ Client ID is only needed if PlayHQ's game page has no line-ups and the association has given you one. **Delete this competition** removes it; deleting the one in use switches the graphics to another.


FirstPass Studio · **Competitions setup**

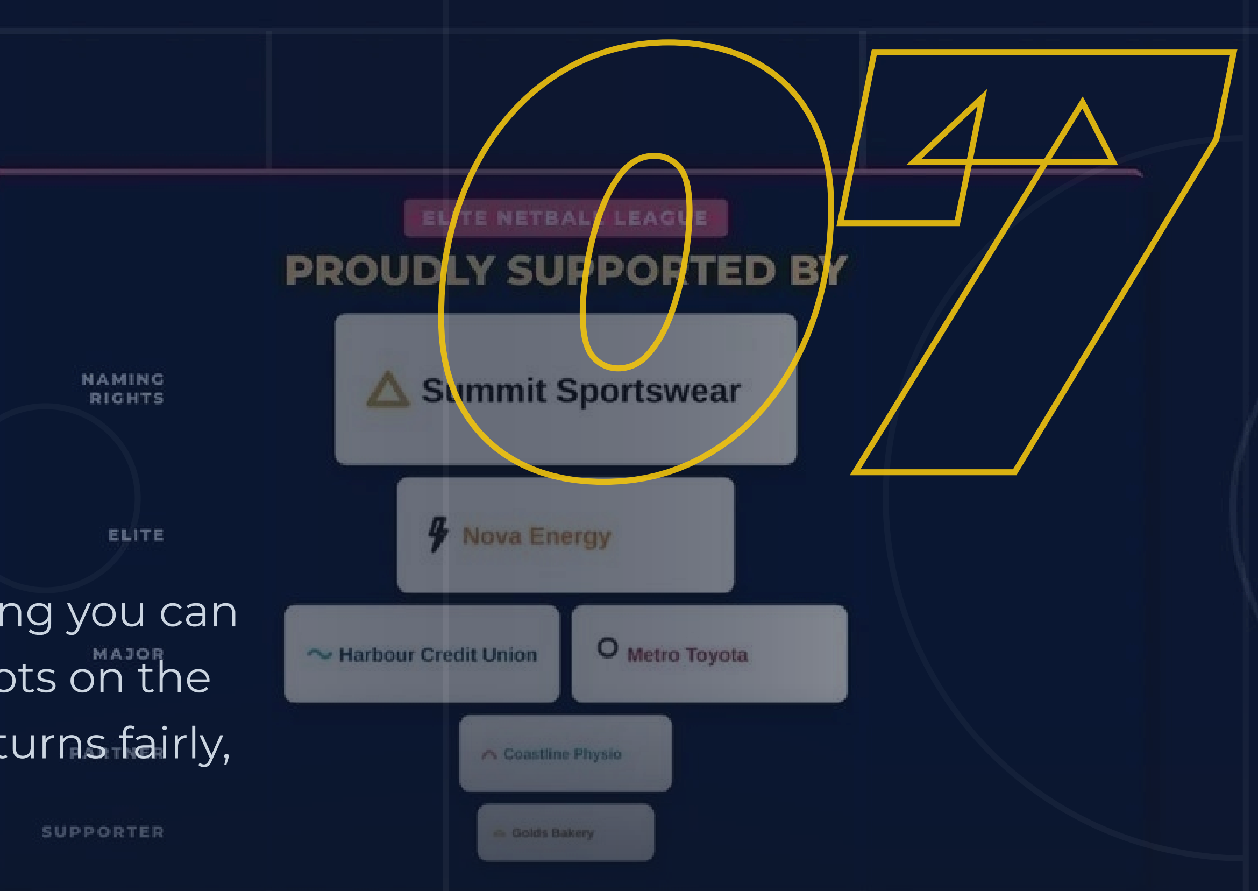
MORE

The emoji either side of the title banner and The Clash's big word are set on each graphic on the Output page.

▸ PlayHQ Client ID (most competitions don't need one)

Export as template saves this competition's settings, team logos, competition logo and sponsors (with their logos; not athlete photos) as one `.json` file, to import on any Studio or keep as a backup. Importing adds its sponsors to the Sponsors page; a sponsor already there by the same name is shown on the new competition too, keeping its own details.

FIG 6.11 The More section: Client ID, export and delete.

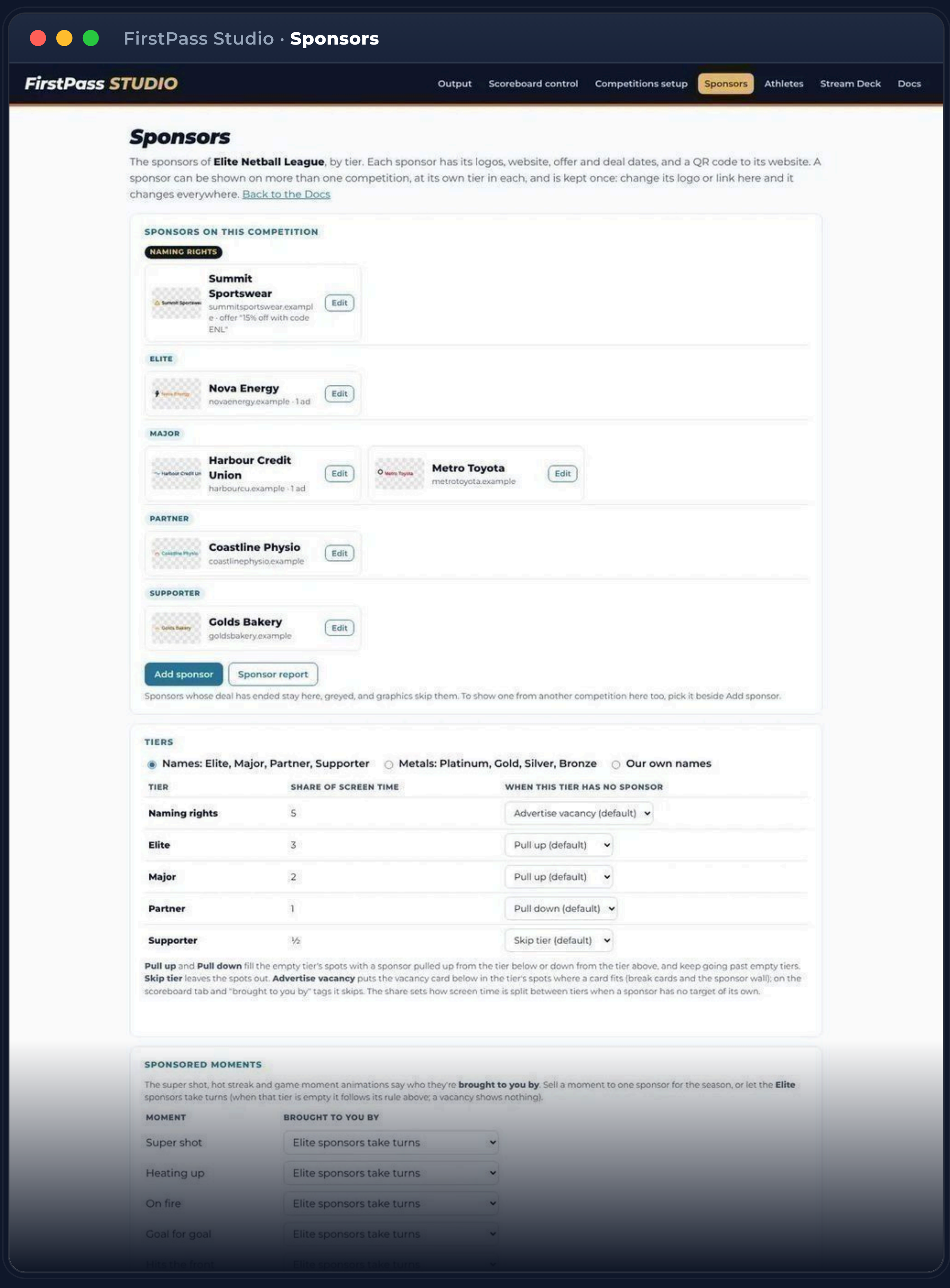


CHAPTER 7 OF 11

Sponsors

The Sponsors tab turns sponsorship into something you can sell, run and prove: a tiered library of sponsors, spots on the graphics that fill themselves, ad breaks that take turns fairly, and a report of what each sponsor got.

1 Tiers
2 A sponsor
3 Spots and moments
4 Proof of play



FirstPass Studio · **Sponsors**

Sponsors

The sponsors of **Elite Netball League**, by tier. Each sponsor has its logos, website, offer and deal dates, and a QR code to its website. A sponsor can be shown on more than one competition, at its own tier in each, and is kept once: change its logo or link here and it changes everywhere. Back to the Docs

SPONSORS ON THIS COMPETITION

NAMING RIGHTS
Summit Sportswear summitsportswear.example · offer "15% off with code ENL" Edit

ELITE
Nova Energy novaenergy.example · 1 ad Edit

MAJOR
Harbour Credit Union harbourcu.example · 1 ad Edit
Metro Toyota metrotoyota.example Edit

PARTNER
Coastline Physio coastlinephysio.example Edit

SUPPORTER
Golds Bakery goldsbakery.example Edit

Add sponsor Sponsor report

Sponsors whose deal has ended stay here, greyed, and graphics skip them. To show one from another competition here too, pick it beside Add sponsor.

TIERS

● Names: Elite, Major, Partner, Supporter ○ Metals: Platinum, Gold, Silver, Bronze ○ Our own names

TIER	SHARE OF SCREEN TIME	WHEN THIS TIER HAS NO SPONSOR
Naming rights	5	Advertise vacancy (default)
Elite	3	Pull up (default)
Major	2	Pull up (default)
Partner	1	Pull down (default)
Supporter	½	Skip tier (default)

Pull up and **Pull down** fill the empty tier's spots with a sponsor pulled up from the tier below or down from the tier above, and keep going past empty tiers. **Skip tier** leaves the spots out. **Advertise vacancy** puts the vacancy card below in the tier's spots where a card fits (break cards and the sponsor wall); on the scoreboard tab and "brought to you by" tags it skips. The share sets how screen time is split between tiers when a sponsor has no target of its own.

SPONSORED MOMENTS

The super shot, hot streak and game moment animations say who they're **brought to you by**. Sell a moment to one sponsor for the season, or let the **Elite** sponsors take turns (when that tier is empty it follows its rule above; a vacancy shows nothing).

MOMENT	BROUGHT TO YOU BY
Super shot	Elite sponsors take turns
Heating up	Elite sponsors take turns
On fire	Elite sponsors take turns
Goal for goal	Elite sponsors take turns
Hits the front	Elite sponsors take turns

FIG 7.1

The Sponsors page for the sample league: sponsors by tier with tonight's exposure meter, the tier rules, sponsored moments and the vacancy card.

7.1 Tiers

Five fixed ranks, named per competition: the **Names** preset (Naming rights, Elite, Major, Partner, Supporter), the **Metals** preset (Naming rights, Platinum, Gold, Silver, Bronze) or your own. Each tier's **share of screen time** decides how turns are split when a sponsor has no target of its own. Each tier gets different places:

TIER	WHERE IT SHOWS
Naming rights	A tab on the scoreboard, under the title banner, first on the sponsor wall, and everything below.
Elite	The sponsored moments (super shot, streaks), ads in the breaks, the break card, and everything below.
Major	Ads and break cards, "presented by" footers on the stats cards, countdown and top shooters.
Partner	A slot in the score ticker and the logo spots.
Supporter	The sponsor wall and the end-of-night strip.

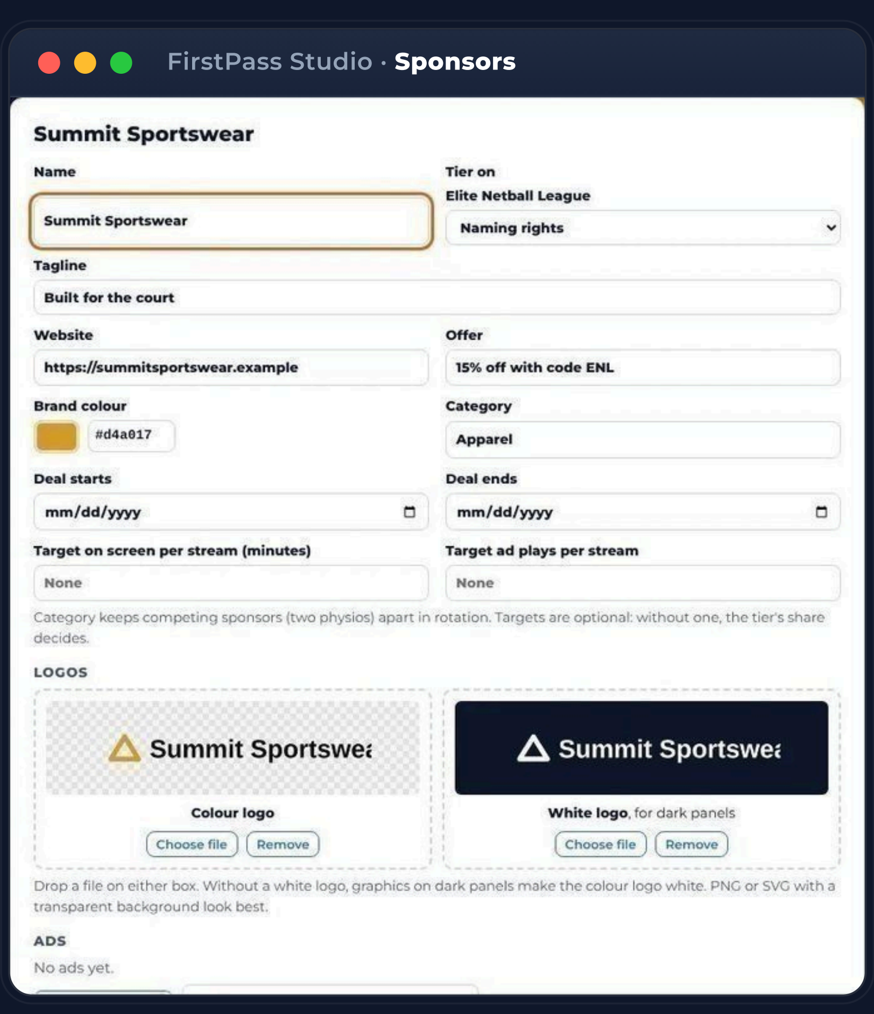
When this tier has no sponsor says what a spot does when its tier is empty: **Pull up** fills it from the tier below, **Pull down** from the tier above, **Skip tier** leaves it out, and **Advertise vacancy** shows the competition's vacancy card wherever a card fits. The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.

> **The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.**
>
> CHAPTER 7 · SPONSORS

7.2 A sponsor

A sponsor has a name, tagline, website, offer, brand colour, category, deal dates, and optional targets per stream (minutes on screen, ad plays). Category keeps competing sponsors apart in rotation. Add a colour logo and a white one for dark panels (without a white one, the colour logo is made white). A sponsor is kept once and can be **shown on** several competitions at its own tier in each; sponsors whose deal has ended stay greyed and the graphics skip them.

Ads are videos or stills (up to 10 per sponsor) in the media folder. A **Sponsor ads** step in a sequence plays the next ads due, taking turns fairly by tier, each video to its end with sound and each still for its seconds; tick **House ad** and the PIVT and FirstPass Studio ad plays when nothing is due. The **QR code** points to the sponsor's website with tracking tags, so visits from the stream show in their analytics; save it as a PNG for the sponsor's own use.



FirstPass Studio · **Sponsors**

Summit Sportswear

Name: Summit Sportswear
Tier on Elite Netball League: Naming rights
Tagline: Built for the court
Website: https://summitsportswear.example
Offer: 15% off with code ENL
Brand colour: #d4a017
Category: Apparel
Deal starts: mm/dd/yyyy
Deal ends: mm/dd/yyyy
Target on screen per stream (minutes): None
Target ad plays per stream: None

Category keeps competing sponsors (two physios) apart in rotation. Targets are optional: without one, the tier's share decides.

LOGOS

Colour logo Choose file Remove
White logo, for dark panels Choose file Remove

Drop a file on either box. Without a white logo, graphics on dark panels make the colour logo white. PNG or SVG with a transparent background look best.

ADS

No ads yet.

FIG 7.2 The sponsor editor: details, logos, ads, the QR code with tracking tags, and the competitions it is shown on.

7.3 Spots and moments

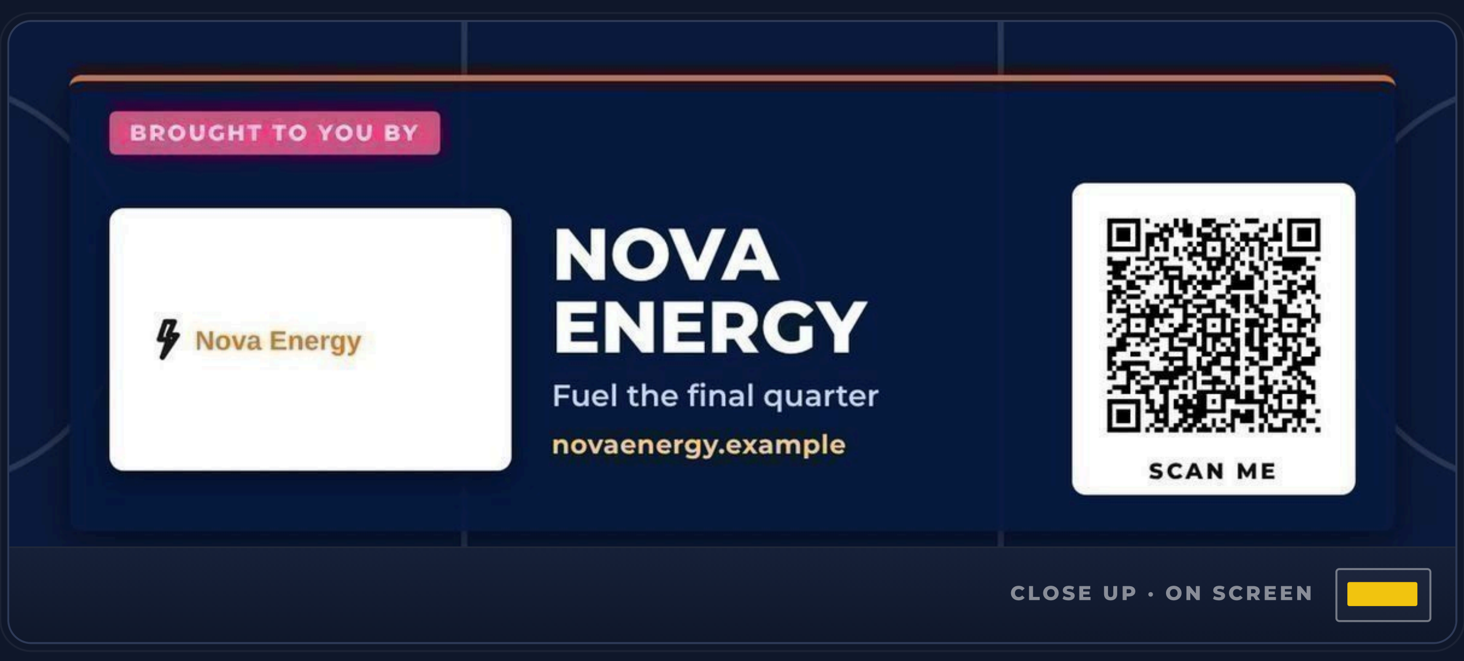


FIG 7.3 The sponsor card in a break, with its QR code.

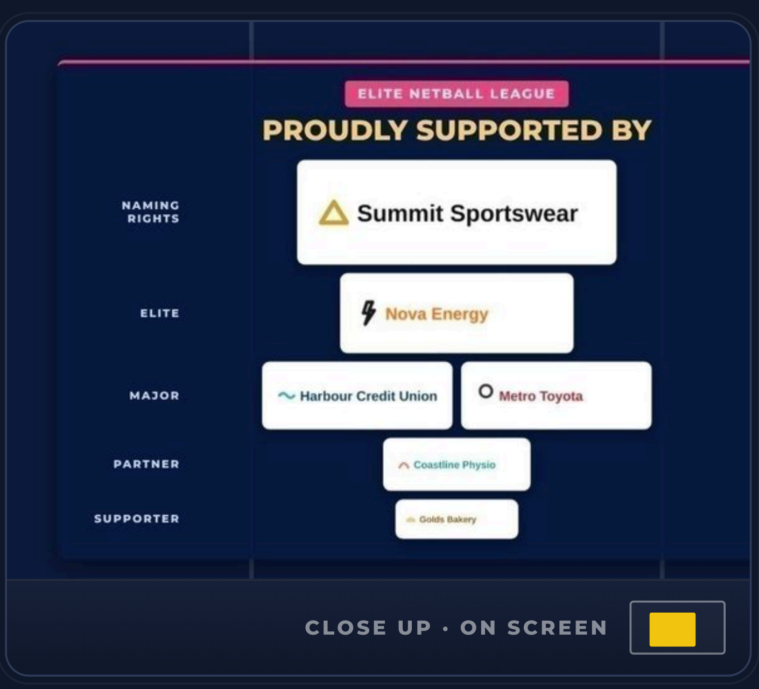
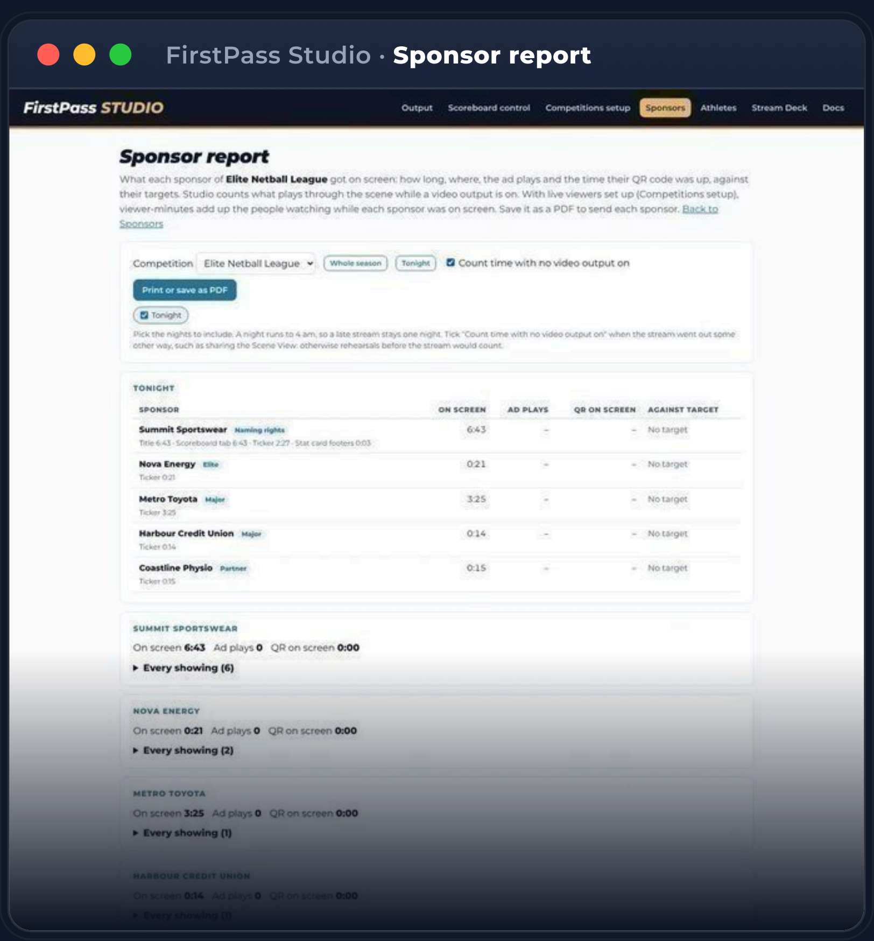


FIG 7.4 The sponsor wall before the game, sized by tier.

Every spot picks its sponsor from the Sponsors page by default. Each graphic's **Sponsor** option can pin one sponsor or turn the spot off. **Sponsored moments** sells the super shot, each hot streak and each game moment to one sponsor for the season, whose name appears in a "brought to you by" tag on the animation; by default the Elite sponsors take turns.

7.4 Proof of play



FirstPass Studio · **Sponsor report**

Sponsor report

What each sponsor of **Elite Netball League** got on screen: how long, where, the ad plays and the time their QR code was up, against their targets. Studio counts what plays through the scene while a video output is on. With live viewers set up (Competitions setup), viewer-minutes add up the people watching while each sponsor was on screen. Save it as a PDF to send each sponsor. Back to Sponsors

Competition: Elite Netball League · Whole season · Tonight · ☑ Count time with no video output on

Print or save as PDF

☑ Tonight

Pick the nights to include. A night runs to 4 am, so a late stream stays one night. Tick "Count time with no video output on" when the stream went out some other way, such as sharing the Scene View: otherwise rehearsals before the stream would count.

TONIGHT

SPONSOR	ON SCREEN	AD PLAYS	QR ON SCREEN	AGAINST TARGET
Summit Sportswear Naming rights Title 6:43 · Scoreboard tab 6:43 · Ticker 2:27 · Stat card footers 0:03	6:43	-	-	No target
Nova Energy Elite Ticker 0:21	0:21	-	-	No target
Metro Toyota Major Ticker 3:25	3:25	-	-	No target
Harbour Credit Union Major Ticker 0:14	0:14	-	-	No target
Coastline Physio Partner Ticker 0:15	0:15	-	-	No target

SUMMIT SPORTSWEAR
On screen **6:43** Ad plays **0** QR on screen **0:00**
▸ Every showing (6)

NOVA ENERGY
On screen **0:21** Ad plays **0** QR on screen **0:00**
▸ Every showing (2)

METRO TOYOTA
On screen **3:25** Ad plays **0** QR on screen **0:00**
▸ Every showing (1)

HARBOUR CREDIT UNION
On screen **0:14** Ad plays **0** QR on screen **0:00**
▸ Every showing (1)

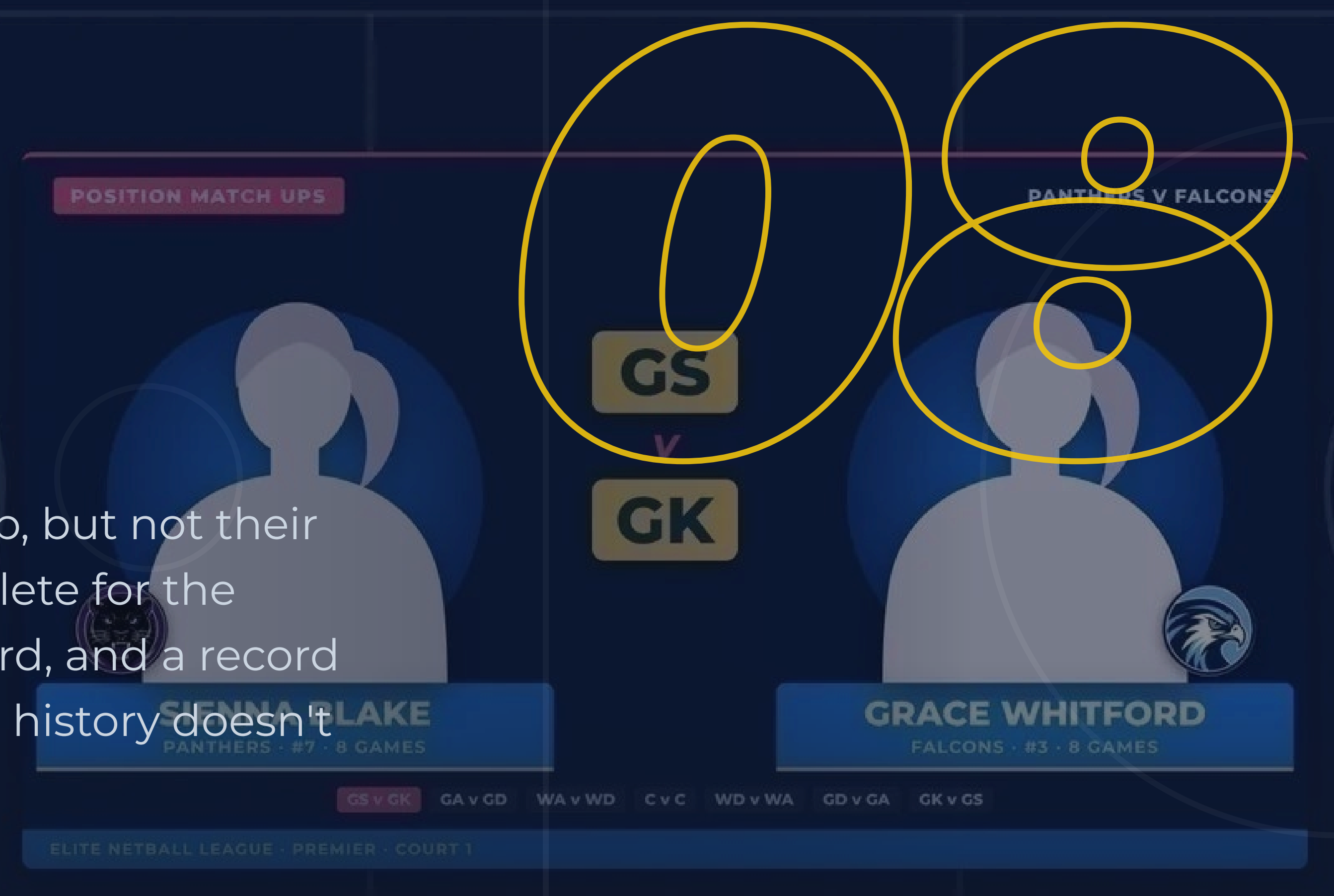
FIG 7.5 The sponsor report for the sample night.

While a video output is on, Studio logs every second each sponsor is on screen and every ad played, matched to the scene's sources, and reads the live viewer count every 30 seconds. **Sponsor report** turns that into a page per night (or a run of nights) with minutes on screen, ad plays, viewer-minutes, peak and average viewers per sponsor, ready to print or save as a PDF for the sponsor. A tick box counts time with no video output on, for a rehearsal. With more than one output, only Output 1 and outputs with **Counts for sponsors** ticked are counted, and each counts as on air only while it's on: an output set to **Web links only** is off.

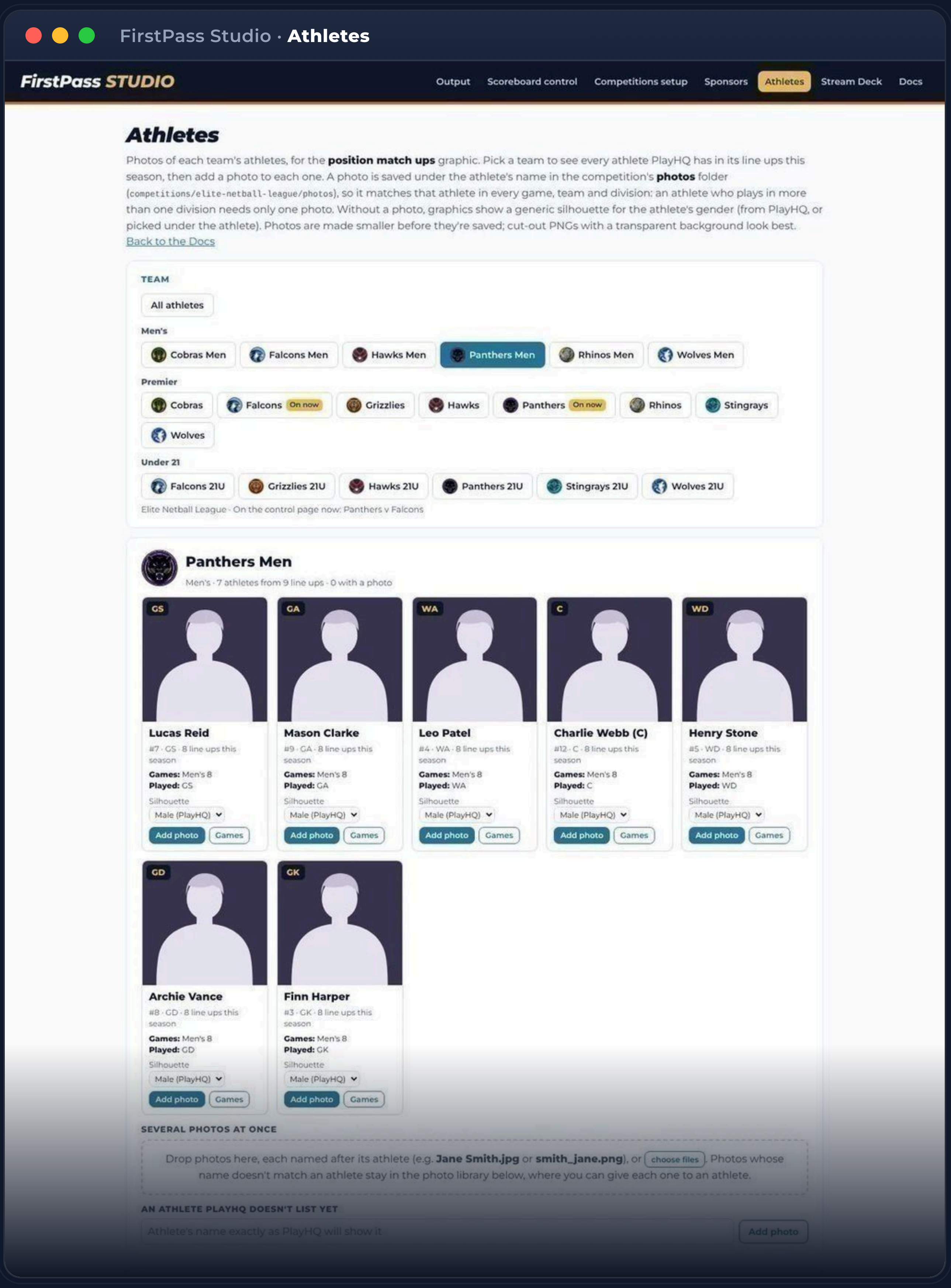
CHAPTER 8 OF 11

Athletes

PlayHQ gives Studio every athlete in a line-up, but not their faces. The Athletes tab holds a photo per athlete for the position match ups and the person lower third, and a record of positions and games played that PlayHQ's history doesn't cover.



① Photos
② Positions and games played



Athletes

Photos of each team's athletes, for the **position match ups** graphic. Pick a team to see every athlete PlayHQ has in its line ups this season, then add a photo to each one. A photo is saved under the athlete's name in the competition's **photos** folder (`competitions/elite-netball-league/photos`), so it matches that athlete in every game, team and division: an athlete who plays in more than one division needs only one photo. Without a photo, graphics show a generic silhouette for the athlete's gender (from PlayHQ, or picked under the athlete). Photos are made smaller before they're saved; cut-out PNGs with a transparent background look best.
Back to the Docs

TEAM

All athletes

Men's: Cobras Men · Falcons Men · Hawks Men · Panthers Men · Rhinos Men · Wolves Men

Premier: Cobras · Falcons On now · Grizzlies · Hawks · Panthers On now · Rhinos · Stingrays · Wolves

Under 21: Falcons 21U · Grizzlies 21U · Hawks 21U · Panthers 21U · Stingrays 21U · Wolves 21U

Elite Netball League · On the control page now: Panthers v Falcons

Panthers Men

Men's · 7 athletes from 9 line ups · 0 with a photo

GS — **Lucas Reid** — #7 · GS · 8 line ups this season — **Games:** Men's 8 — **Played:** GS — Silhouette: Male (PlayHQ) — Add photo · Games

GA — **Mason Clarke** — #9 · GA · 8 line ups this season — **Games:** Men's 8 — **Played:** GA — Silhouette: Male (PlayHQ) — Add photo · Games

WA — **Leo Patel** — #4 · WA · 8 line ups this season — **Games:** Men's 8 — **Played:** WA — Silhouette: Male (PlayHQ) — Add photo · Games

C — **Charlie Webb (C)** — #12 · C · 8 line ups this season — **Games:** Men's 8 — **Played:** C — Silhouette: Male (PlayHQ) — Add photo · Games

WD — **Henry Stone** — #5 · WD · 8 line ups this season — **Games:** Men's 8 — **Played:** WD — Silhouette: Male (PlayHQ) — Add photo · Games

GD — **Archie Vance** — #8 · GD · 8 line ups this season — **Games:** Men's 8 — **Played:** GD — Silhouette: Male (PlayHQ) — Add photo · Games

GK — **Finn Harper** — #3 · GK · 8 line ups this season — **Games:** Men's 8 — **Played:** GK — Silhouette: Male (PlayHQ) — Add photo · Games

SEVERAL PHOTOS AT ONCE

Drop photos here, each named after its athlete (e.g. **Jane Smith.jpg** or **smith_jane.png**), or choose files. Photos whose name doesn't match an athlete stay in the photo library below, where you can give each one to an athlete.

AN ATHLETE PLAYHQ DOESN'T LIST YET

Athlete's name exactly as PlayHQ will show it — Add photo

FIG 8.1

The Athletes page: teams grouped by division, every athlete PlayHQ has in a line-up this season, and the photo library.

8.1 Photos

Pick a team and add a photo to each athlete, or drop a folder of photos named after the athletes ("Jane Smith.jpg" or "smith_jane.png") on **Several photos at once**; any that don't match wait in the photo library to be given to someone. A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo. Studio also remembers each athlete's PlayHQ id, so if PlayHQ changes a player's name during the season, their photo still finds them. Photos are made smaller when saved; cut-out PNGs with a transparent background look best. Without a photo the graphics show a silhouette for the athlete's gender, taken from the division (men's, women's; mixed and open are neutral) and changeable under the athlete. **An athlete PlayHQ doesn't list yet** adds one ahead of their first game.

A photo only goes on the stream and the Socials pictures once **Approved for photo use** is ticked under the athlete. Tick it when the athlete, or their parent or guardian, has agreed to their image being used. Until then the graphics show the silhouette. **Approve all for photo use** ticks every athlete listed for the team at once.

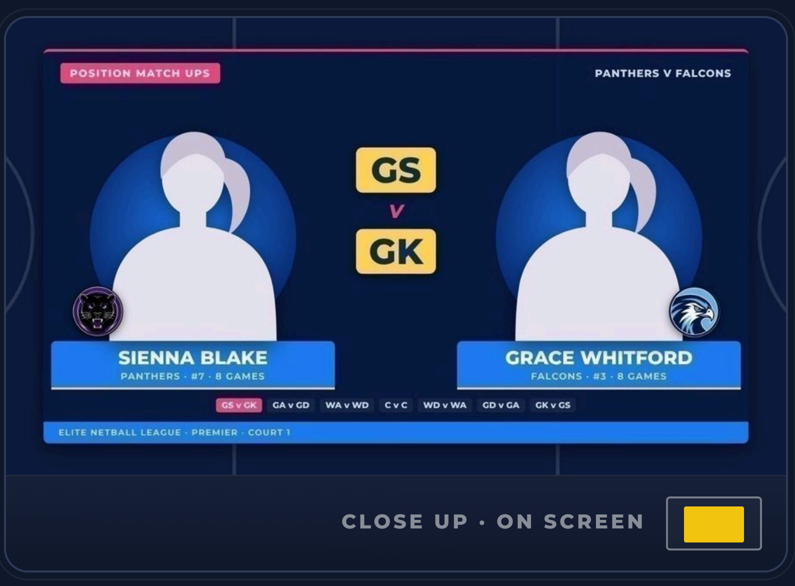


FIG 8.2 Position match ups use the photos, pairing by pairing.

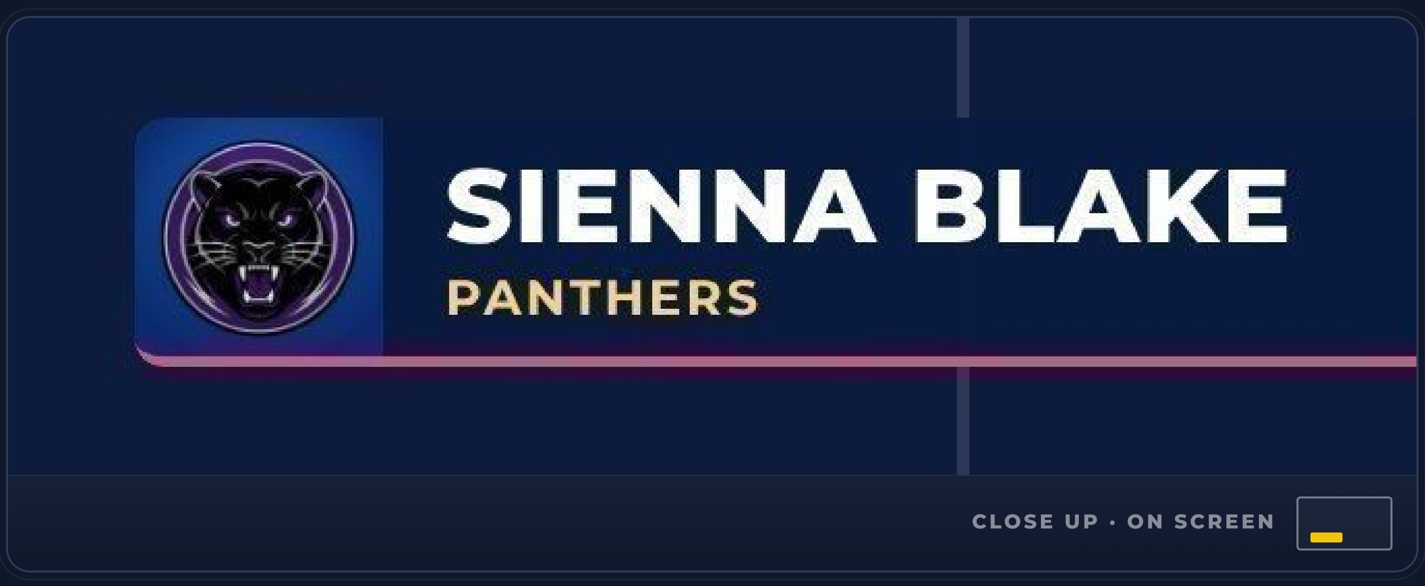
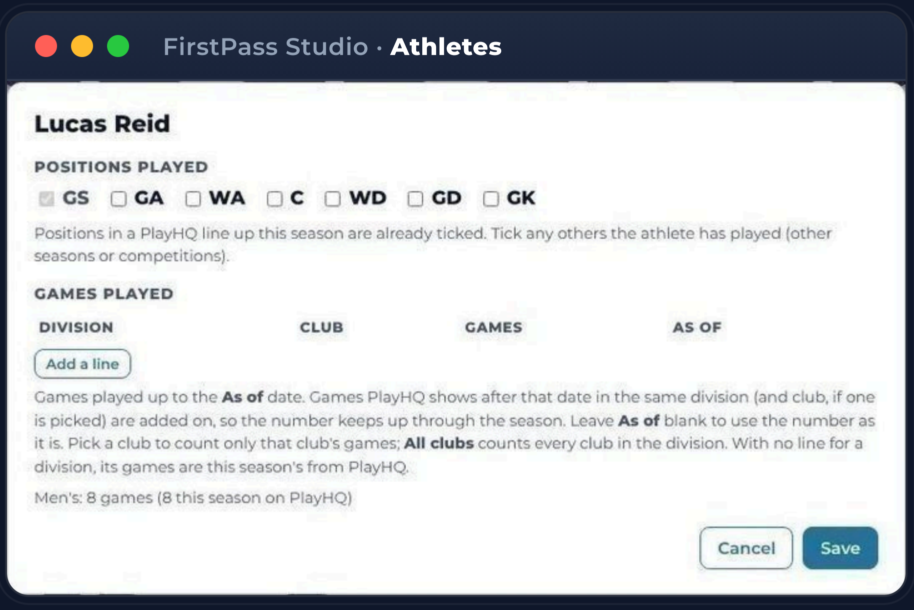


FIG 8.3 The athlete lower third, named from the register as you type on the Output page.

> ***A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo.***
>
> CHAPTER 8 · ATHLETES

8.2 Positions and games played

Games under an athlete opens their record. Positions played in a PlayHQ line-up this season are ticked already; tick any others from other seasons or competitions. Games played takes a line per division (and club): the number as of a date, to which Studio adds the games PlayHQ shows after it, so milestones keep up through the season. With no line, a division's games are this season's from PlayHQ.



Lucas Reid

POSITIONS PLAYED

☑ GS ☐ GA ☐ WA ☐ C ☐ WD ☐ GD ☐ GK

Positions in a PlayHQ line up this season are already ticked. Tick any others the athlete has played (other seasons or competitions).

GAMES PLAYED

DIVISION	CLUB	GAMES	AS OF

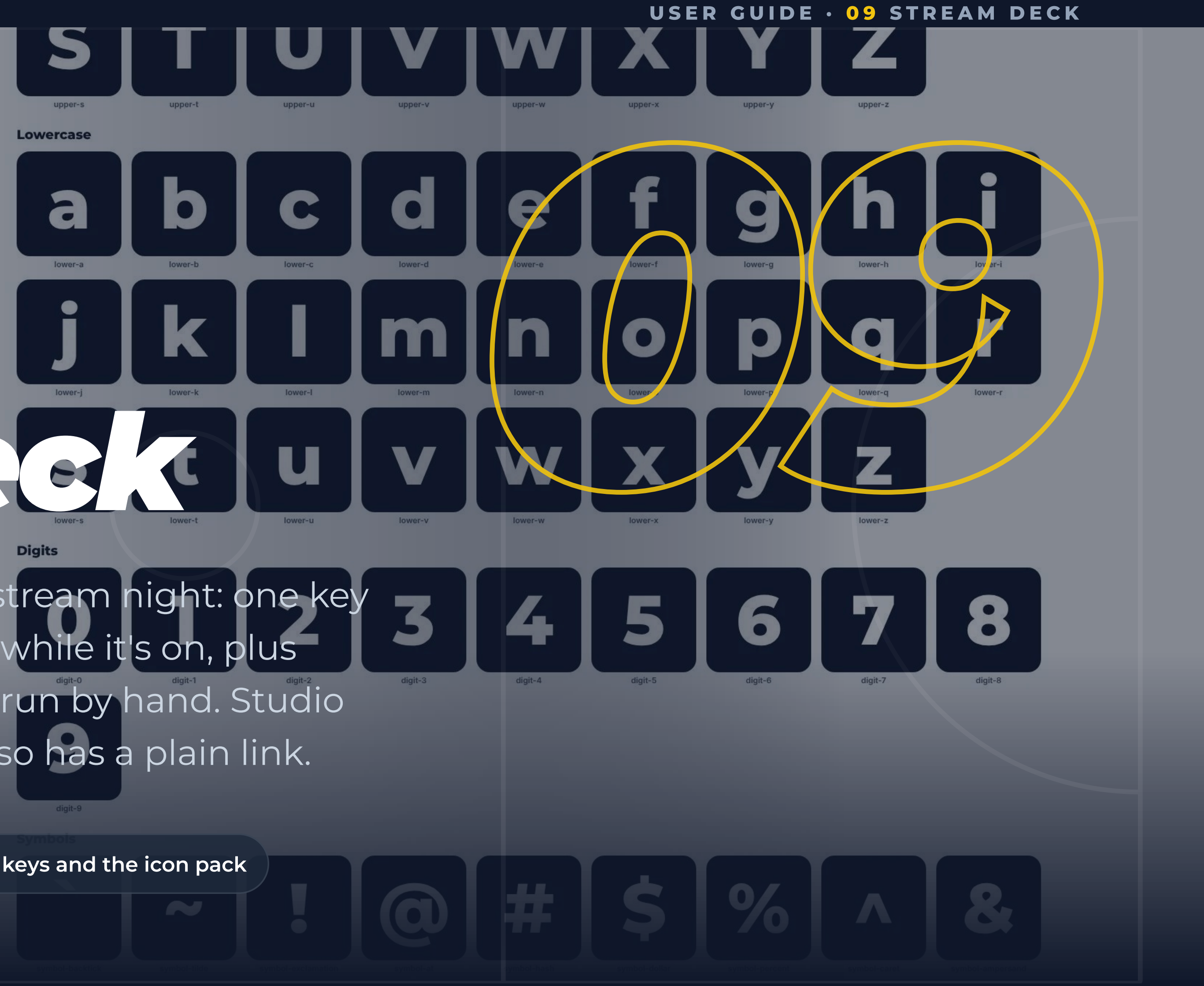
Add a line

Games played up to the **As of** date. Games PlayHQ shows after that date in the same division (and club, if one is picked) are added on, so the number keeps up through the season. Leave **As of** blank to use the number as it is. Pick a club to count only that club's games; **All clubs** counts every club in the division. With no line for a division, its games are this season's from PlayHQ.

Men's: 8 games (8 this season on PlayHQ)

Cancel · Save

FIG 8.4 An athlete's positions and games played.

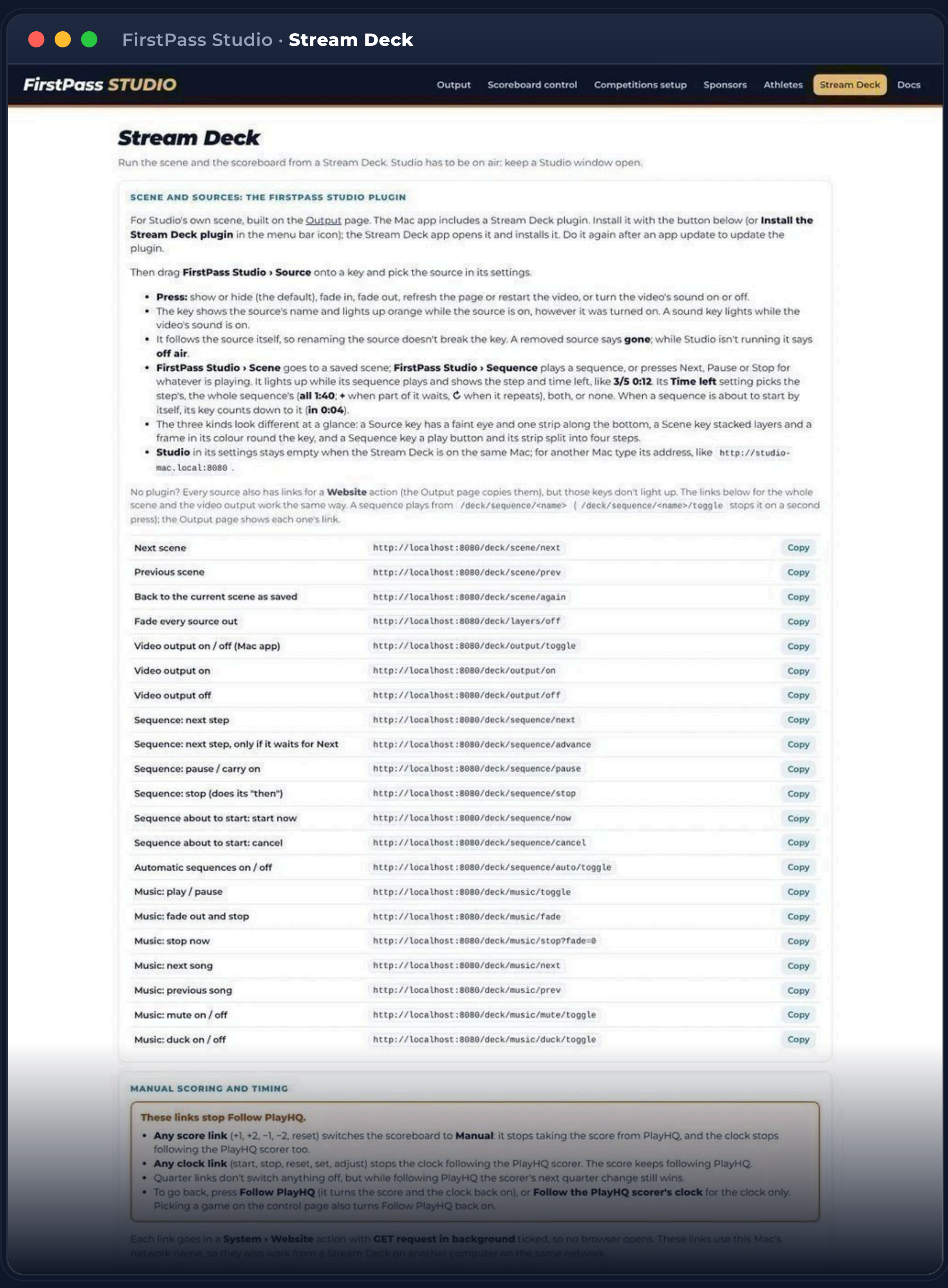


CHAPTER 9 OF 11

Stream Deck

A Stream Deck is the control surface for stream night: one key per source, scene and sequence, each lit while it's on, plus score, clock and quarter keys for a game run by hand. Studio ships its own plugin, and every control also has a plain link.

1 The plugin
2 Links for everything else
3 Timed keys and the icon pack



Stream Deck

Run the scene and the scoreboard from a Stream Deck. Studio has to be on air: keep a Studio window open.

SCENE AND SOURCES: THE FIRSTPASS STUDIO PLUGIN

For Studio's own scene, built on the Output page. The Mac app includes a Stream Deck plugin. Install it with the button below (or **Install the Stream Deck plugin** in the menu bar icon); the Stream Deck app opens it and installs it. Do it again after an app update to update the plugin.

Then drag **FirstPass Studio › Source** onto a key and pick the source in its settings.

- **Press:** show or hide (the default), fade in, fade out, refresh the page or restart the video, or turn the video's sound on or off.
- The key shows the source's name and lights up orange while the source is on, however it was turned on. A sound key lights while the video's sound is on.
- It follows the source itself, so renaming the source doesn't break the key. A removed source says **gone**; while Studio isn't running it says **off air**.
- **FirstPass Studio › Scene** goes to a saved scene; **FirstPass Studio › Sequence** plays a sequence, or presses Next, Pause or Stop for whatever is playing. It lights up while its sequence plays and shows the step and time left, like **3/5 0:12**. Its **Time left** setting picks the step's, the whole sequence's (**all 1:40**; **+** when part of it waits, ↻ when it repeats), both, or none. When a sequence is about to start by itself, its key counts down to it (**in 0:04**).
- The three kinds look different at a glance: a Source key has a faint eye and one strip along the bottom, a Scene key stacked layers and a frame in its colour round the key, and a Sequence key a play button and its strip split into four steps.
- **Studio** in its settings stays empty when the Stream Deck is on the same Mac; for another Mac type its address, like `http://studio-mac.local:8080`.

No plugin? Every source also has links for a **Website** action (the Output page copies them), but those keys don't light up. The links below for the whole scene and the video output work the same way. A sequence plays from `/deck/sequence/<name>` (`/deck/sequence/<name>/toggle` stops it on a second press); the Output page shows each one's link.

Next scene	`http://localhost:8080/deck/scene/next`	Copy
Previous scene	`http://localhost:8080/deck/scene/prev`	Copy
Back to the current scene as saved	`http://localhost:8080/deck/scene/again`	Copy
Fade every source out	`http://localhost:8080/deck/layers/off`	Copy
Video output on / off (Mac app)	`http://localhost:8080/deck/output/toggle`	Copy
Video output on	`http://localhost:8080/deck/output/on`	Copy
Video output off	`http://localhost:8080/deck/output/off`	Copy
Sequence: next step	`http://localhost:8080/deck/sequence/next`	Copy
Sequence: next step, only if it waits for Next	`http://localhost:8080/deck/sequence/advance`	Copy
Sequence: pause / carry on	`http://localhost:8080/deck/sequence/pause`	Copy
Sequence: stop (does its "then")	`http://localhost:8080/deck/sequence/stop`	Copy
Sequence about to start: start now	`http://localhost:8080/deck/sequence/now`	Copy
Sequence about to start: cancel	`http://localhost:8080/deck/sequence/cancel`	Copy
Automatic sequences on / off	`http://localhost:8080/deck/sequence/auto/toggle`	Copy
Music: play / pause	`http://localhost:8080/deck/music/toggle`	Copy
Music: fade out and stop	`http://localhost:8080/deck/music/fade`	Copy
Music: stop now	`http://localhost:8080/deck/music/stop?fade=0`	Copy
Music: next song	`http://localhost:8080/deck/music/next`	Copy
Music: previous song	`http://localhost:8080/deck/music/prev`	Copy
Music: mute on / off	`http://localhost:8080/deck/music/mute/toggle`	Copy
Music: duck on / off	`http://localhost:8080/deck/music/duck/toggle`	Copy

MANUAL SCORING AND TIMING

These links stop Follow PlayHQ.

- **Any score link** (+1, +2, −1, −2, reset) switches the scoreboard to **Manual**: it stops taking the score from PlayHQ, and the clock stops following the PlayHQ scorer too.
- **Any clock link** (start, stop, reset, set, adjust) stops the clock following the PlayHQ scorer. The score keeps following PlayHQ.
- Quarter links don't switch anything off, but while following PlayHQ the scorer's next quarter change still wins.
- To go back, press **Follow PlayHQ** (it turns the score and the clock back on), or **Follow the PlayHQ scorer's clock** for the clock only. Picking a game on the control page also turns Follow PlayHQ back on.

Each link goes in a **System › Website** action with **GET request in background** ticked, so no browser opens. These links use this Mac's network name, so they also work from a Stream Deck on another computer on the same network.

FIG 9.1

The Stream Deck tab: the plugin, the scoring and clock links, and the full reference below them.

9.1 The plugin

Install it from the 1P menu (**Install the Stream Deck plugin**) or the button on the Output page; the Stream Deck app opens and installs it. Do the same after an app update. The **Studio** field in a key's settings stays empty when the Stream Deck is on the same Mac; for another Mac, type its address, like `http://studio-mac.local:8080`. Four kinds of key, each a different shape at a glance:

KEY	LOOKS	PRESS	SHOWS
Source	A faint eye and one strip along the bottom	Show or hide (default), fade in, fade out, refresh the page or restart the video, video sound on or off	The source's name; lit in its output's colour while it's on, however it was turned on
Scene	Stacked layers and a frame in its colour	Go to a scene, next or previous scene, or back to the current scene as saved	Lit while its scene is on, with a yellow dot when sources have changed since
Sequence	A play button and a strip in four steps	Play or stop a sequence, play from the first step, or Next, Next when waiting, Pause and Stop for whatever is playing	Step and time left, like **3/5 0:12**, the whole sequence's time, both or none; **in 0:04** when one is about to start by itself
Switcher	A camera and a strip in two halves	A camera to preview (default) or program, stinger to a camera, Cut, Auto, an upstream or downstream key, Fade to black, run a macro	The camera's name from the switcher; red on program and green on preview, a key or Fade to black red while on air; a yellow dot while the switcher isn't answering

Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them; a removed one says **gone**, and while Studio is off air they say **off air**. Colour groups keys by the part of the stream: green pre-game, blue live, purple quarter time, grey post-game, yellow ads, or the scene's or sequence's own colour.

With more than one output, each key has an **Output** setting: the output it shows a source on, plays a scene on or plays a sequence on (a Switcher key has none: there is one switcher). A lit key takes its output's colour (Output 1 orange, 2 pink, 3 blue, 4 green) with a black title, so a commentators' Stream Deck of Commentary keys lights pink while the stream's keys light orange. A key for a removed output says **no output**.

> **“ Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them.**
>
> CHAPTER 9 · STREAM DECK

9.2 Links for everything else

Every control is also a link, for a **System > Website** action (tick **GET request in background**), a Multi Action, or any program on the network. Scoreboard control, the Output page and the Stream Deck tab copy them, with this Mac's Stream Deck key after the port (`…:8080/k/key/deck/…`) so no web page can press them. The Studio plugin needs no key: the first time you press one of its keys the Mac asks you to **Allow** it, and from then on it signs every press with its own key pair. A link without it is refused, so copy any older Website keys' links again from Studio. With more than one Stream Deck, give each its own link set: pick **New link set…** in **Copy links for** on the Stream Deck tab, name it after the Stream Deck, and copy its links from there. If a link gets out, choose **Devices…** in the menu bar icon and press **Make a new key** on that set: its links stop working and you copy them again, while the other Stream Decks carry on. A program on another computer can ask for a key of its own: the Mac asks you to **Allow** or **Deny** it by name and address, and Devices… can take that one away without touching your other keys. The useful ones:

GROUP	LINKS
Scores	Home and away +1, +2, −1, −2; reset to 0; play a super shot or any streak without changing the score.
Scoring mode	Follow PlayHQ, Manual, toggle. Any score key switches to Manual.
Clock	Start, stop, toggle, reset, set to a time, add or take off any amount, follow the PlayHQ clock on or off, quarter length.
Quarter	Q1 to Q4, extra time, next, previous. The clock is separate; pair them in a Multi Action.
Game	Automatic selection on, off, toggle; the court to follow.
Ticker	Faster, slower, normal, a set speed.
Sources	Show, hide, toggle, restart, video sound; every source out; video output on or off.
Scenes	Go to a scene, next, previous, back to the scene as saved.
Sequences	Play, toggle, next, advance (only when waiting), pause, stop, start the one about to start now, cancel it, automatic sequences on or off.
Music	Play, pause, resume, stop, fade out, next, previous, load a song or playlist, mute, duck; add `?fade=2` for a 2-second fade.
Sponsors	Play the next sponsor ad due.
Switcher	Cut, auto, stinger (to a camera or what's on preview), fade to black, a camera to preview or program, an upstream or downstream key on, off or toggled, run a macro.

A LAYOUT THAT WORKS Put the scene keys in a row across the top (Pre-game, Live game, Clear), the sequence keys under them with Next when waiting and Stop at the end, and a page of Source keys for the lower thirds and stats. A game run by hand gets a second page: home scores left, away scores right, the clock and quarter keys down the middle.

9.3 Timed keys and the icon pack

A second plugin, **Timed Sequence**, arms a key for a time of day and then presses a list of links with delays between them: reset the board at 18:55 and start the clock at 19:00, say. It is built from the repository. The **FirstPass Studio Characters** icon pack gives Stream Deck keys letters, numbers and symbols in the Studio style, for labelling keys by hand.

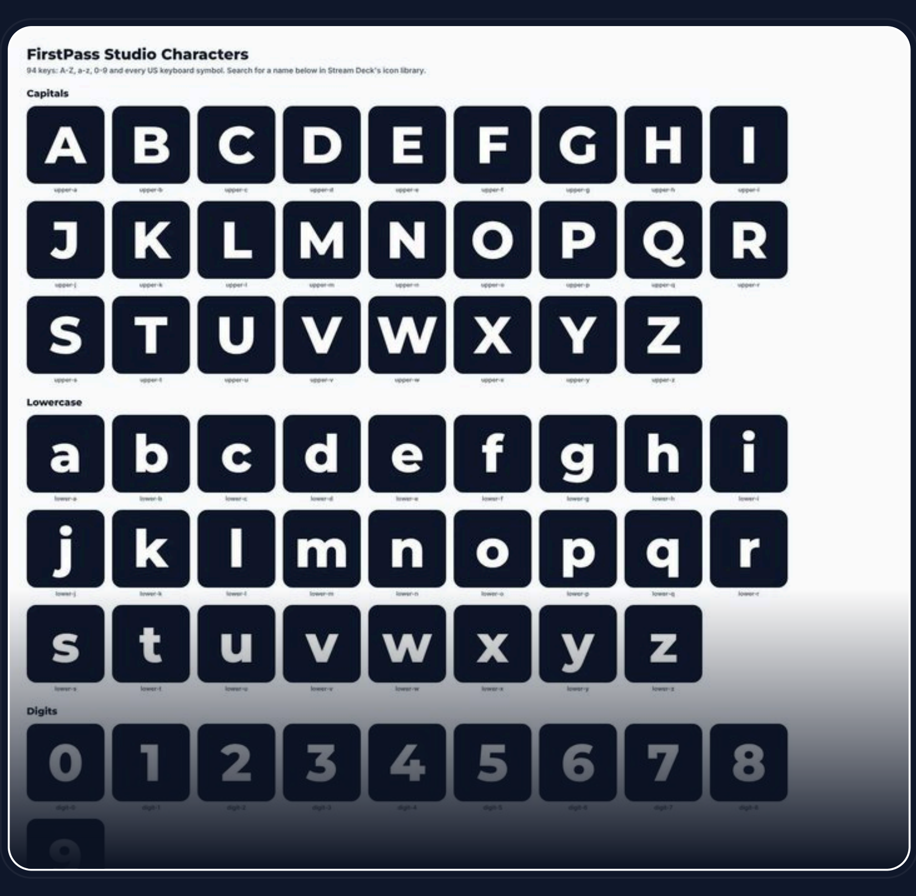

FirstPass Studio Characters
94 keys: A-Z, a-z, 0-9 and every US keyboard symbol. Search for a name below in Stream Deck's icon library.
Capitals
Lowercase
Digits

FIG 9.2 The Characters icon pack: A to Z, a to z, 0 to 9 and the usual symbols on navy.

CHAPTER 10 OF 11

Settings and care

Where Studio keeps its settings, how to carry them between Macs and back them up, how updates arrive, and what to do when something isn't right.

- 1 The 1P menu
- 2 The data folder
- 3 Settings in iCloud Drive
- 4 Backup and restore
- 5 Updates
- 6 Links and other devices
- 7 When something isn't right
- 8 Data and bandwidth

10.1 *The 1P menu*

The menu bar icon is Studio's home. From top to bottom: the version and whether Studio is on air; **Stop Studio** (while on air); **Open Studio**, Scoreboard control, Competition setup and **Copy the Studio link**; **Devices...**; **Install the Stream Deck plugin**; **Open the settings folder**, **Settings folder** (iCloud), **Back up...** and **Restore...**; **Open the debug log**, **Open the app log**, **Restart Studio**, **Check for updates**; **Open at login**, **Only this Mac can change things**, **What's new...** and Quit.

10.2 *The data folder*

Everything Studio saves lives in one folder outside the app, `~/Library/Application Support/FirstPass Studio`, so updates never touch it. In it: **competitions/** (a folder per competition with its settings, logos, font, athlete photos and register), **sponsors/**, the scene, sequences, music and playlists, the **media** folder, **control.json** (Scoreboard control's choices), **scoreboard-state.json** (the game, scores and clock, saved every few seconds so a restart mid-game loses nothing), **config.json** for this Mac's own settings, **logs/** and **backups/**.

10.3 *Settings in iCloud Drive*

Settings folder > Move to iCloud Drive... moves the settings into a folder you choose in iCloud Drive, in a subfolder named after this Mac, so each Mac on the same Apple account keeps its own and none saves over another's. The debug log, scores and clock, caches and window positions always stay on the Mac. **Restore from another Mac...** copies another Mac's settings over this one's (competitions, scenes, sequences, music, control choices), backing this Mac's up first. **Move back to this Mac...** undoes the move. Studio goes off air for a moment for each and comes straight back.

10.4 *Backup and restore*

Back up... saves every setting in one zip named with the date and time, with notes you type and, if you choose, the media folder; beside it goes a text summary listing the competitions, sequences, playlists, sources, scenes and sponsors inside. **Restore...** shows when and where a backup was made and what it holds, backs up this Mac's settings first, and replaces them only once the whole zip has unpacked and checked out. This Mac keeps its own **Only this Mac can change things** choice, port and link name whatever the backup says. Make a backup after setting up a season and before trying anything new on the Output page.

10.5 *Updates*

Updates download by themselves and install when the app quits, never mid-stream; **Check for updates** looks now, and **Restart to update** appears when one is ready. If a newer version has come out since that one downloaded, it asks whether to get the newer one first or install the one it named. The first launch after an update shows **What's new**, the changes since the version you last opened. When an update changes how settings are saved, Studio backs the data folder up to **backups/** first; settings saved by a newer Studio open read-only in an older one.

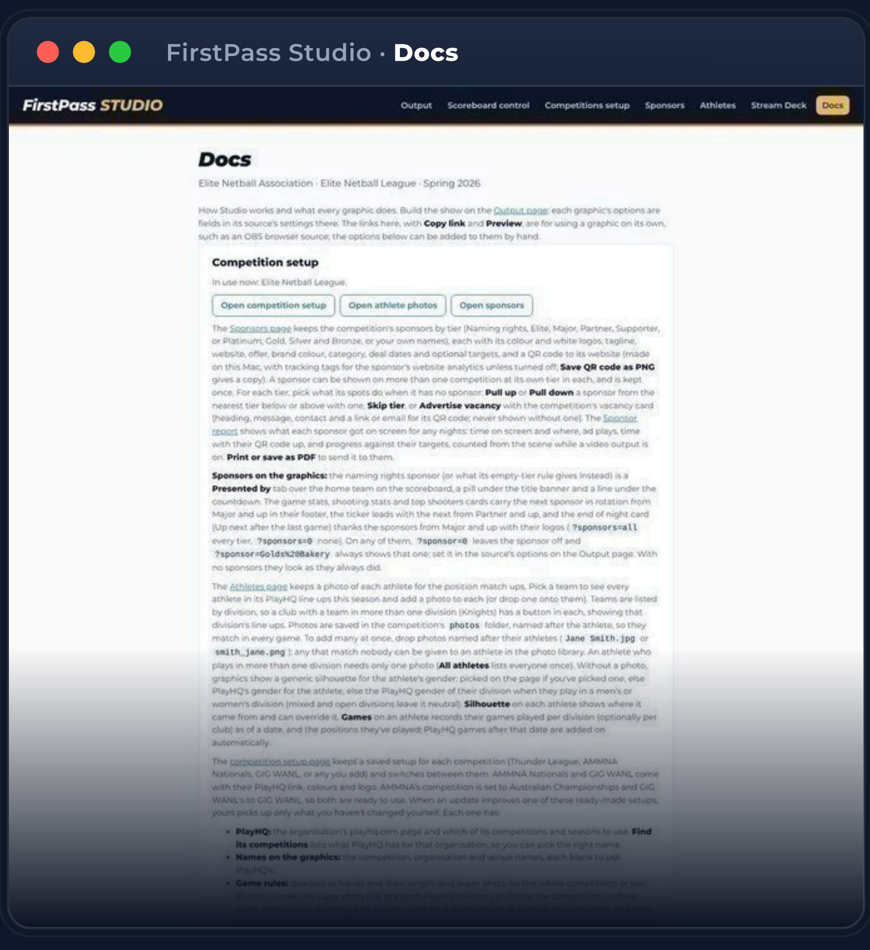


FIG 10.1 The Docs tab: every page with its link, Copy and Preview, the page options, and the notes this guide draws on.

> ***Updates download by themselves and install when the app quits, never mid-stream.***
>
> CHAPTER 10 · SETTINGS AND CARE

10.6 *Links and other devices*

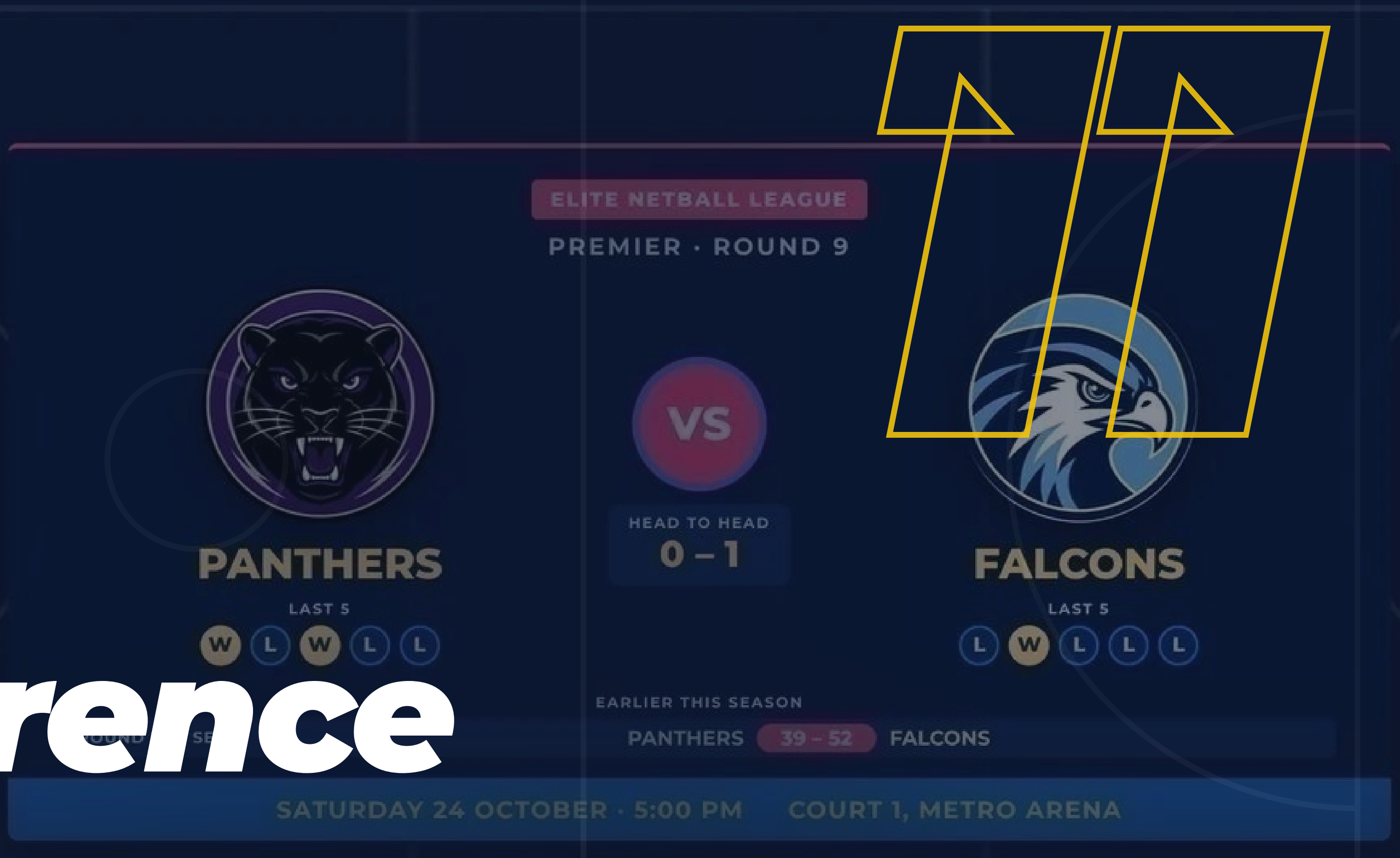
Studio's links use this Mac's Bonjour name (the Local hostname in System Settings > General > Sharing), so the same link works from a phone, another computer or a Stream Deck on the network; `localhost` still works on the Mac. Anything on the network can open the graphics. Only this Mac can change scores, the scene, sequences and settings: **Only this Mac can change things** is ticked every time Studio opens. To score from a phone or let another computer drive, pair it instead: open `/pair.html` on it (the Studio link with `/pair.html` on the end), type a name and press **Ask the Mac**, then choose **Allow scoring only** or **Allow everything** on the Mac. A paired device works with the box ticked, and **Devices...** in the menu bar icon lists it, changes what it may do, or removes it. Unticking the box lets every device on the network in until Studio quits; keep that for emergencies. On a network you don't control, leave it ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

10.7 *When something isn't right*

WHAT YOU SEE	WHAT TO DO
The board doesn't move before the game	Normal: nothing is read before the electronic scoring window opens. Scoreboard control says when it will start checking.
The board stops mid-game	PlayHQ isn't answering. Switch to Manual and run it by hand; Follow PlayHQ picks it up again when PlayHQ is back.
The wrong night or game is on	Pick the game night on Scoreboard control and pick the game; tick automatic selection again afterwards.
A key says off air	No Studio window is open. Open Studio from the 1P menu.
A graphic looks stale	Press its Restart key or turn it off and on; with **Load the page fresh** on it reloads each time.
Studio itself stops	**Restart Studio** in the 1P menu brings it back with the scores and clock it had.
Anything else	**Open the debug log** in the 1P menu. Everything Studio does and every PlayHQ answer is in it, 15 MB at most. Send it with a note of the time.

10.8 *Data and bandwidth*

The graphics only ever talk to Studio on the Mac. Studio alone contacts PlayHQ: one small request every 5 seconds for the followed game (or as set, down to one a second), the ticker's other courts every 30 seconds, the fixture about once a minute for automatic selection, and the rest once when a graphic loads. The Production Planner and the Socials tab read the same fixture, results and ladders through the same cache, and finished games' line ups once, so they add nothing; the Fixture board reads other courts' live scores the ticker's way, so it adds nothing while the ticker is on and at most one request a minute per live game when it isn't. Commentator stats' auto-refresh asks Studio only; its team lists are read at the start and at each quarter break. Finished games are read once and kept. Extra outputs read the same data from Studio, so they add nothing here. It all comes to well under a megabyte an hour, so a phone hotspot is enough for Studio; the stream itself is the bandwidth.



CHAPTER 11 OF 11

Quick reference

1. Keyboard
2. Stream night in nine lines
3. Where things are set

11.1 Keyboard

WHERE	KEYS
Scoreboard control	A / Z home +1 / −1, K / M away +1 / −1, Shift for +2 / −2, Space starts or stops the clock.
Any Studio window	⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Stream Deck, Docs. ⌘N new window, ⇧⌘O open in the browser, ⇧⌘C copy the page link, ⌘[back, ⌥⌘W close every Studio window.

11.2 Stream night in nine lines

01 Open Studio from the 1P menu and keep the Output page open all night.

02 Check the competition in use and the court on Competitions setup.

03 On Scoreboard control, the first game on your court is picked with Follow PlayHQ on.

04 Turn on the video output your switcher takes and show one graphic to see it land.

05 Play a few seconds of music and an animation with sound.

06 Press a Source key and watch the Output page follow it; put scoring back to Follow PlayHQ after testing.

07 Go to the Live game scene and let the sequences run the breaks.

08 Pick finals winners as they happen; turn on lower thirds when people appear.

09 Press Stream close, turn the output off, and send the sponsor report.

11.3 Where things are set

TO CHANGE	GO TO
Colours, font, logos, rules, finals format, court	Competitions setup
Which game, scoring mode, score, clock, finals winners	Scoreboard control
What's on screen, scenes, sequences, playlists, music, mics, outputs	Output
A graphic's options (heading, position, division, animate away)	Output, under the source's address
Sponsors, tiers, ads, moments, the report	Sponsors
Photos, positions, games played	Athletes
Each night's court, the commentators and who calls each game	Production Planner
Keys and links	Stream Deck
Pictures of a game night for social media	Socials
Backups, iCloud, updates, the debug log	The 1P menu

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.

FirstPass STUDIO

MADE BY PIVT

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